

Lecture 16

Climate Risk and Financial Instruments

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AEM 4510

Roadmap

1. **Weather markets (Schlenker and Taylor, 2021):** do traders forecast weather and climate?
2. **Municipal bond markets (Painter, 2020):** is sea level rise capitalized into municipal financing costs?
 - Marginal damages
3. **Prediction markets (Meng, 2017):** what is the probability of environmental regulation?
4. **Equity markets (Meng, 2017):** what is the financial impact of expected environmental regulation?
 - Marginal abatement costs

Weather markets

Betting on the weather

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How?

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How?

Studying the market for weather derivatives

Betting on the weather

Weather derivatives are a way for weather-exposed firms to manage climate risk

- Which kind of firms?

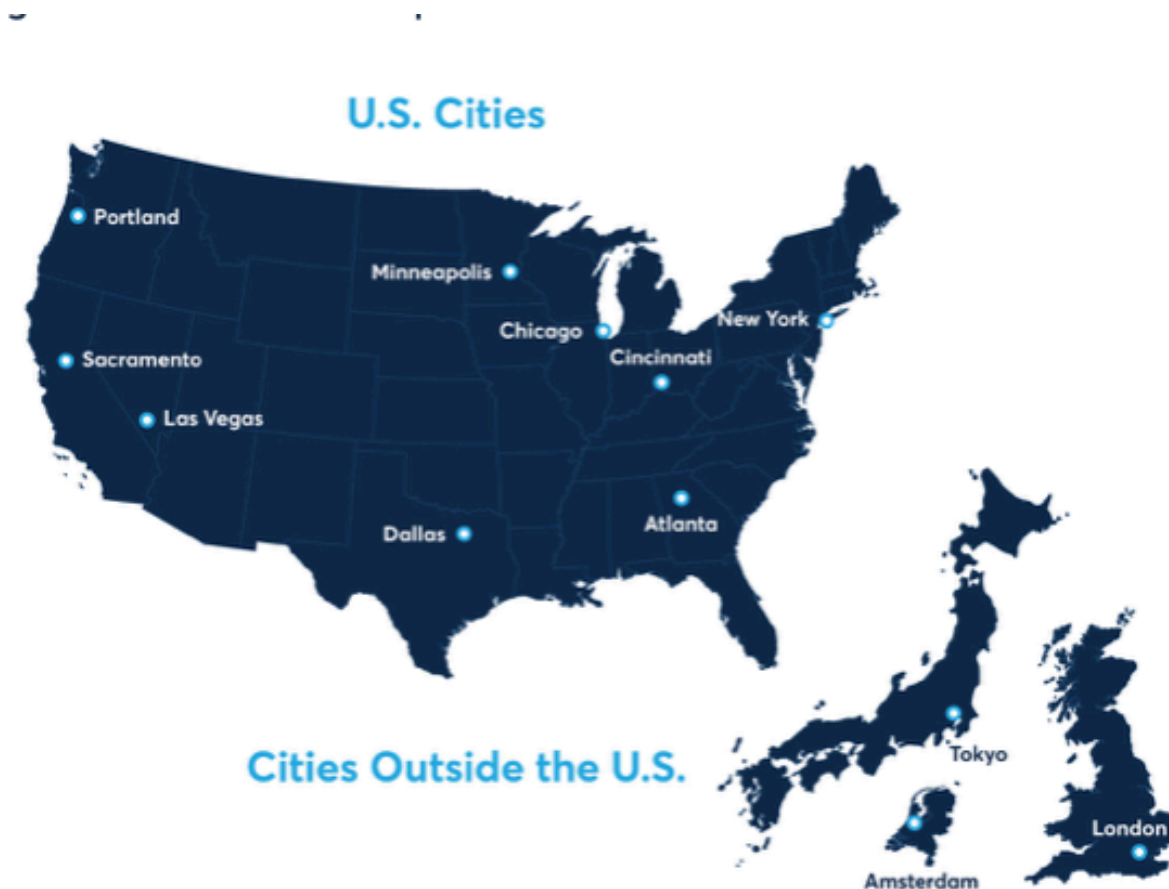


Betting on the weather

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CME offers contracts based on weather indices in 13 cities (mostly US)



Betting on the weather

Winter contracts are based on **heating degree days (HDD)**

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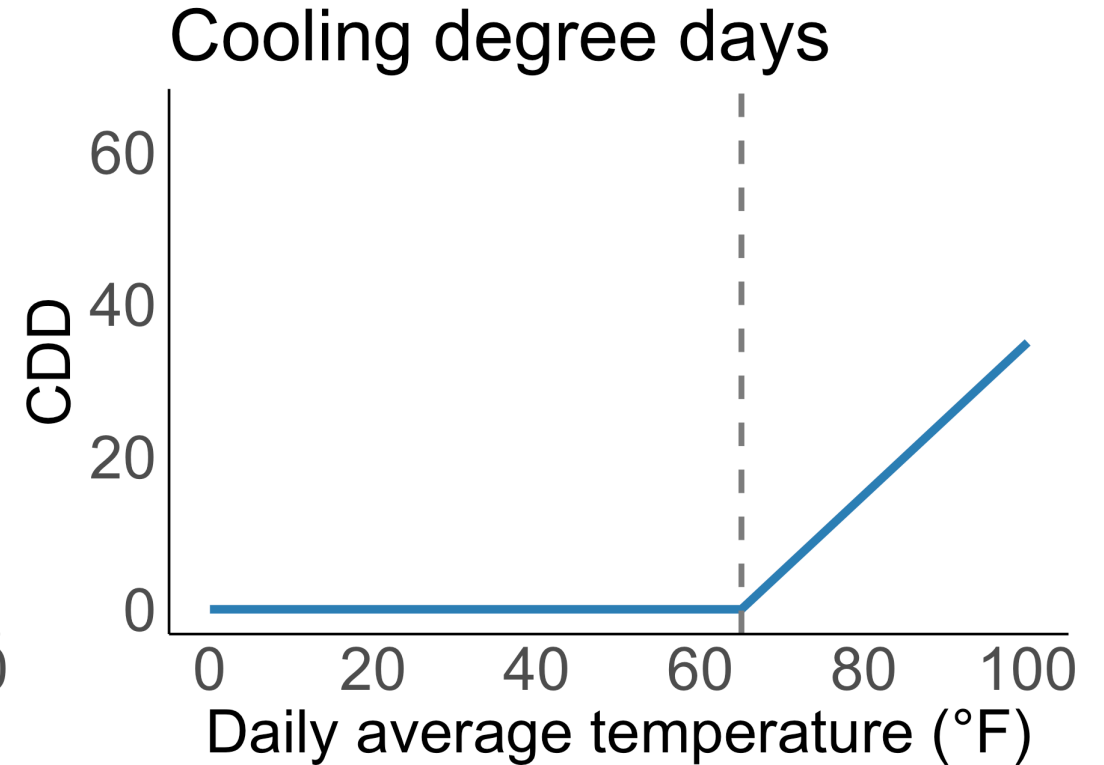
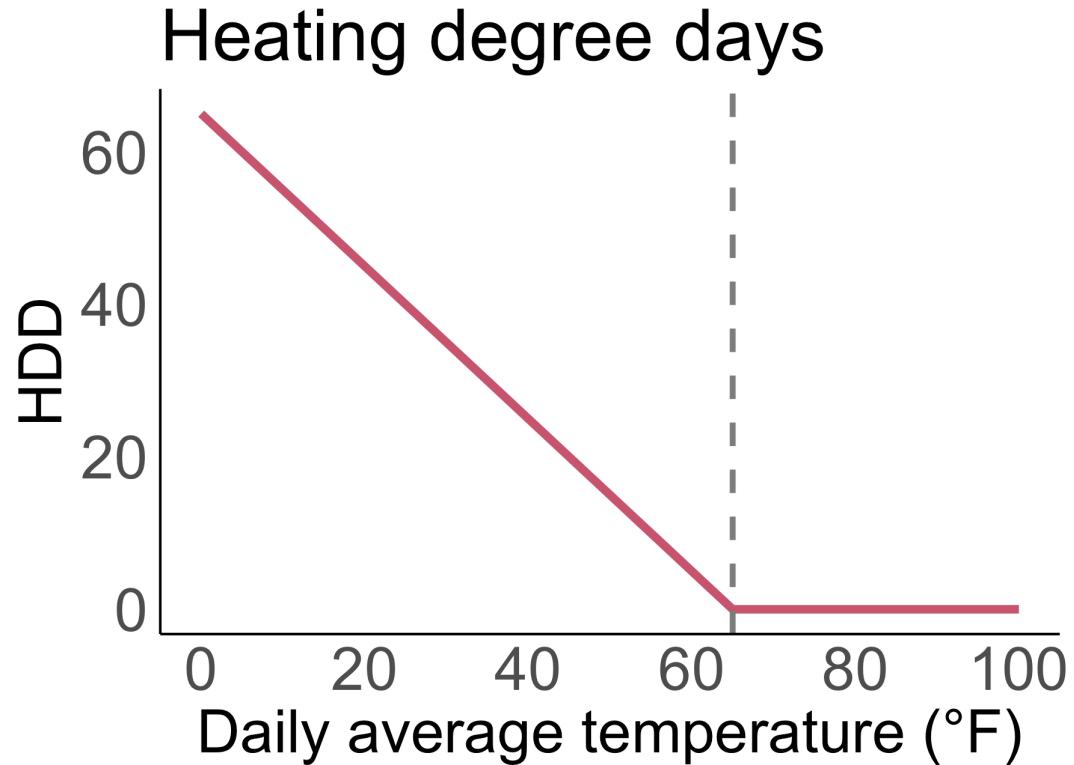
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Idea is 65°F is about where you would heat or cool a building to during the day

Betting on the weather



At 65°F both contracts are at zero

Below 65 only HDDs matter, above 65 only CDDs matter

Betting on the weather

How do the contracts work?

Betting on the weather

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The contracts are bought and sold like regular assets, but the settlement price is explicitly based on the realized HDD/CDD and are sold in terms of months

For example for July:

$$CDD_{july} = \sum_{d=1}^{31} CDD_{july,d} = \sum_{d=1}^{31} \max(0, 65 - T_{july,d})$$

where we sum over the days of July d

Betting on the weather

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If *actual* July CDDs happened to be 330, a buy-side trader profits:
 $20 \times (330 - 300) = 600$

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Sell, Why?

Mitigating risk: simple example

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Futures profits increase in HDD_{actual} , natural gas profits decrease

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How many contracts should NYSEG sell if it wants to eliminate all risk?

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Eliminate risk by setting sum of futures market profit and HDD-related natural gas profit to zero

$$Y \times 20 \times (1250 - HDD_{actual}) + 80,000 \times (HDD_{actual} - 1250) = 0$$

$$Y = 4,000 \text{ contracts}$$

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The contract looks **undervalued** so the trader buys July CDD contracts

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Profit per contract is:

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Higher demand pushes the July CDD price up: 300 → 308 → 323 → 330

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As the price rises, expected profit on the remaining trades gets smaller

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Higher demand pushes the July CDD price up: $300 \rightarrow 308 \rightarrow 323 \rightarrow 330$

As the price rises, expected profit on the remaining trades gets smaller

The trader keeps buying until price is close to the forecast-implied value

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Trading does two things at once:

1. Informed traders make profits on their private information
2. Prices become more informative about expected HDD/CDD through their trades

Betting on the weather

Who else might participate in these markets (summer or winter)?

- Farmers
- Amusement parks
- Electricity utilities
- Snow plow services
- People who think they have better private information
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If traders actually internalize climate information, we should see weather derivative prices respond to weather and climate forecasts

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How do they do it?

1. Compute the **weather anomaly**: how much warmer or cooler a day is relative to its average (accounting for overall warming over time)
2. Compute whether the change in the price from open to close on a given day is associated with weather anomalies

Betting on the weather

Which weather forecast provides the biggest impact on weather futures prices?

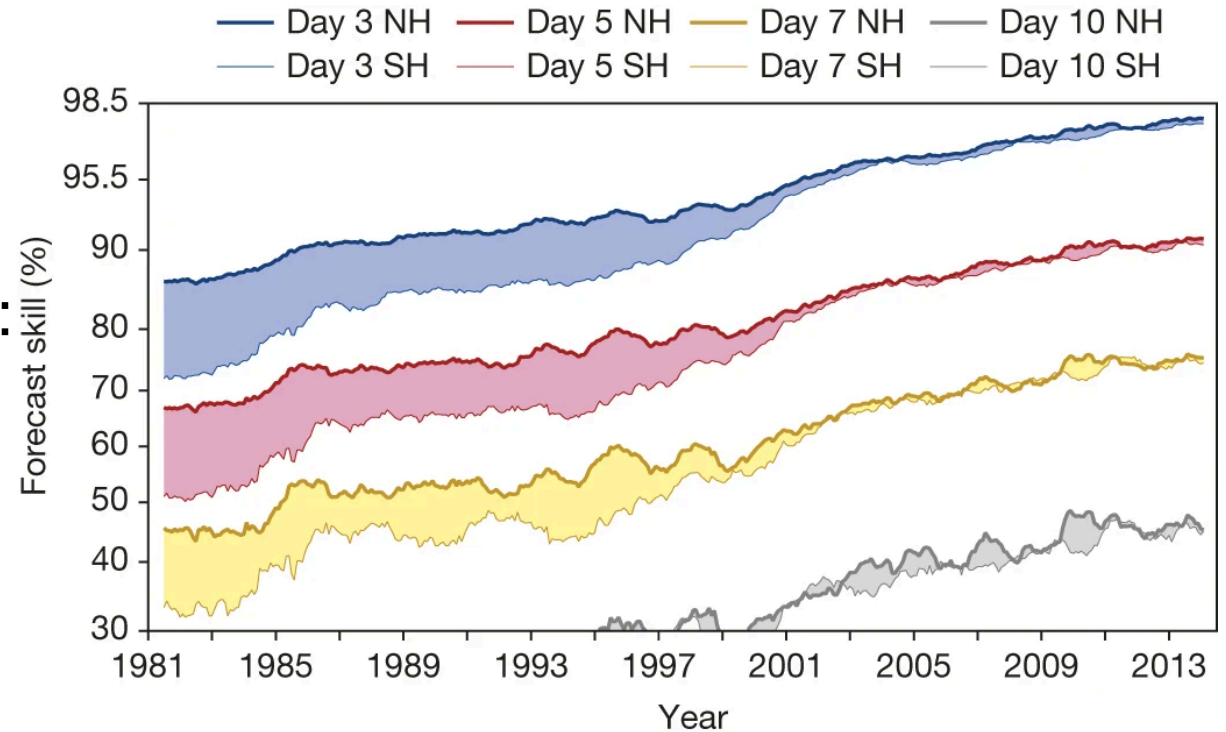
Another way of asking this: how many days ahead is a weather forecast providing the most **new** information about weather

First lets see how good forecasts are at different lead times from 3 to 10 days ahead

Betting on the weather

1-3 day forecasts are essentially perfect now

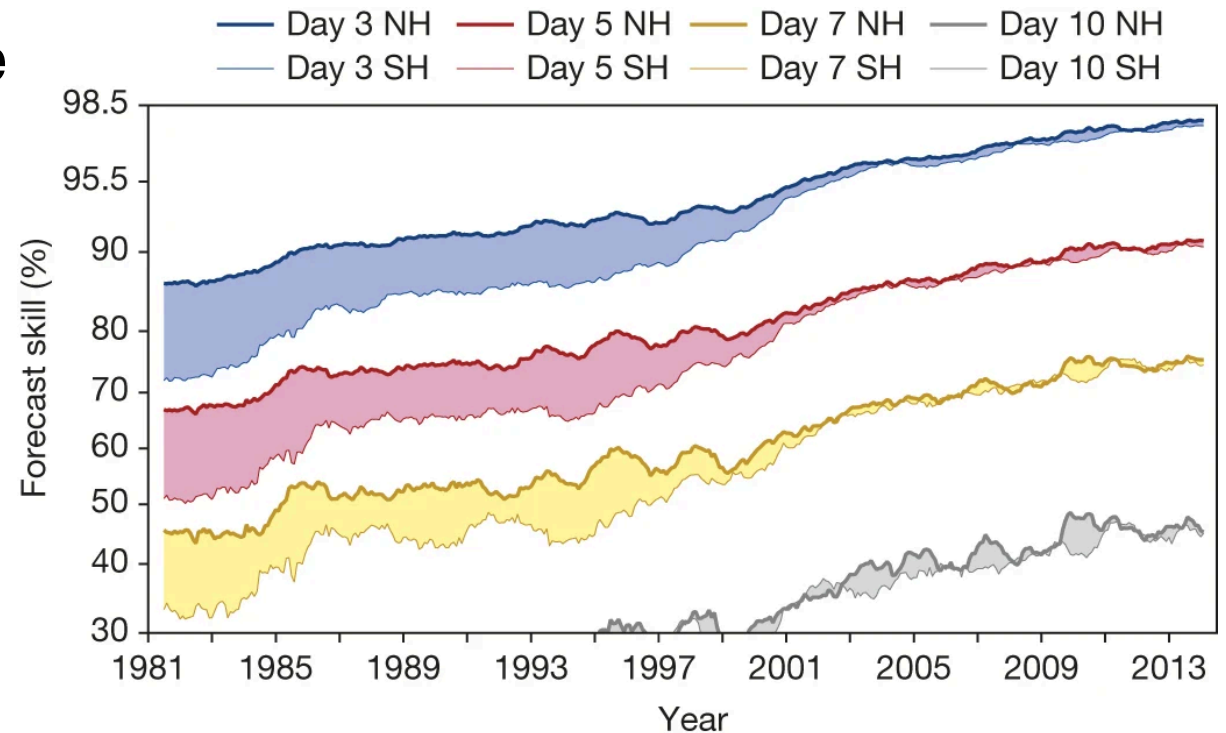
10 day forecasts have 40% "skill":
40% smaller error than if you just assumed temperature would be its long-run average



Betting on the weather

Forecasts >10 days out have little skill → little information value

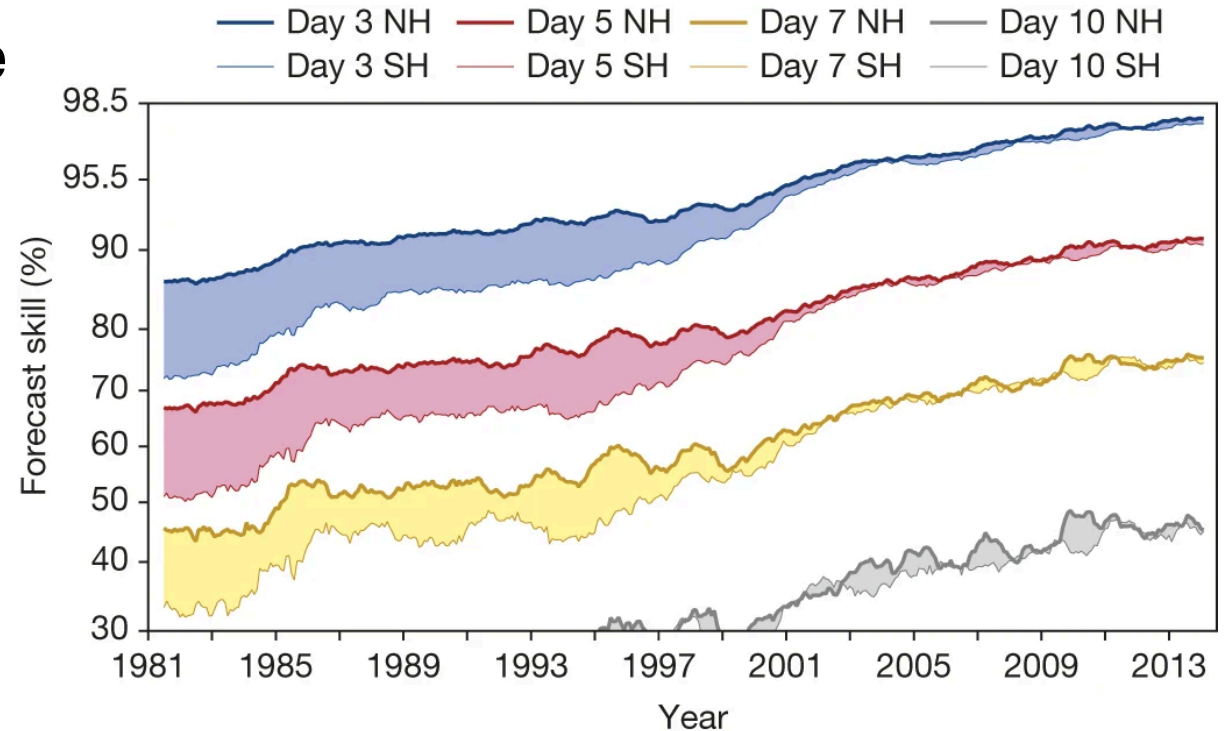
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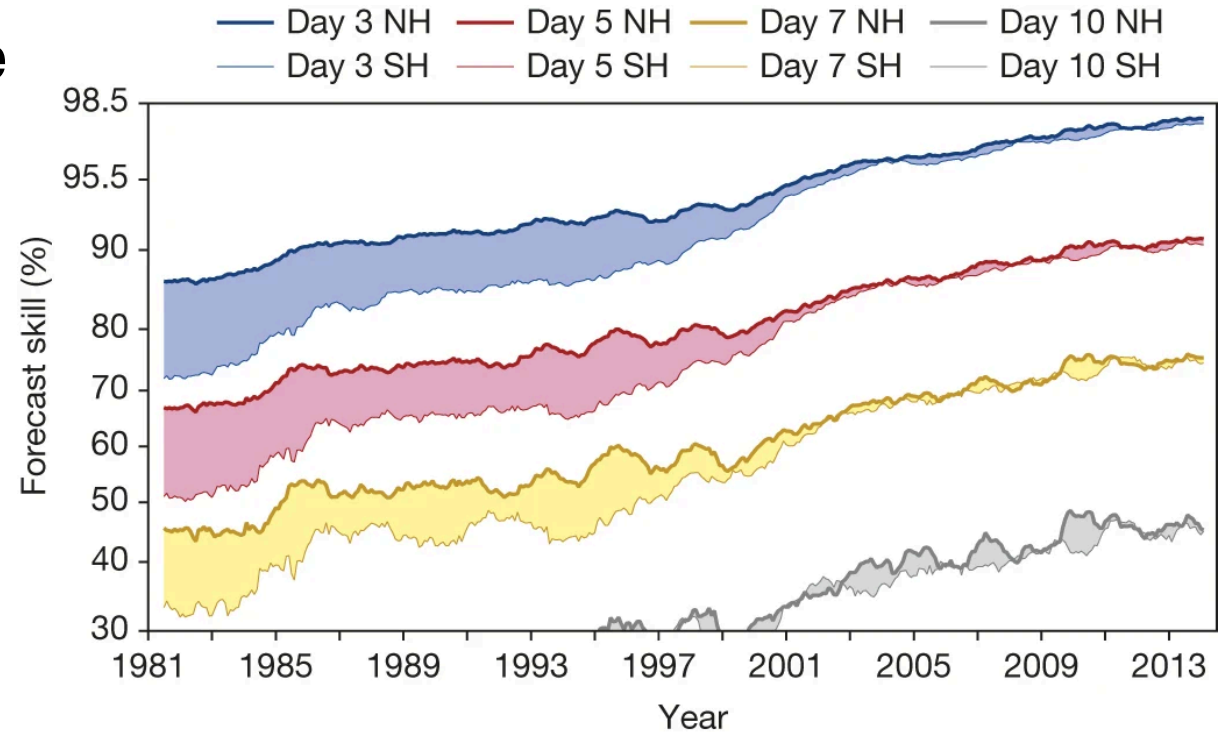
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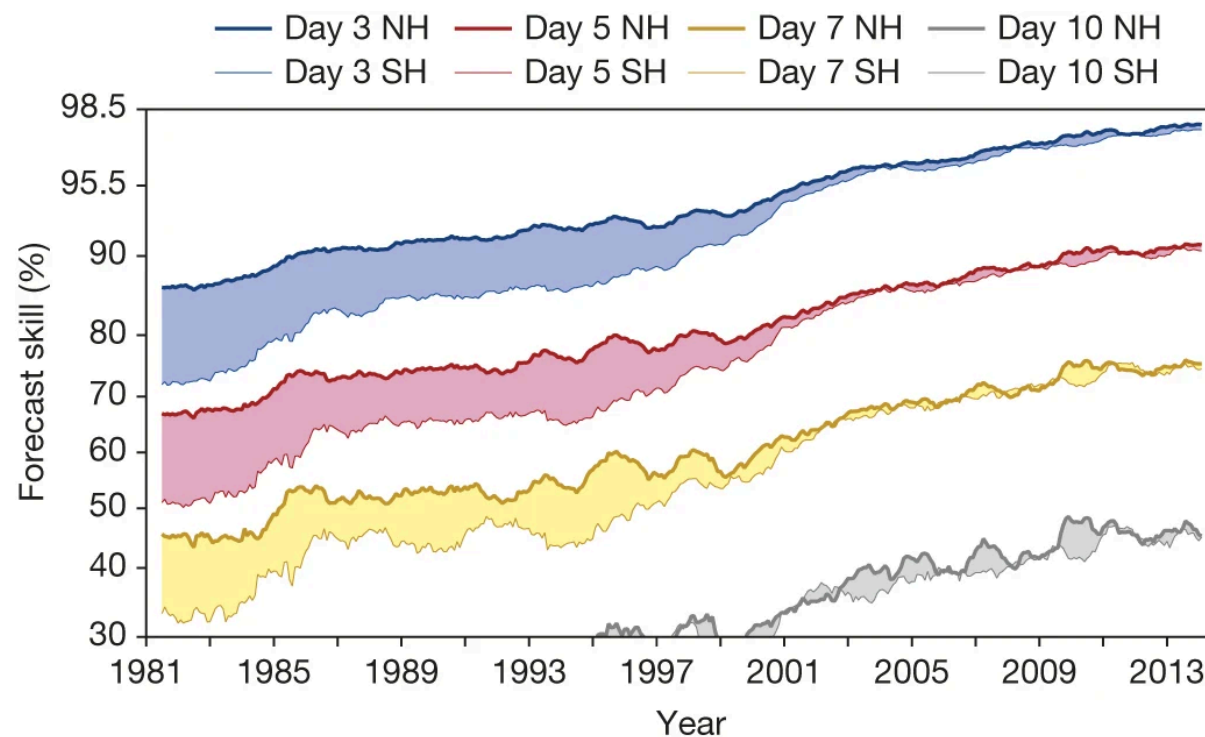
Forecasts 1-3 days out have near-perfect skill, but their information is probably already capitalized by prior forecasts 5-7 days ago



Betting on the weather

We should expect forecasts 3-10 days out to matter the most for contract prices

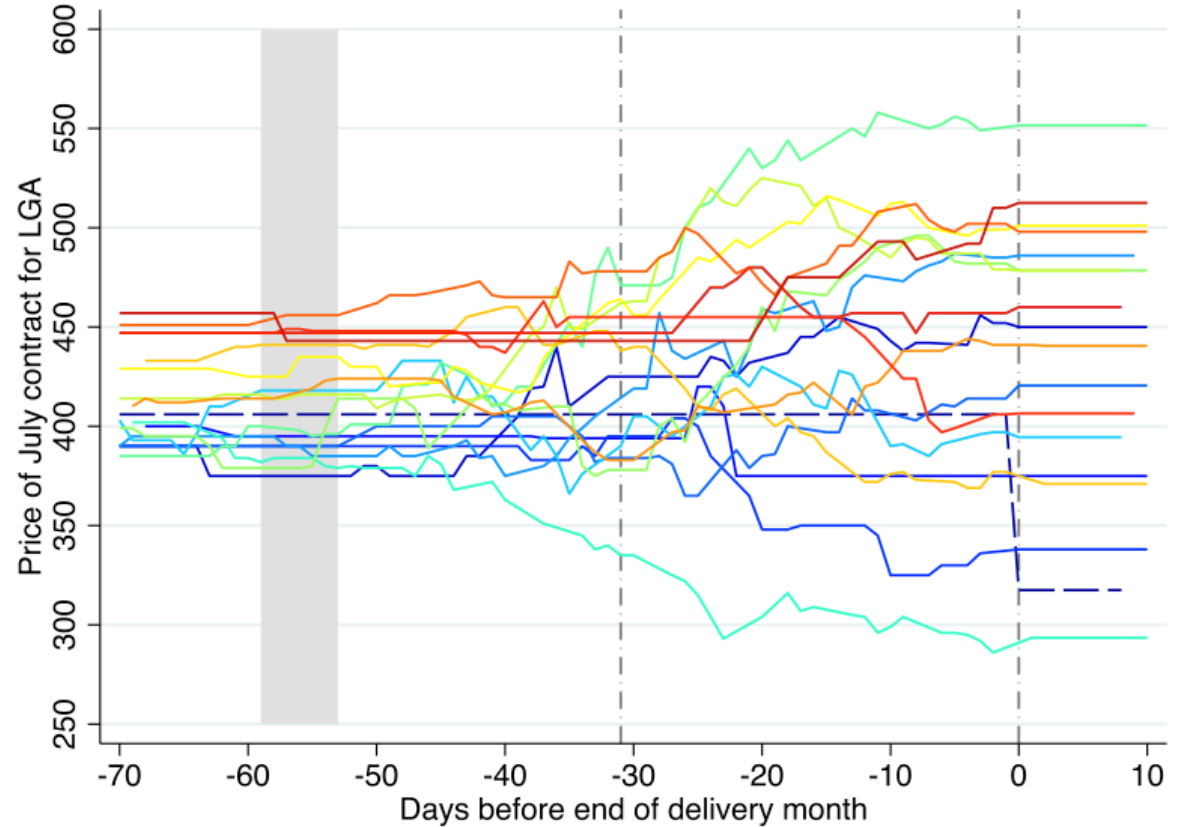
We should expect little to zero effect of 1-3 day forecasts, and 10+ day forecasts



July CDD prices for Lagaardia airport

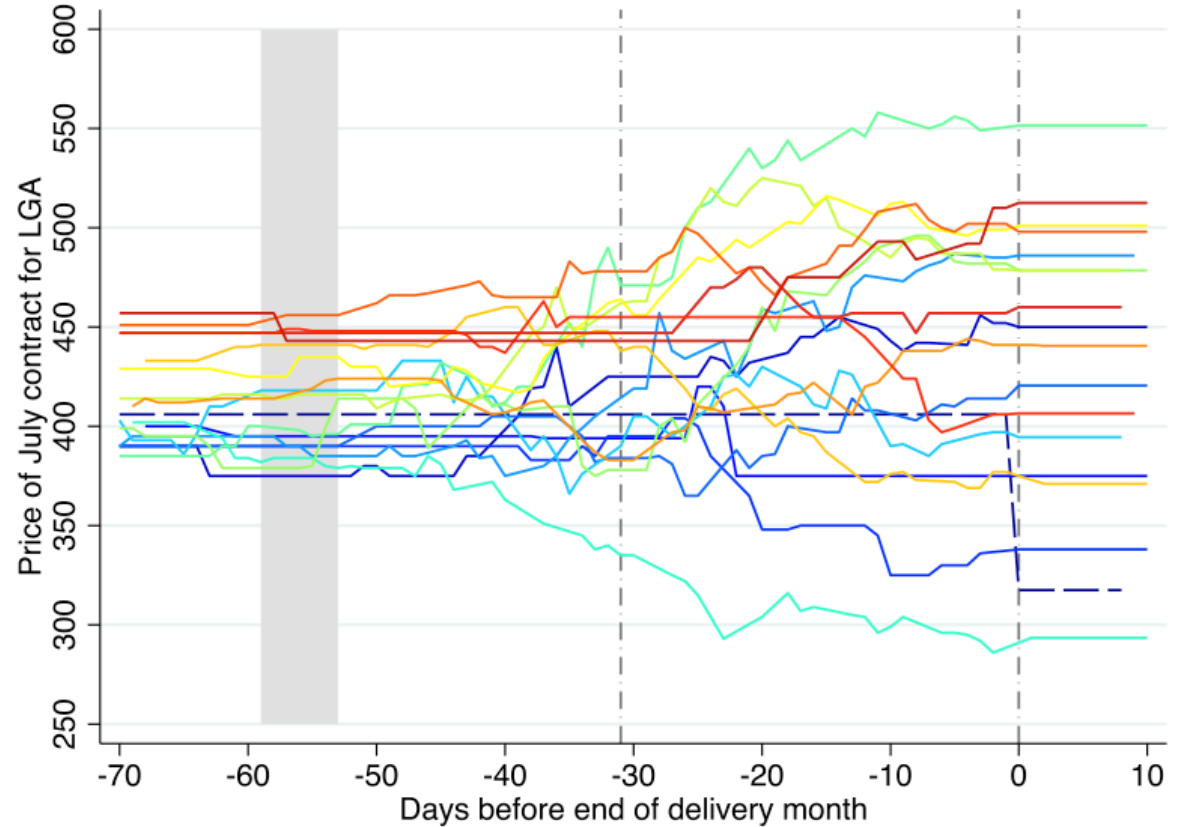
July in NY averages about 400 CDDs

Each year from 2001-2020 (different color lines) differs in terms of **actual CDDs** (price at 0), and **expected CDDs** (prices to the left of 0)



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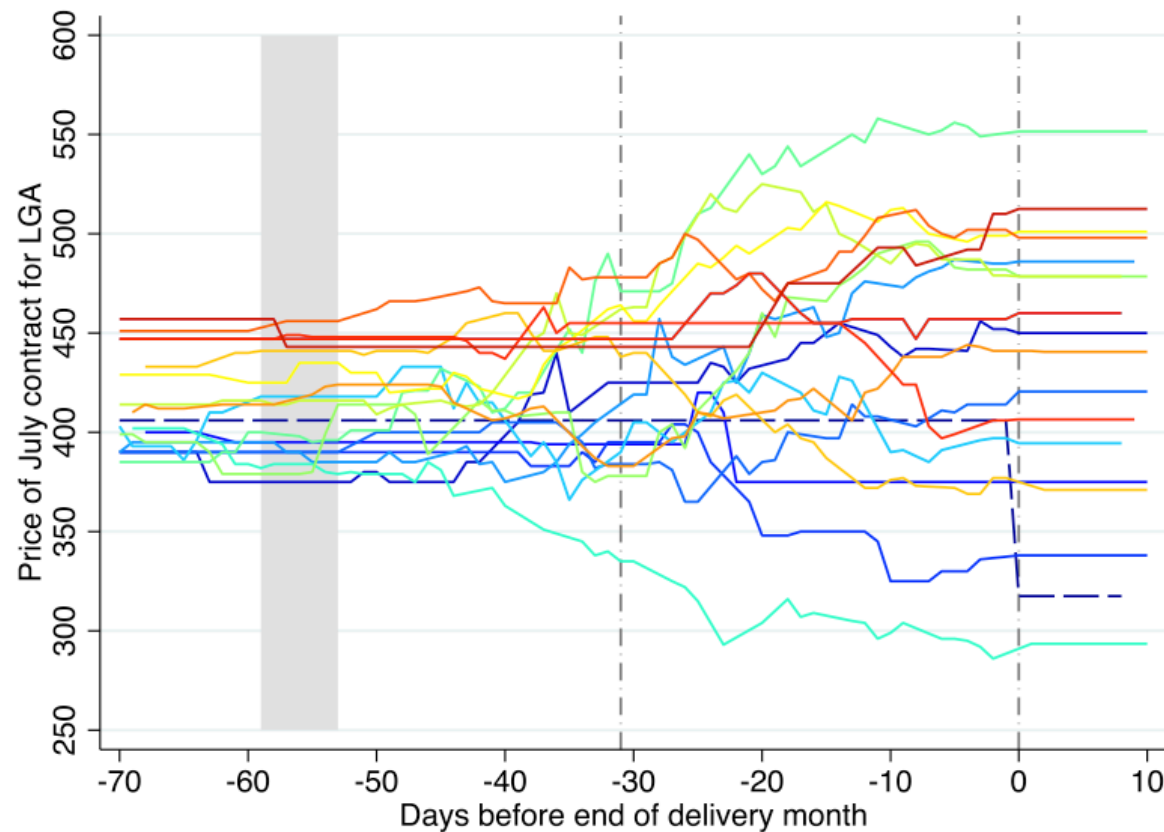
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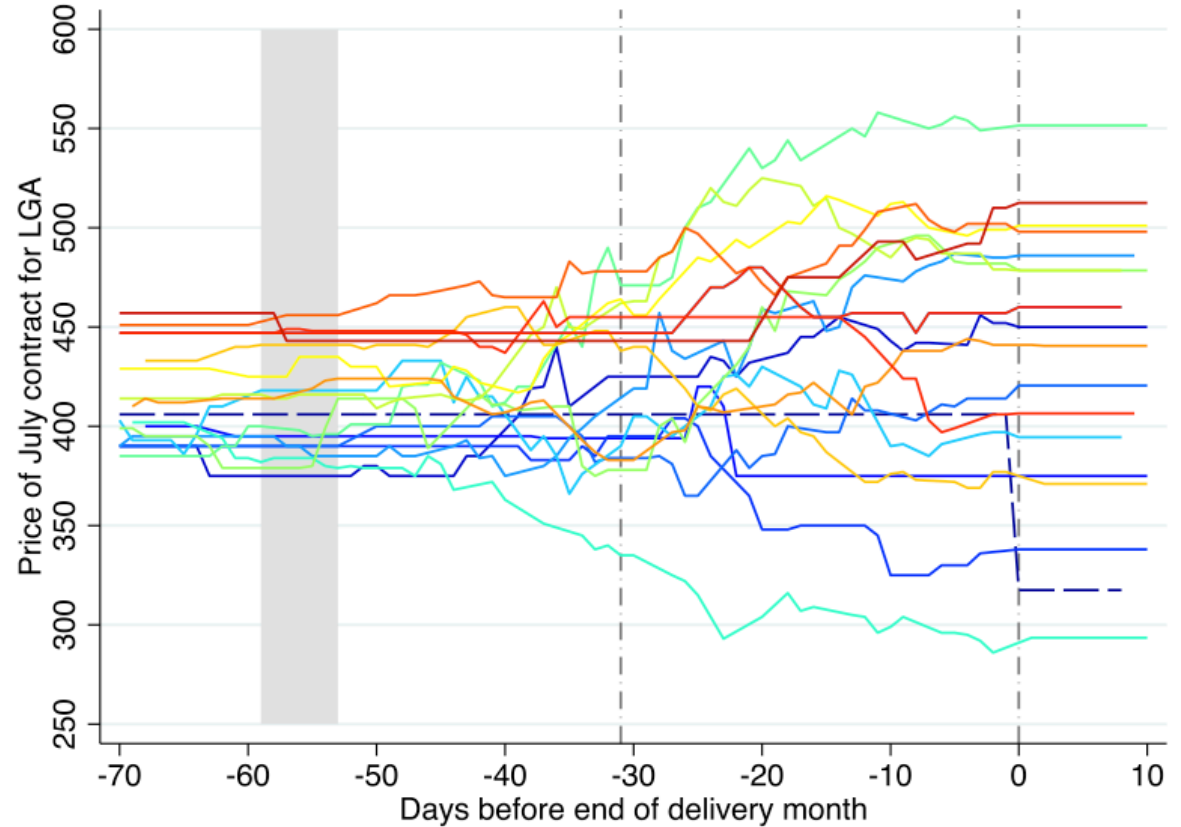
What do you see?

In general, prices don't move much further than 10 days before the start of July → consistent with short-run 10+ day forecasts not containing a lot of info



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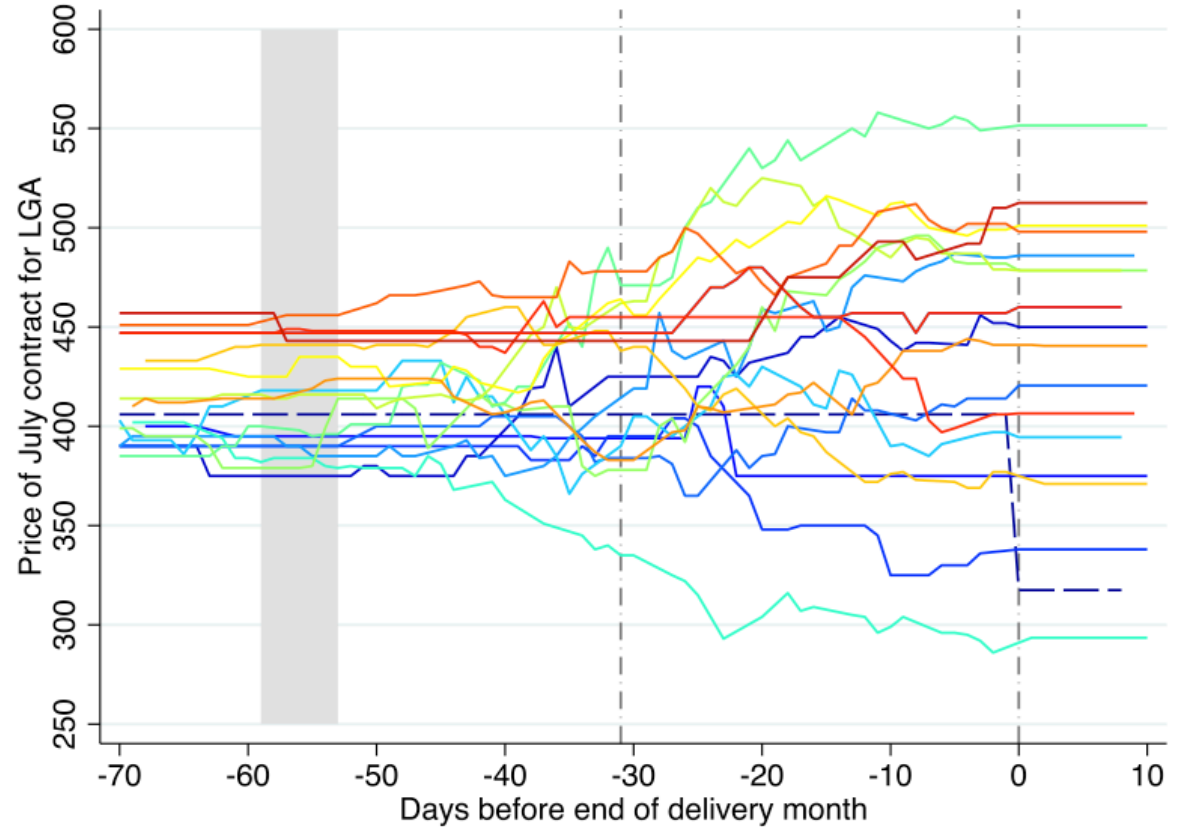
But if that's true, why are there differences across years?



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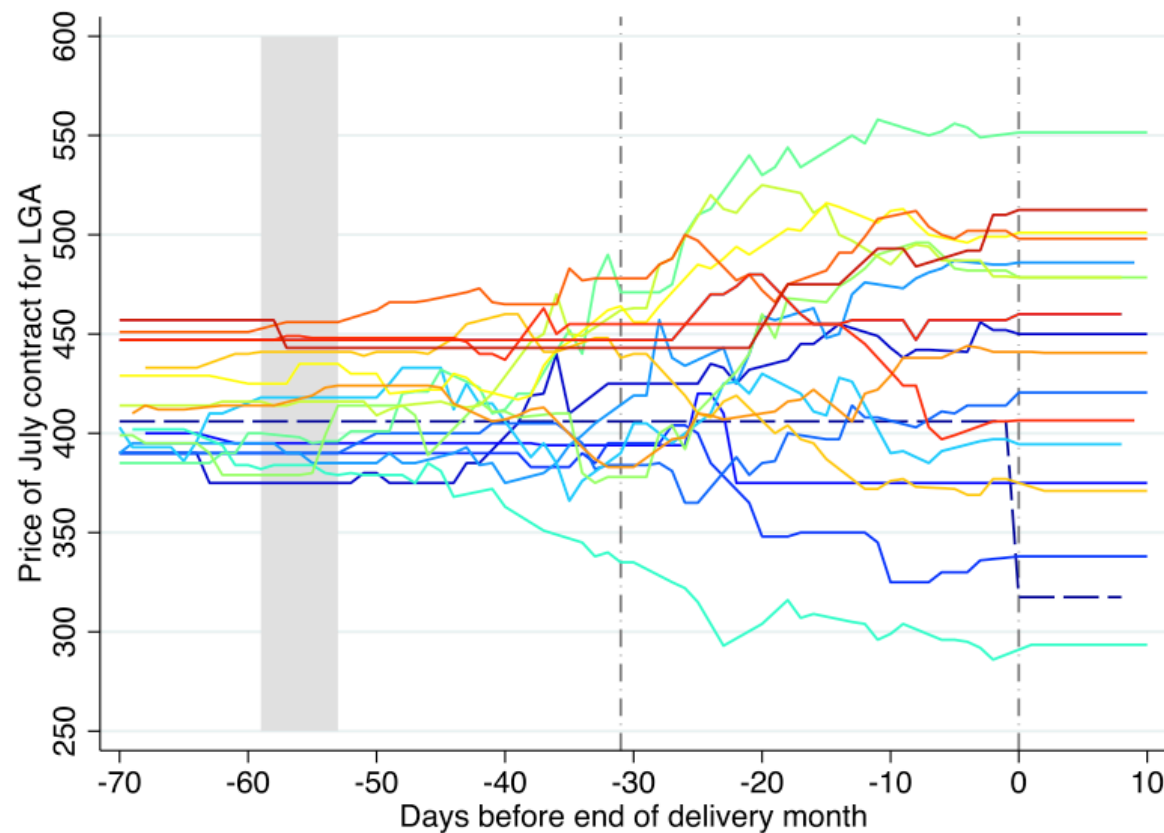
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Differences across years can be from long-run trends, El Nino, etc



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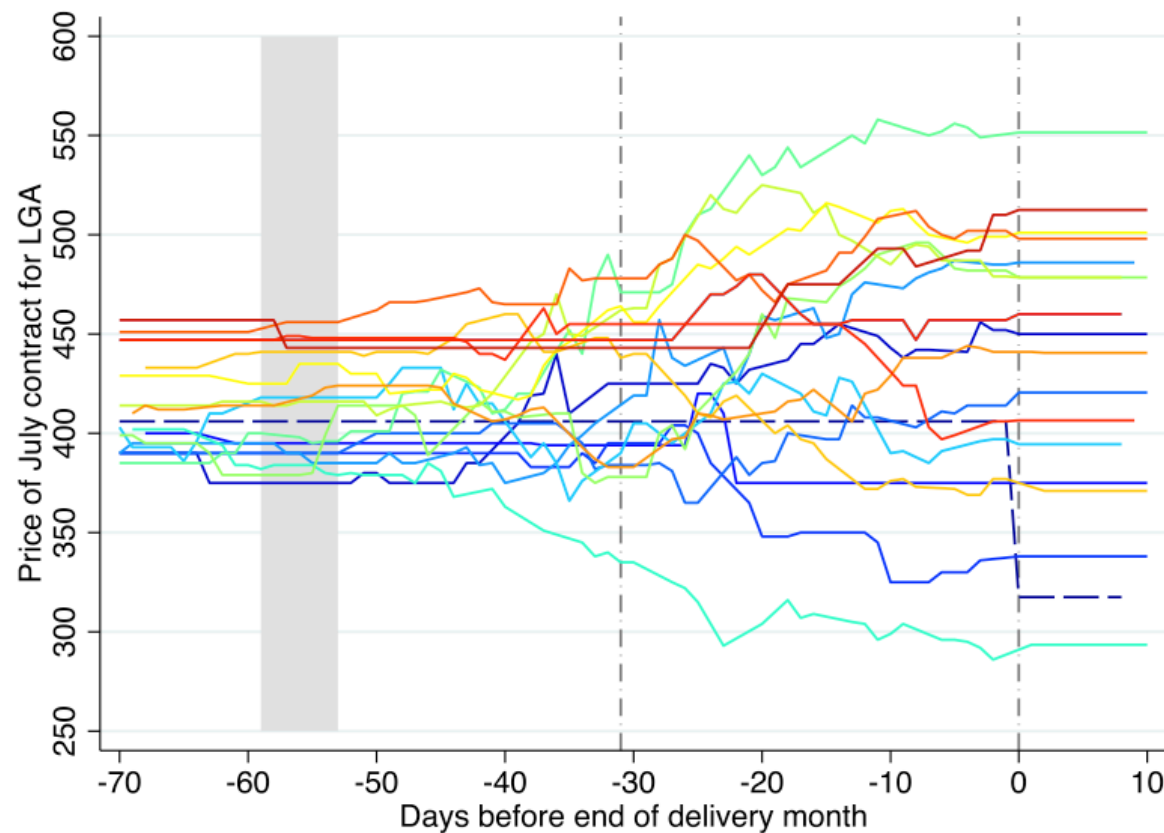
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They start trending toward their realized values (at 0)

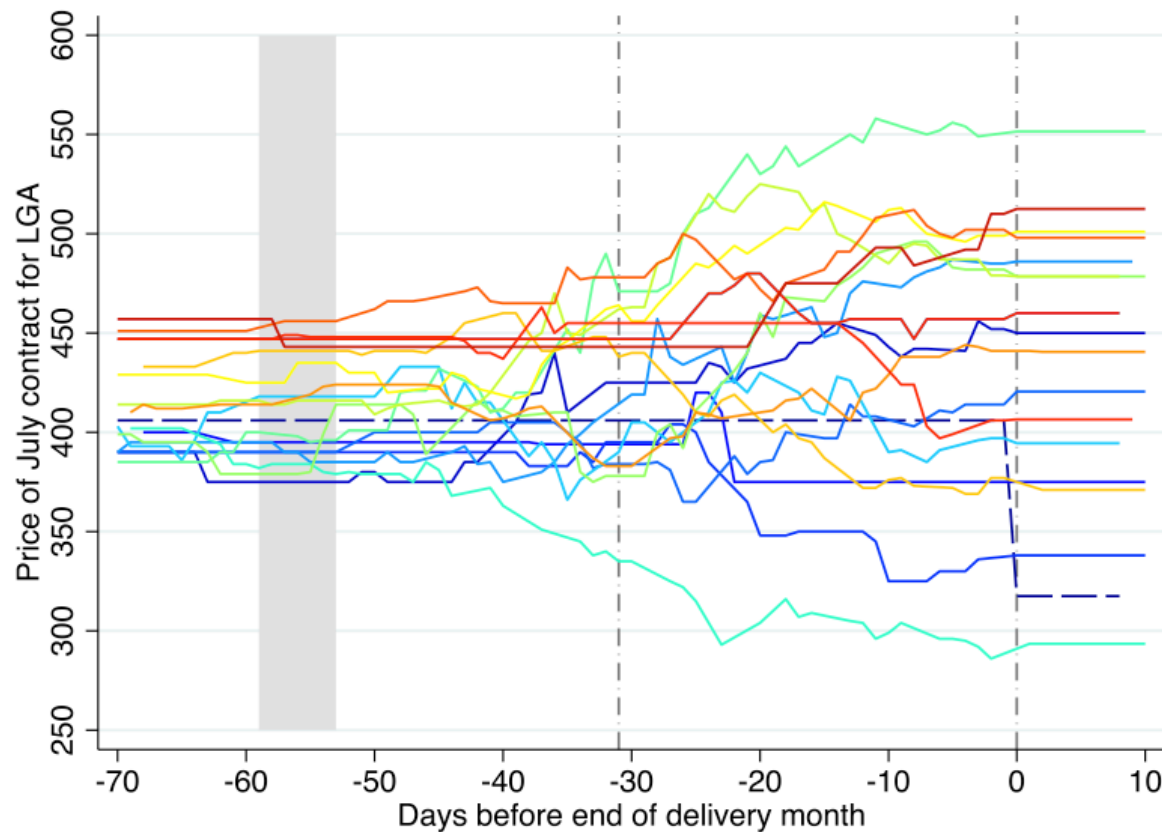


July CDD prices for Lagaardia airport

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From -30 to 0 we are **in** the actual month and observe some of the realized CDDs



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Let's think through the intuition before seeing the results

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No! It should have already been priced in

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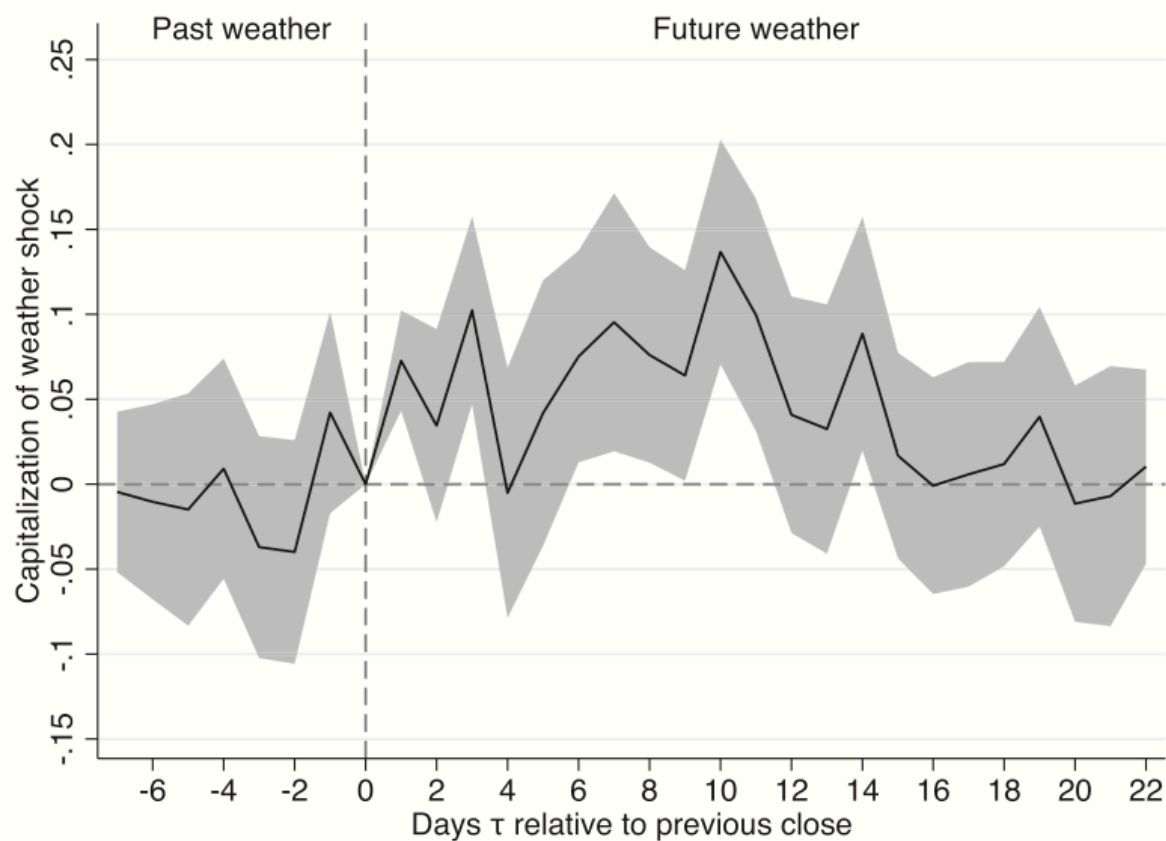
Yes! Skillful forecasts should predict future weather anomalies, if traders use these forecasts then future weather anomalies should affect the current price change

- Suggests forecasts 10+ days ahead might not affect the price

Futures prices predict future weather

X-axis: days before (left) and after (right) current trading day

Y-axis: change in contract price given a 1°C higher CDD anomaly

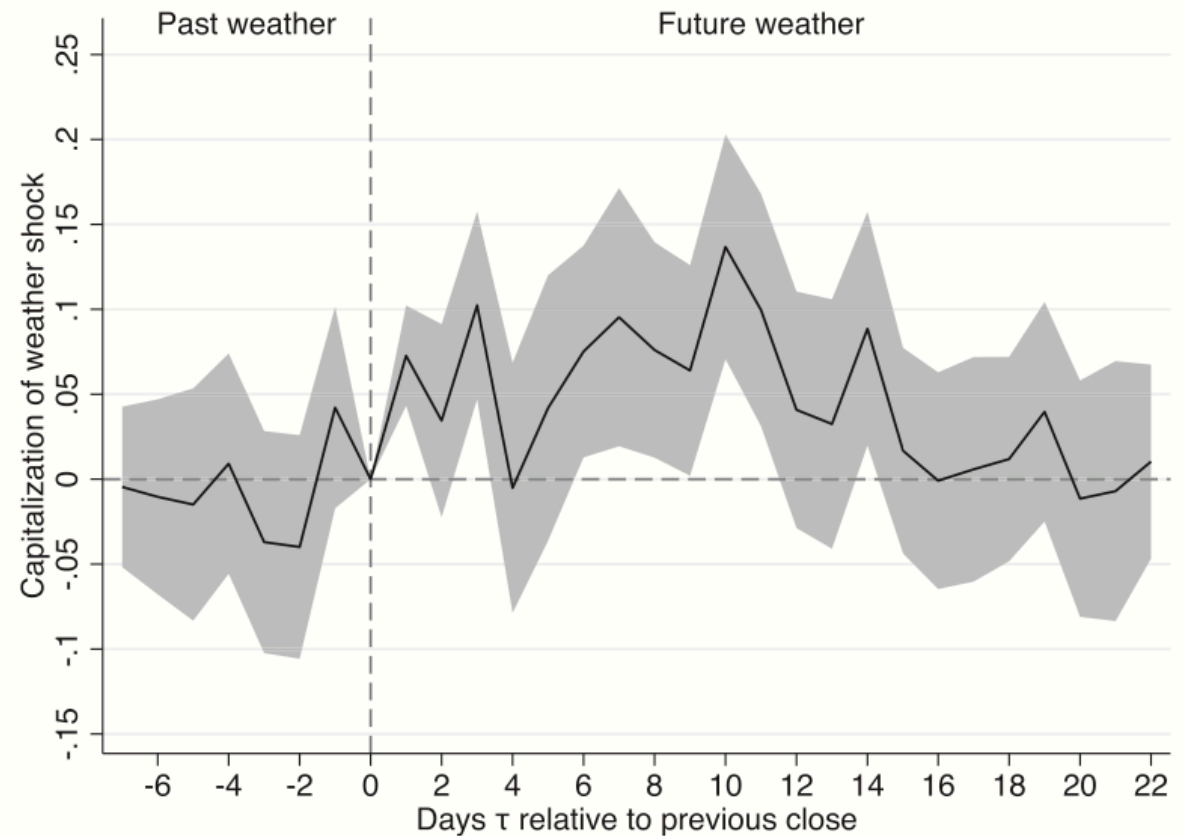


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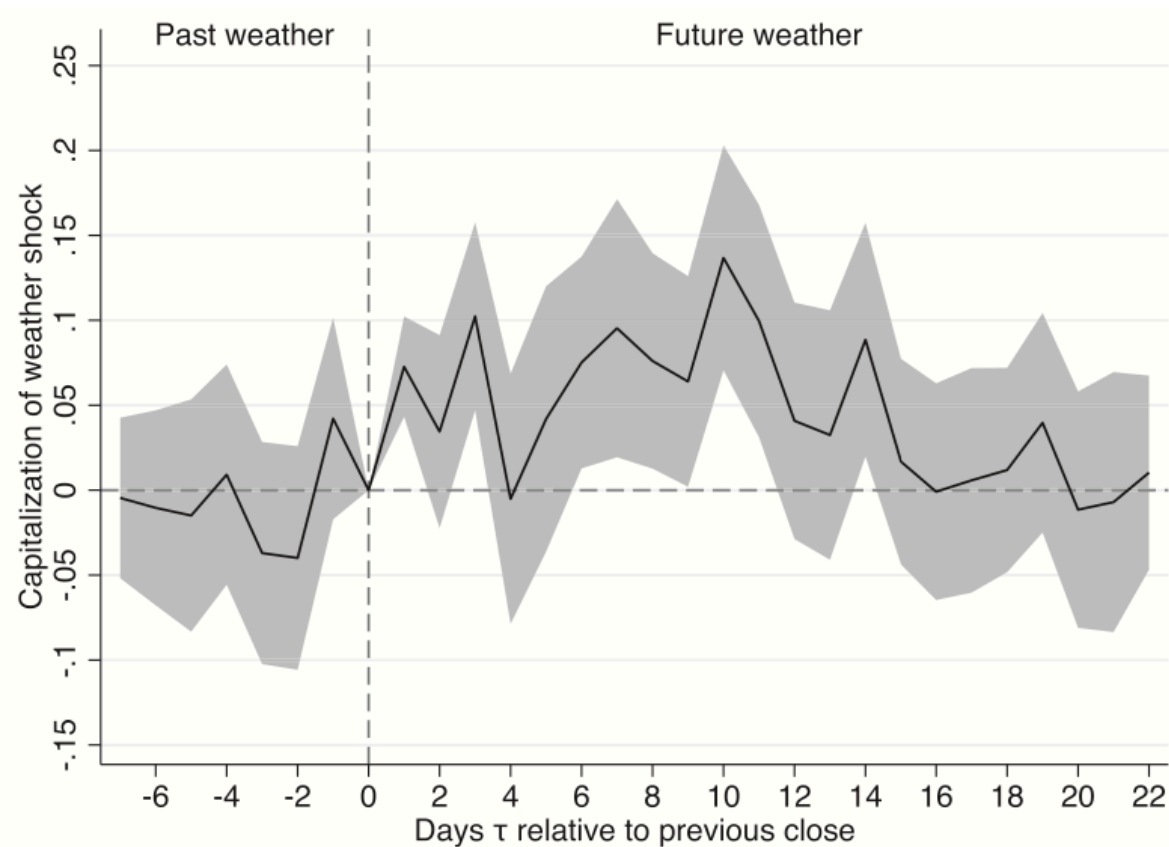
Black line: change in today's contract price given 1 CDD more in a day in the past (left of dashed line) or future (right)



Negative τ s are past days / weather anomalies, positive τ s are future days / weather anomalies

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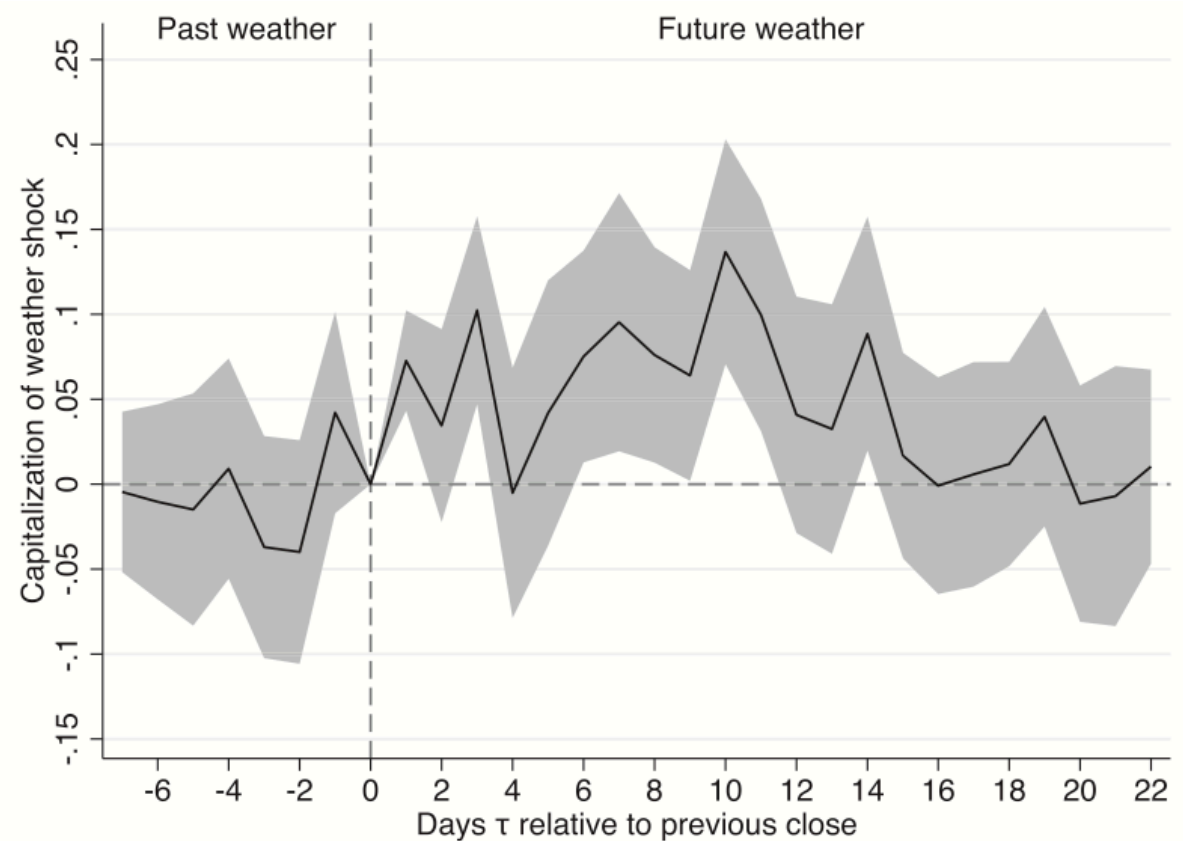
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No! Capitalization is close to 0 for all $\tau < 0$

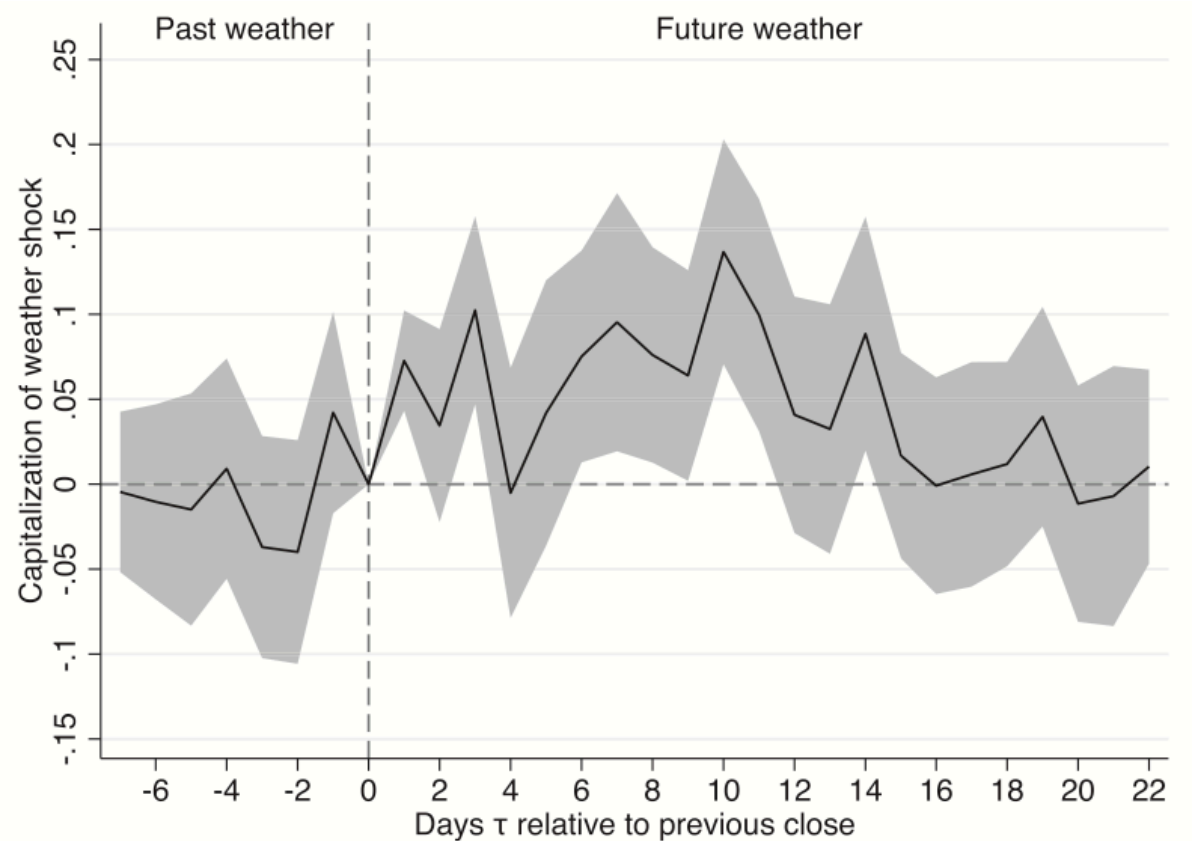


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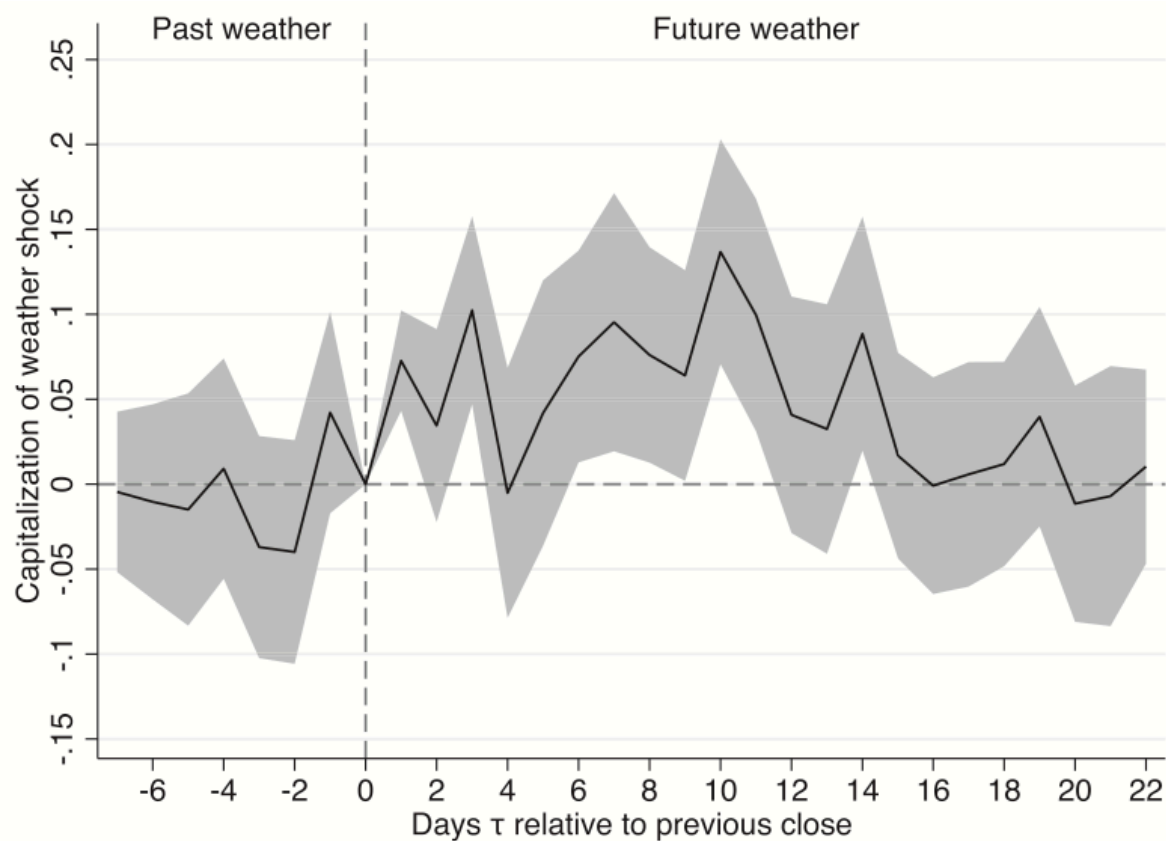
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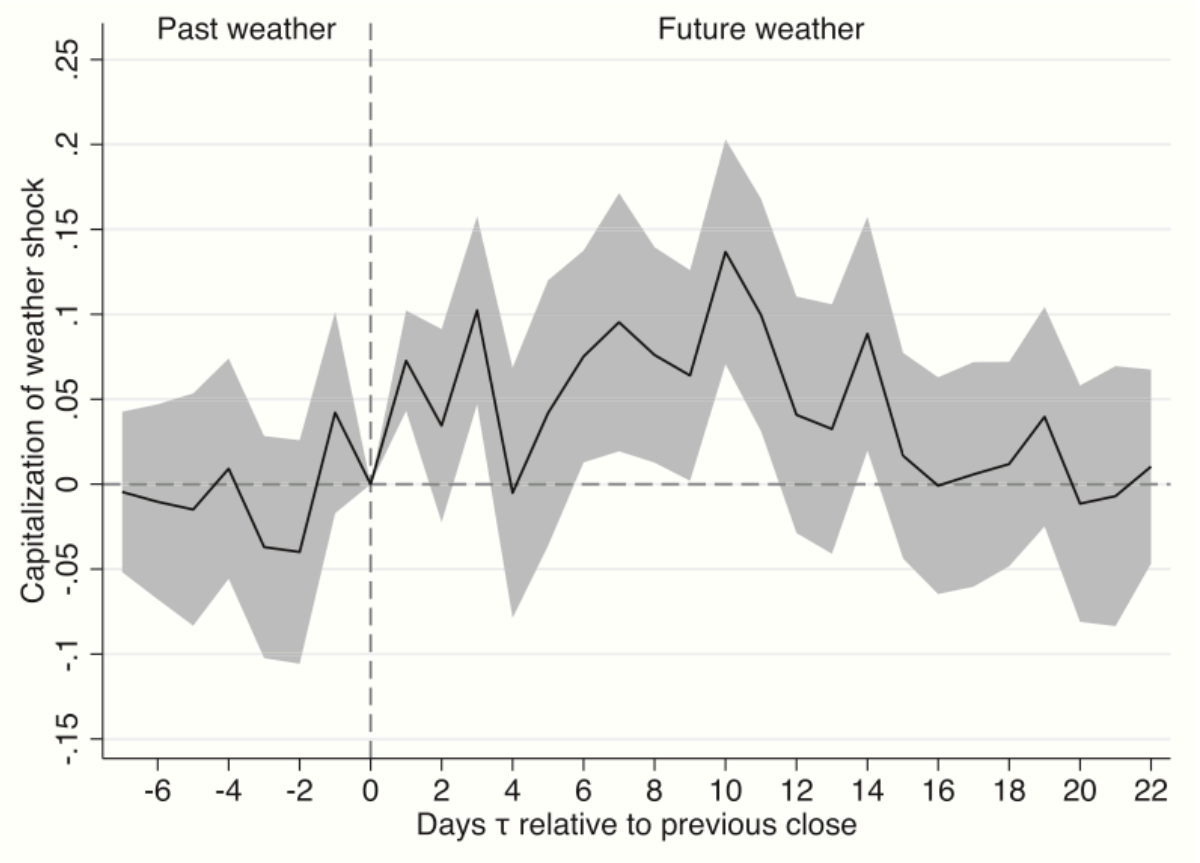
Does future weather affect changes in current prices?

Yes! Up to about 2 weeks into the future



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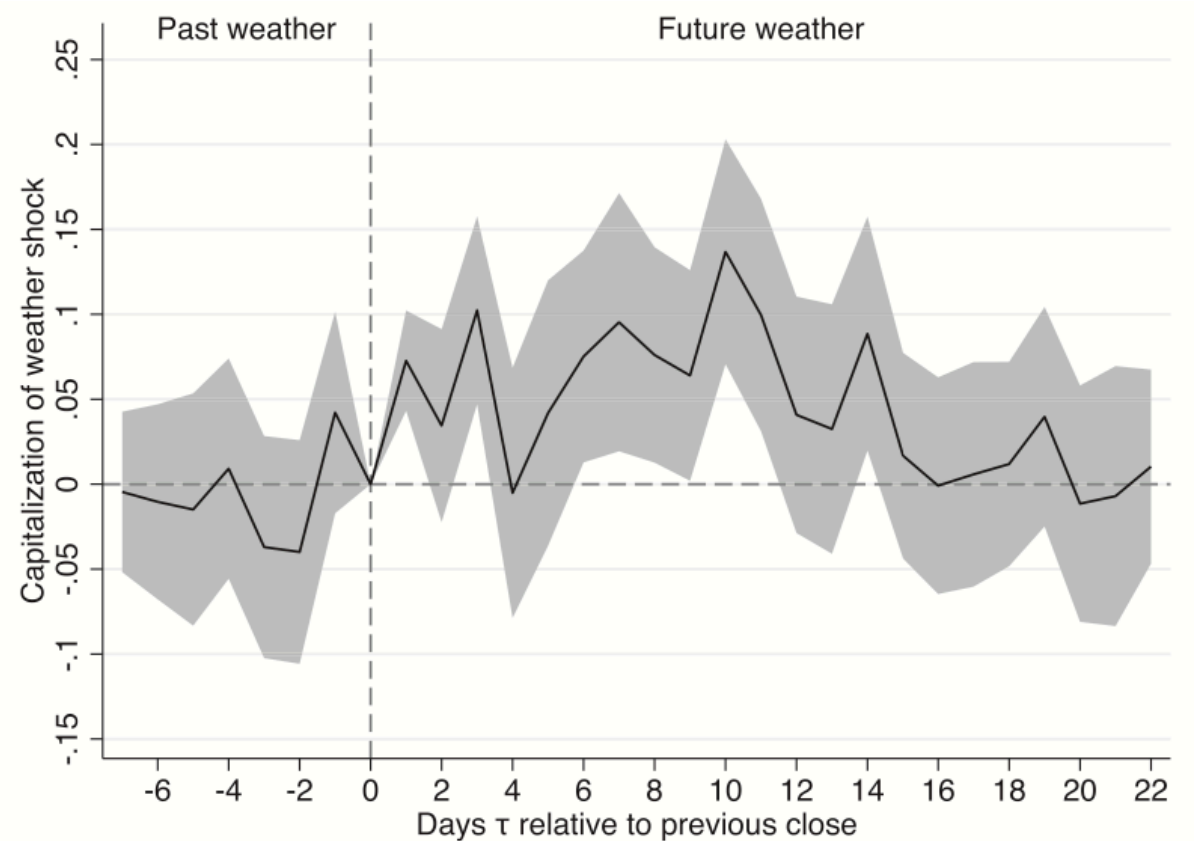
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How can future weather affect today's prices or expectations?

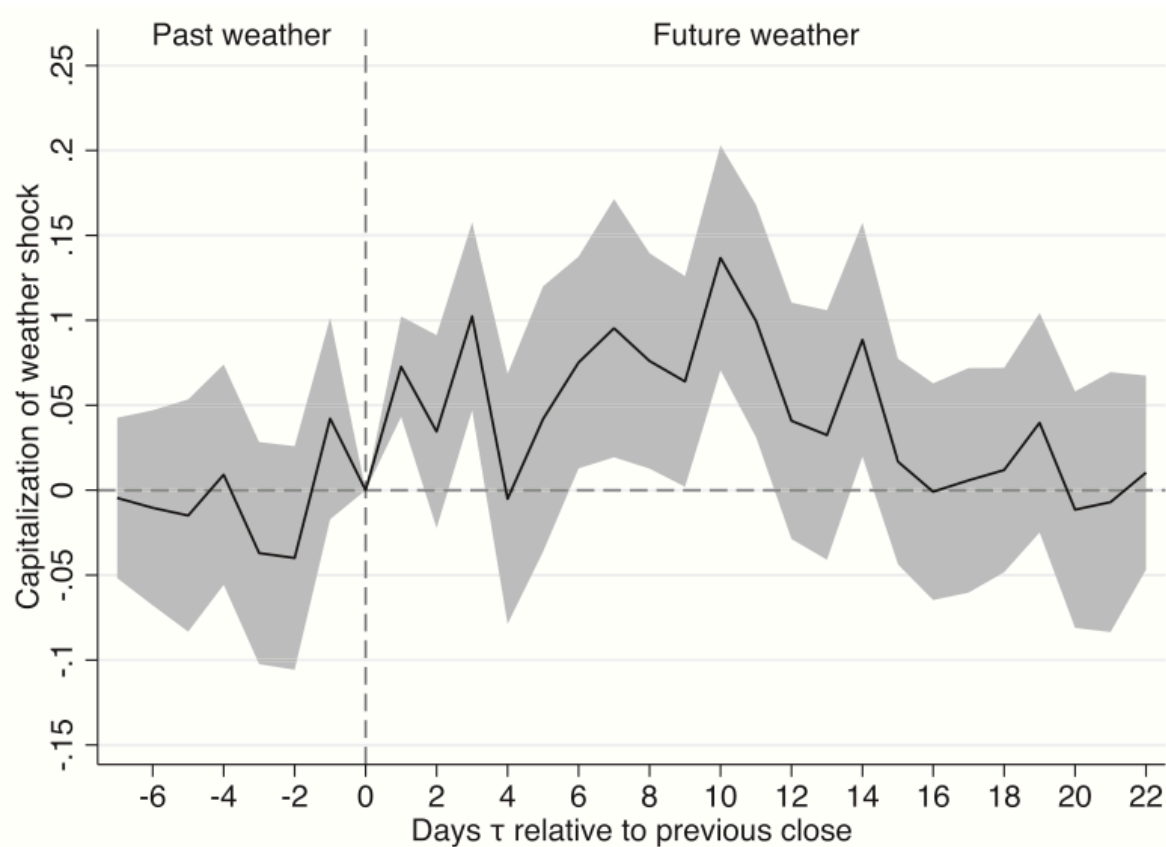


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Traders must respond to weather forecasts



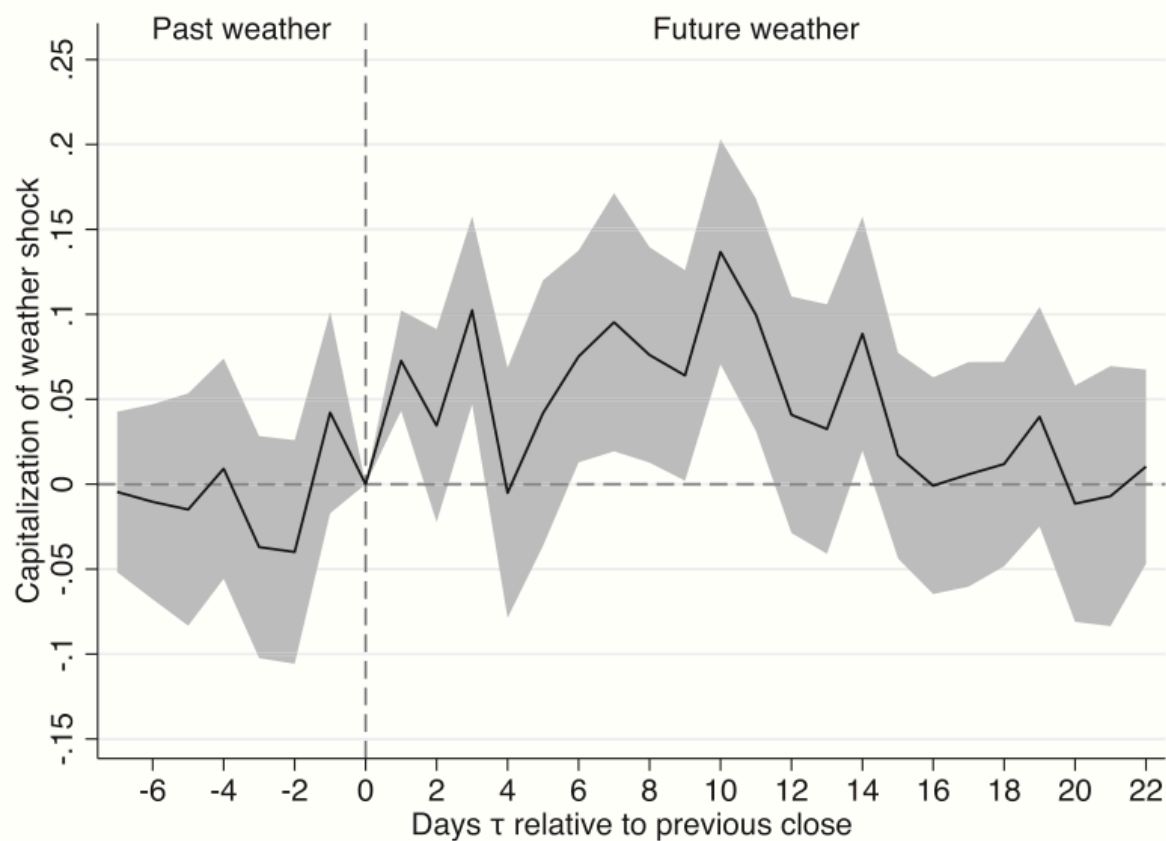
Futures prices predict future weather

What does all this mean?

How can future weather affect today's prices or expectations?

Traders must respond to weather forecasts

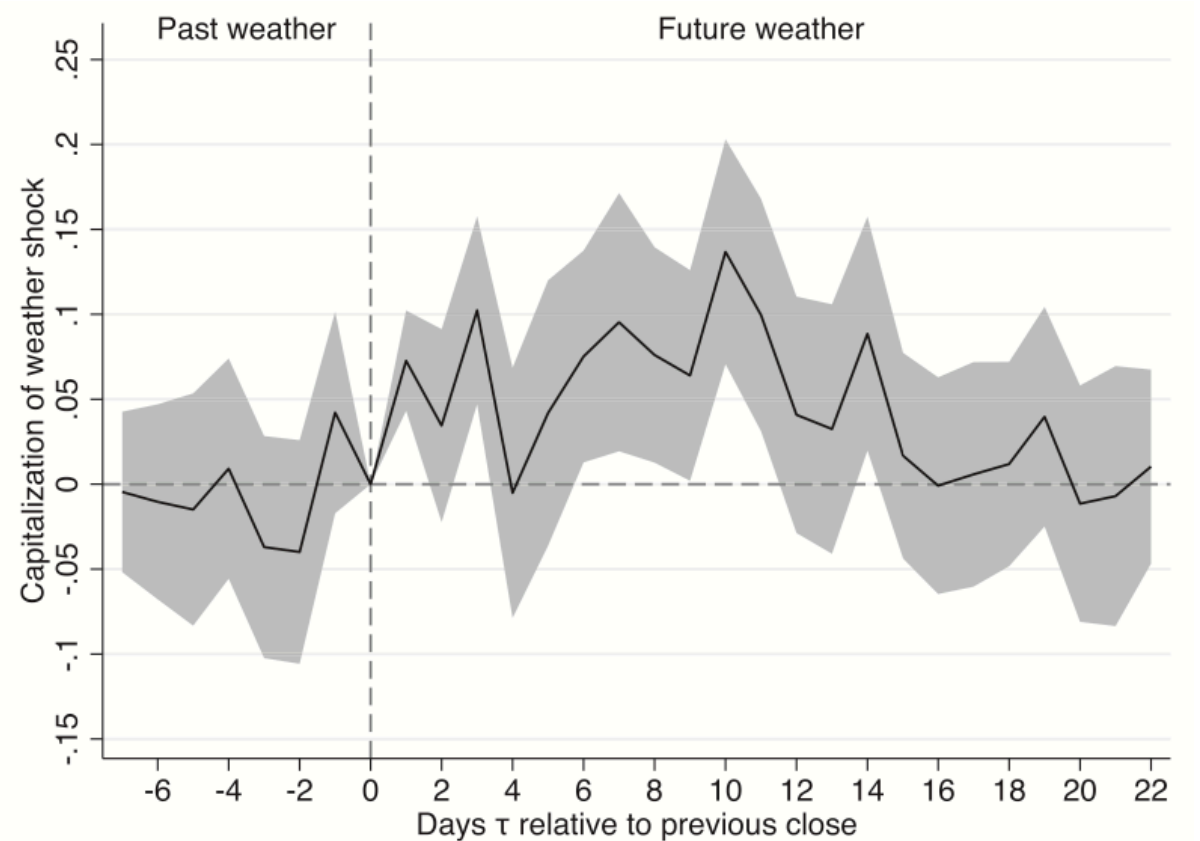
Traders also are using the info correctly because prices are **positively** associated with weather anomalies



Futures prices predict future weather

Does the weather anomaly get fully capitalized?

- Is the market accurately pricing in future weather?

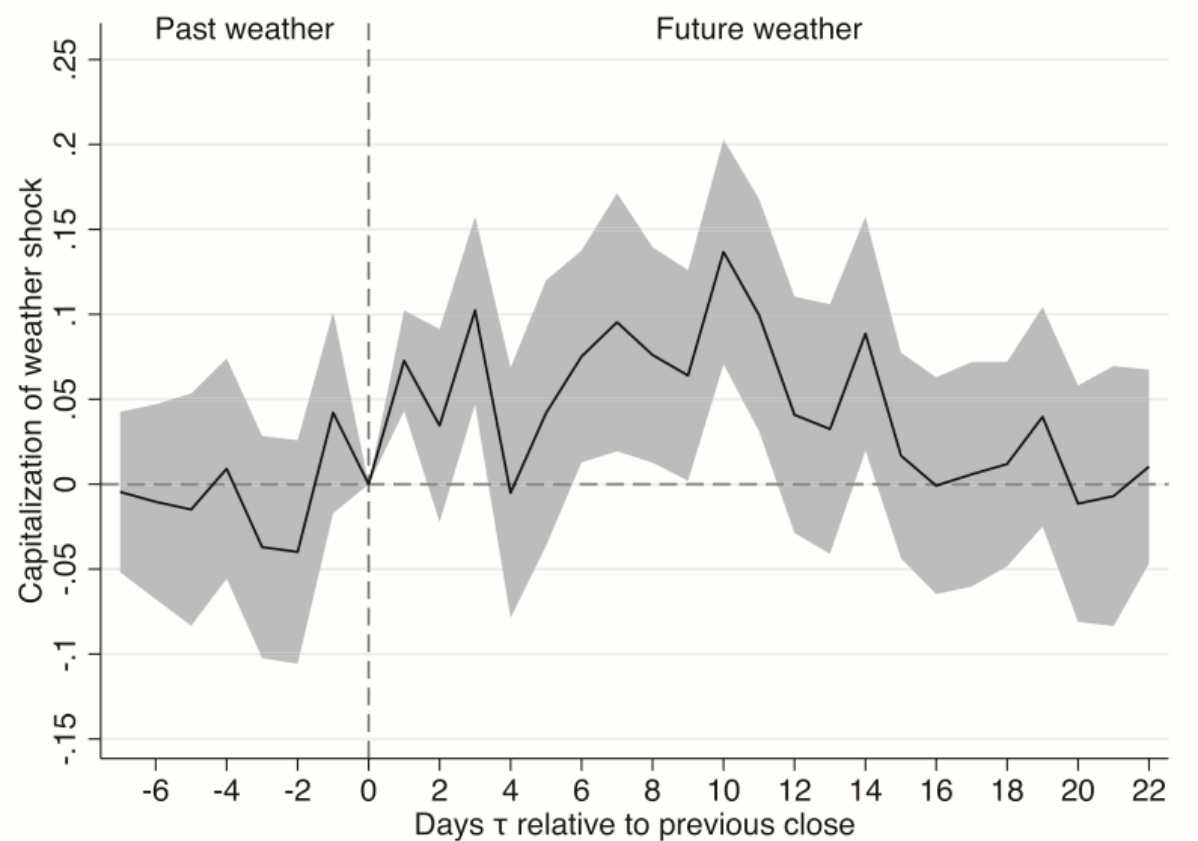


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If so the **total capitalization** of a 1 CDD anomaly should add up to 1

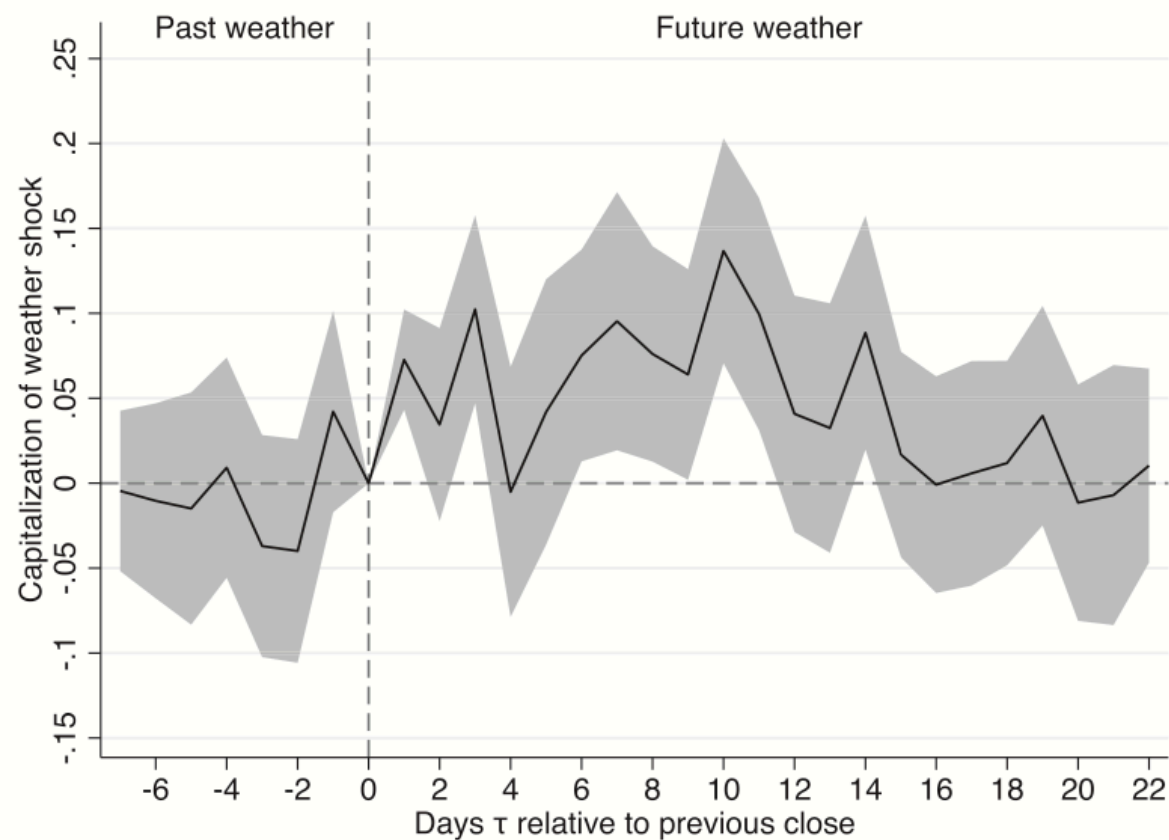


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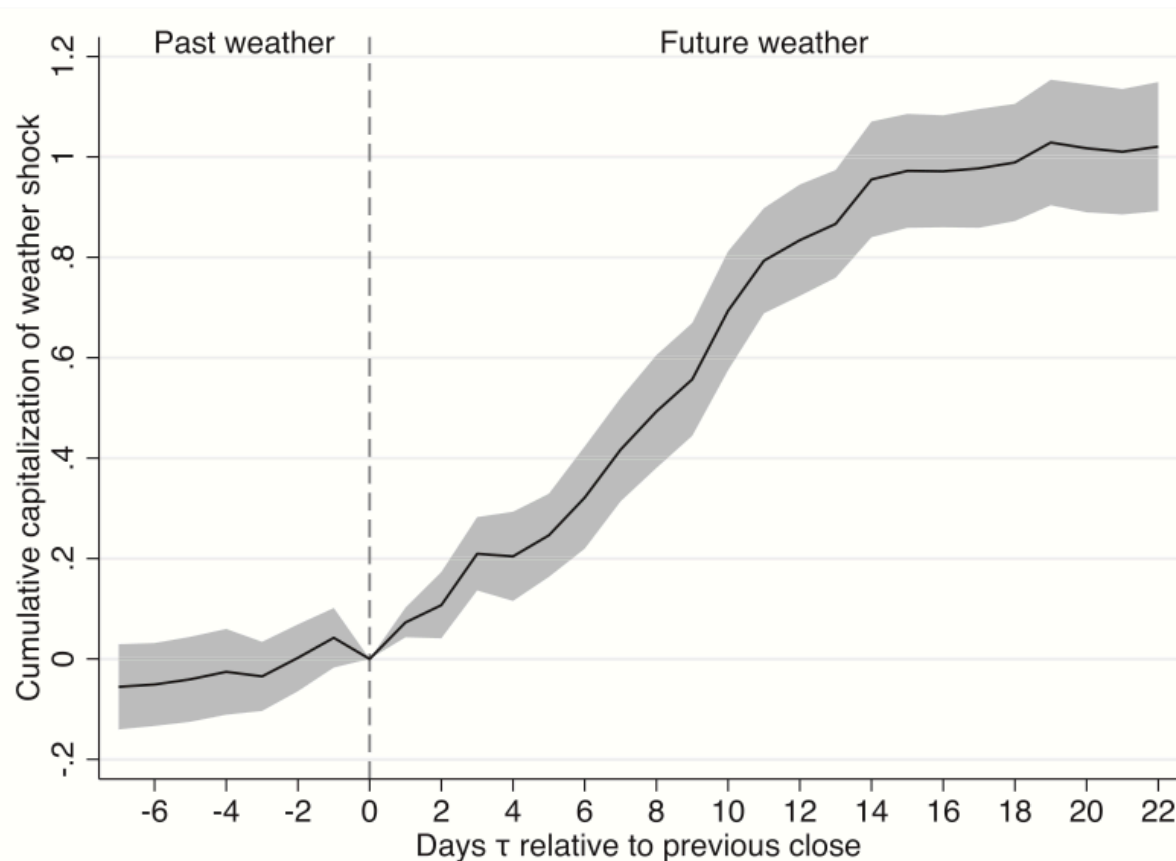
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The sum (integral) of the values of the black line over all τ equals 1 if the market is pricing correctly

Futures prices predict future weather

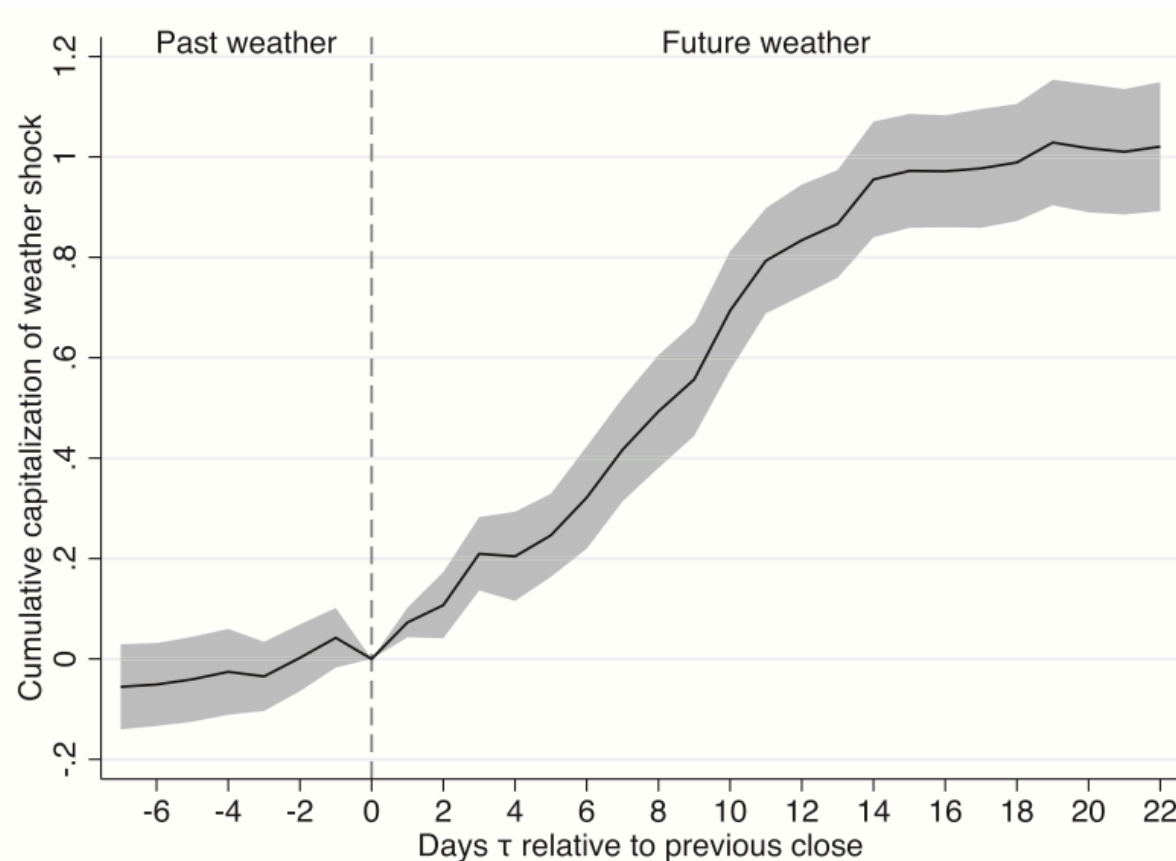
Black line: the **cumulative** sum of the previous black line



Futures prices predict future weather

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Adding up over all days gives a sum of **1**

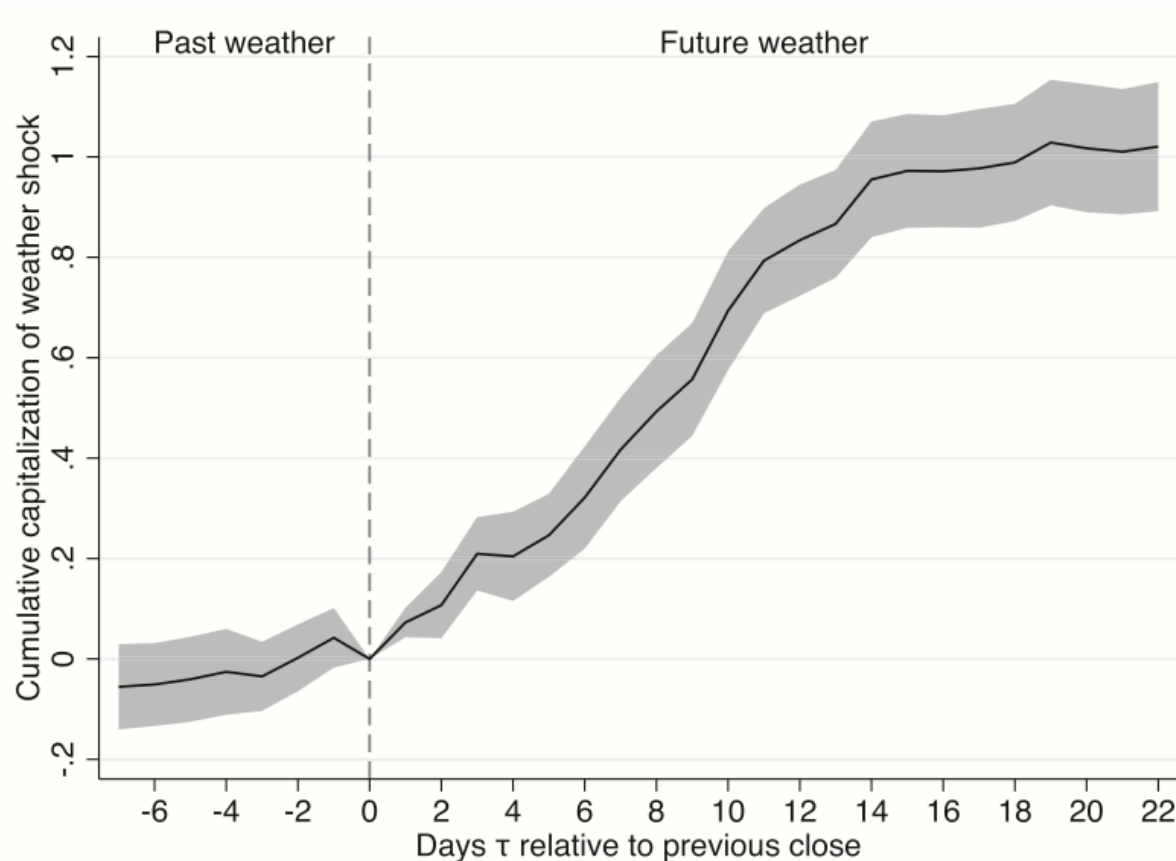


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The market fully internalizes short-run weather!



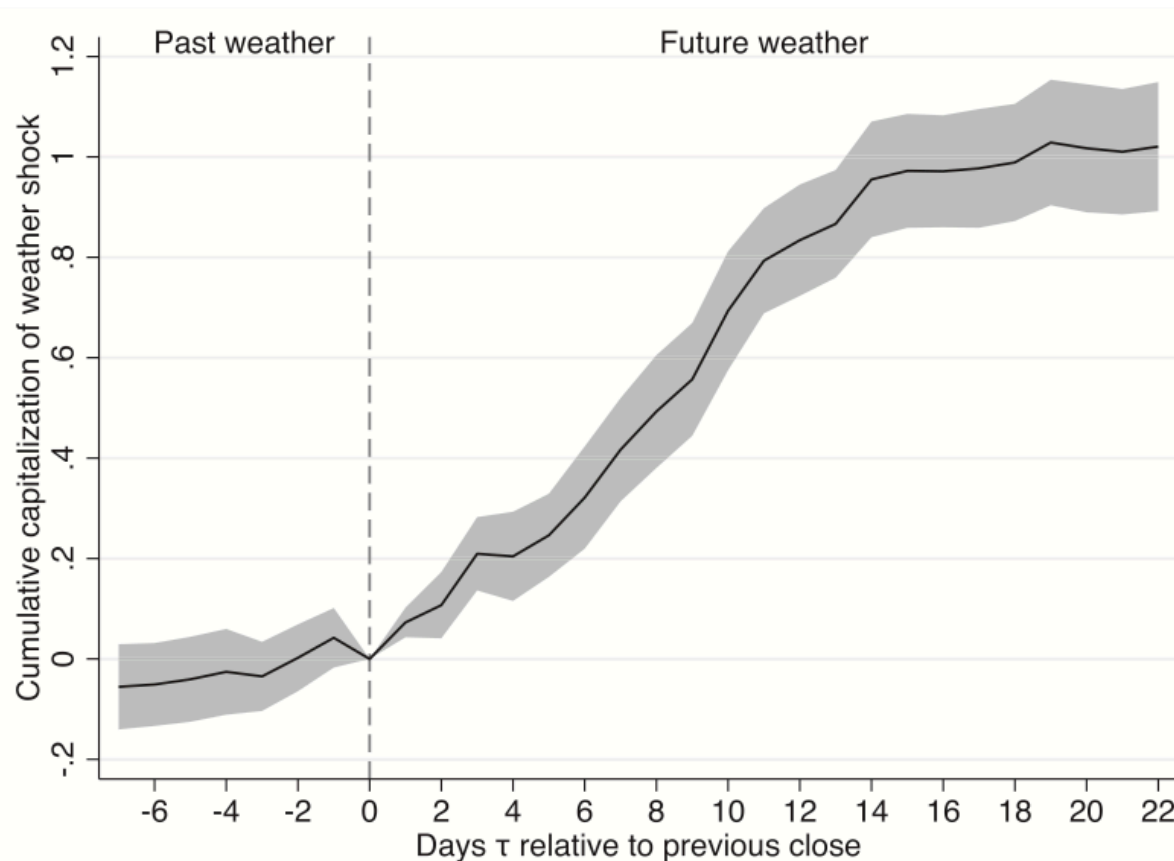
Futures prices predict future weather

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Adding up over all days gives a sum of **1**

The market fully internalizes short-run weather!

It's fully internalized using forecasts up to 14 days ahead



Futures prices and long-run climate

Weather futures capitalize short-run weather

Futures prices and long-run climate

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What about long-run changes in climate?

Futures prices and long-run climate

Weather futures capitalize short-run weather

What about long-run changes in climate?

If so, the long run trends in futures prices should match either:

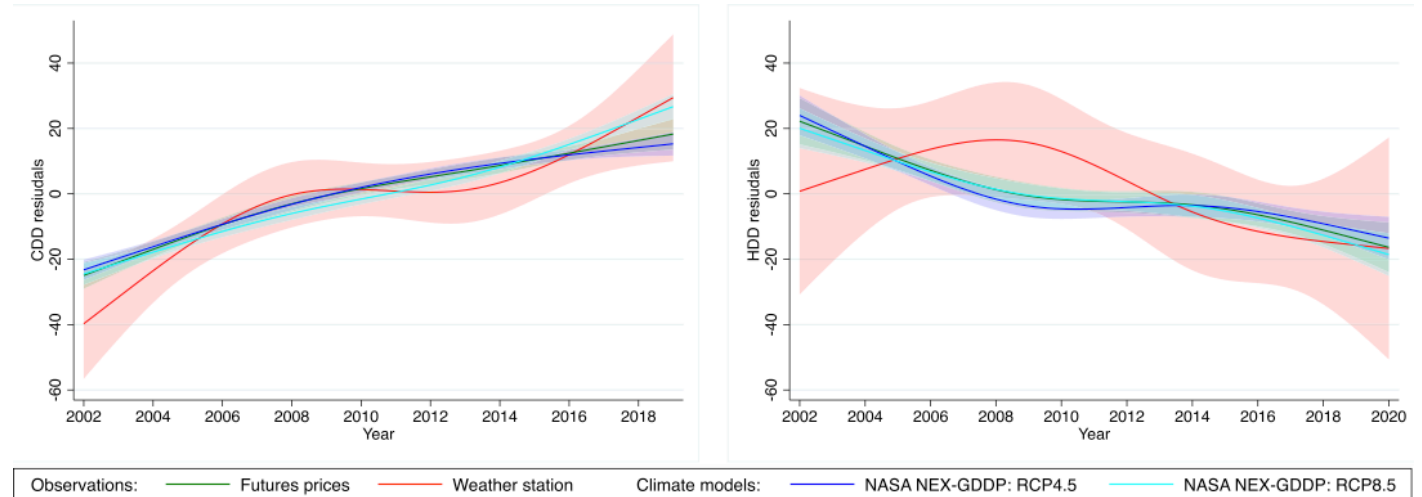
1. Long run trends in weather
2. Predicted trends from climate models

Futures prices predict long-run climate

Y-axis: CDD/HDD relative to the city average (0 is average)

Lines: contract price (dark green), actual weather

CDDs/HDDs (red), climate model predicted CDD/HDDs (blue/neon)

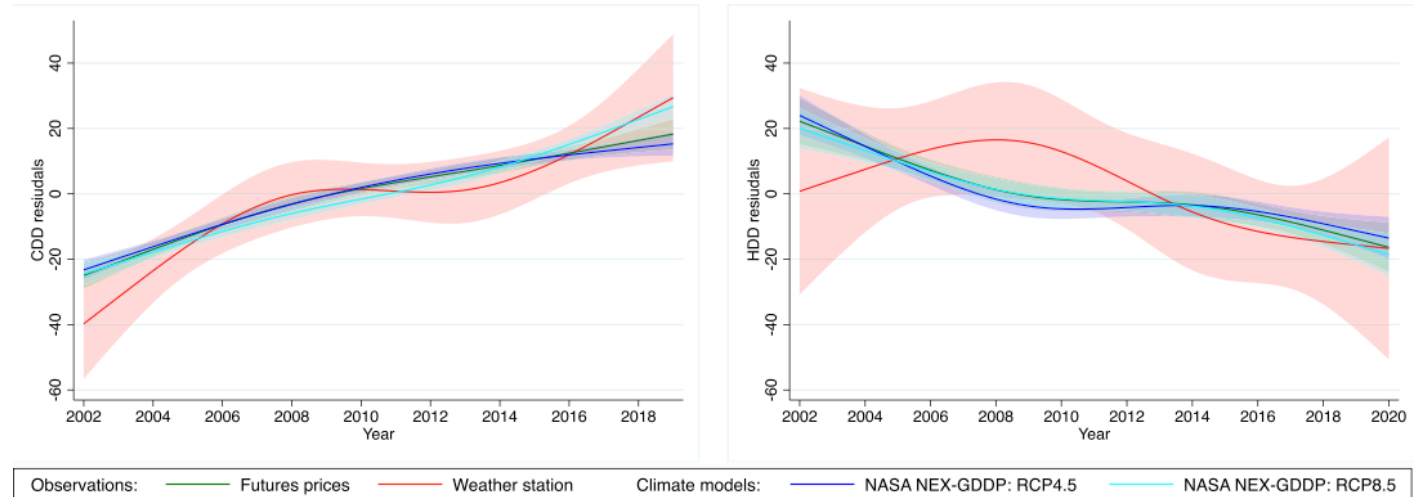


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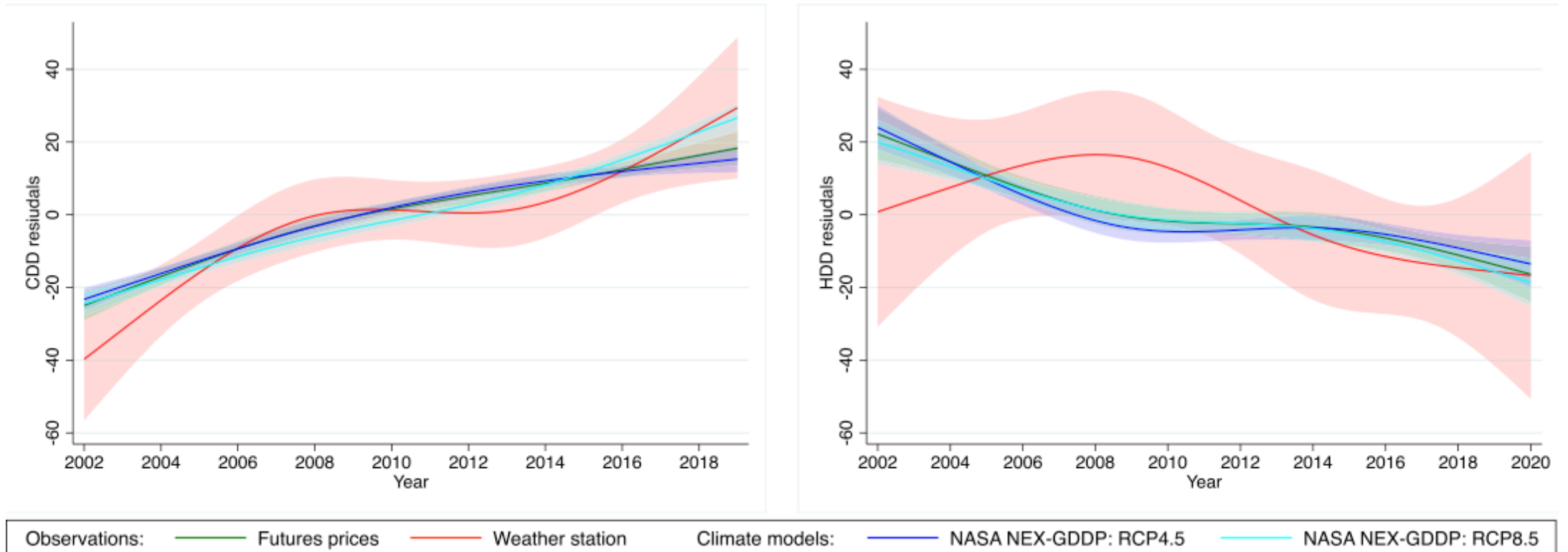
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What stands out?



Futures prices predict long-run climate

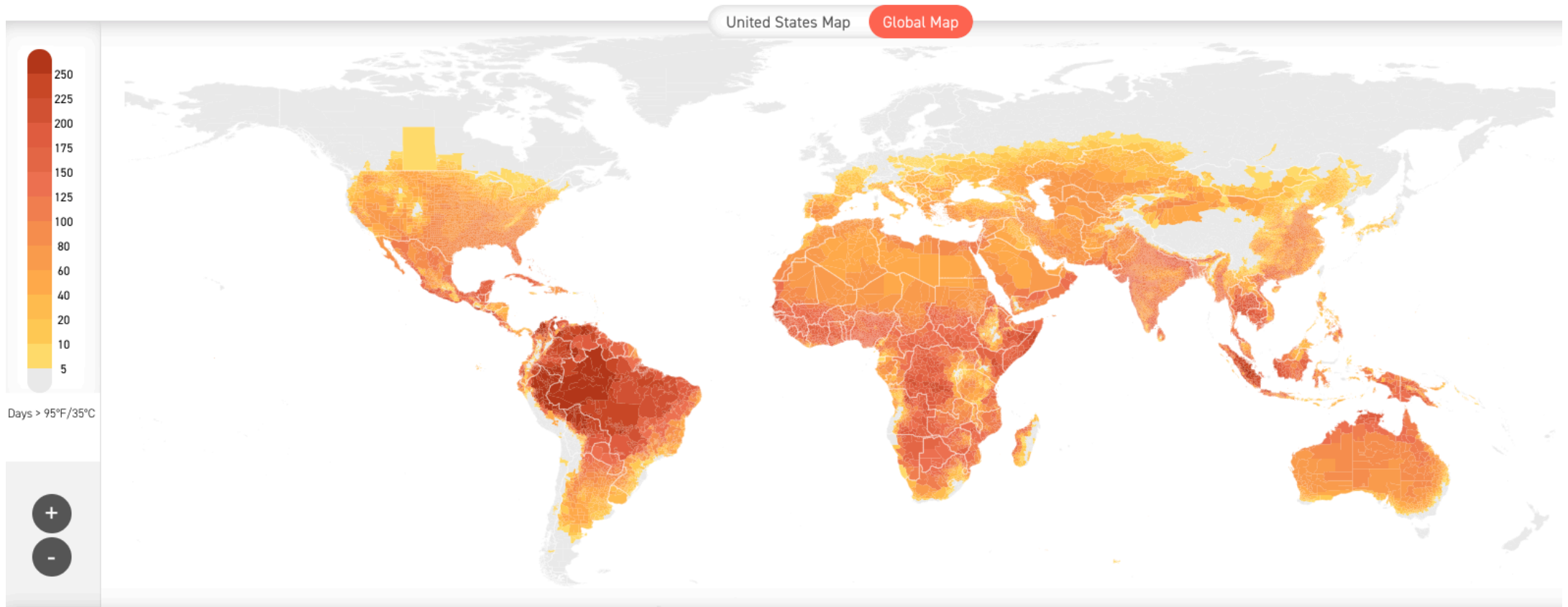


Futures prices are strongly correlated with **climate model predictions**,
weather to a lesser extent

Discounting

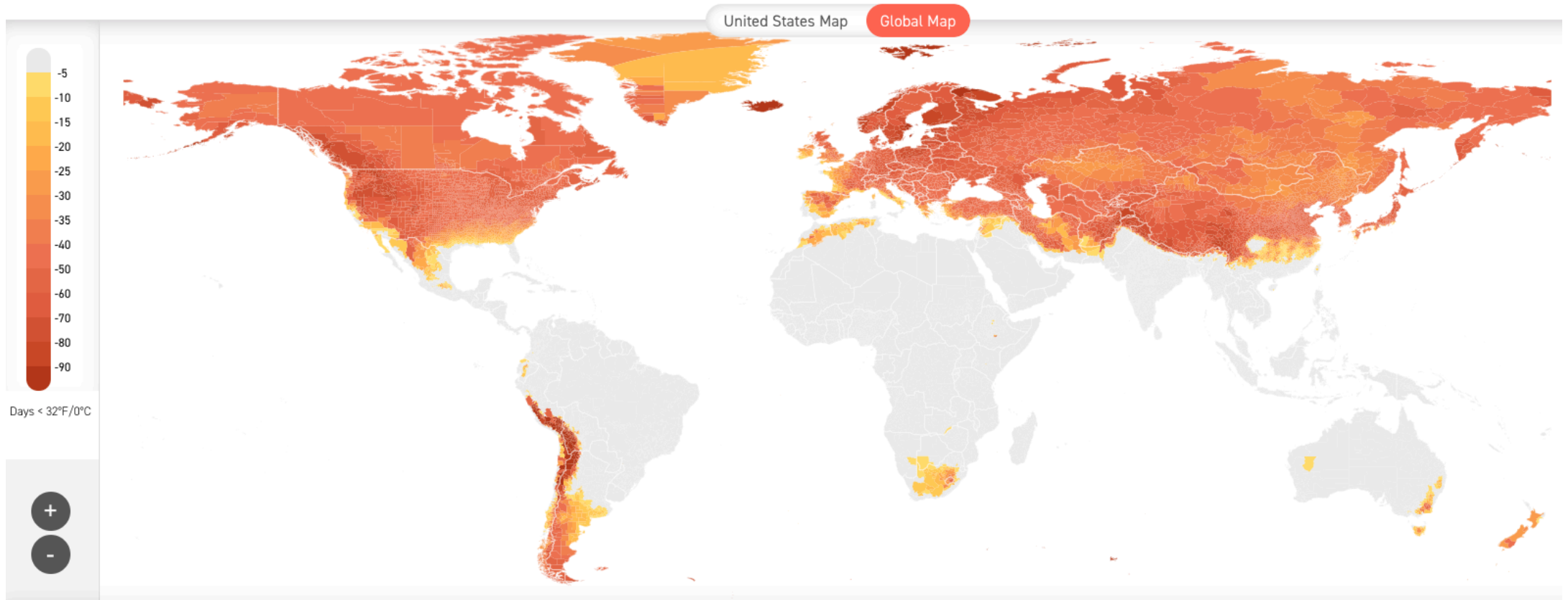
Motivating discounting: <http://impactlab.org/map>

At the end of the century we will have much more hot days in some places



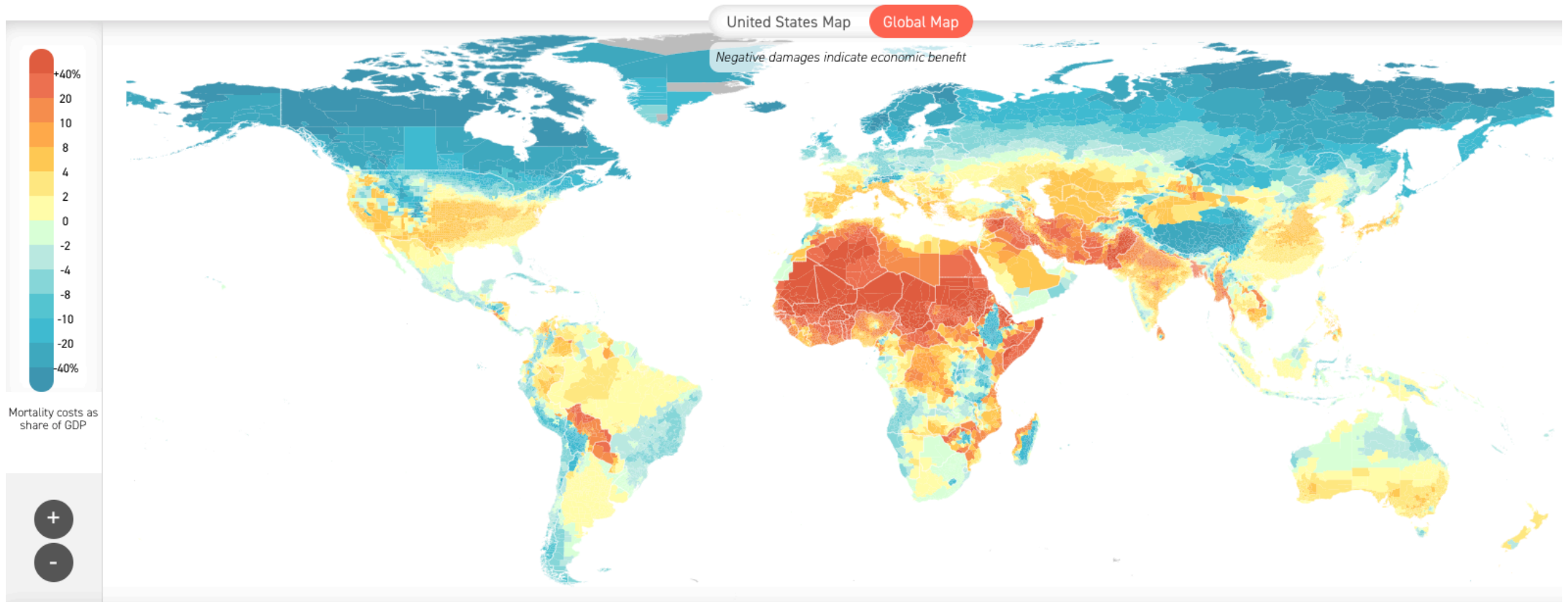
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At the end of the century we will have much fewer freezing days in others



Motivating discounting: <http://impactlab.org/map>

This has massive implications for mortality



Motivating discounting

Some places are expecting to have huge gains in GDP from mortality risk

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How do we compare these costs and benefits to those incurred today?

Motivating discounting

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How do we compare these costs and benefits to those incurred today?

We use a **discount rate**: a value that tells us how much future dollars are worth in today's terms

A simple example

Let r be the discount rate, so $\beta = \frac{1}{1+r}$ is the discount factor

Suppose we are considering two different projects that have costs and benefits that accrue differently over time

Year	Project A Cost	Project A Benefit	Project B Cost	Project B Benefit
0	10000	0	6000	0
1	1000	4000	0	1000
2	0	4000	0	3000
3	0	4000	0	3000

Project A has higher costs and benefits in nominal terms

A simple example

Year	Project A Cost	Project A Benefit	Project B Cost	Project B Benefit
0	10000	0	6000	0
1	1000	4000	0	1000
2	0	4000	0	3000
3	0	4000	0	3000

Project A:

$$PV_A = \frac{4000}{1.05^1} + \frac{4000}{1.05^2} + \frac{4000}{1.05^3} - \frac{10000}{1.05^0} - \frac{1000}{1.05^1} = -59.39$$

Project B:

$$PV_B = \frac{1000}{1.05^1} + \frac{3000}{1.05^2} + \frac{3000}{1.05^3} - \frac{6000}{1.05^0} = 264.98$$

What if the discount rate was 3%?

Year	Project A Cost	Project A Benefit	Project B Cost	Project B Benefit
0	10000	0	6000	0
1	1000	4000	0	1000
2	0	4000	0	3000
3	0	4000	0	3000

Project A:

$$PV_A = \frac{4000}{1.03^1} + \frac{4000}{1.03^2} + \frac{4000}{1.03^3} - \frac{10000}{1.03^0} - \frac{1000}{1.03^1} = 343.57$$

Project B:

$$PV_B = \frac{1000}{1.03^1} + \frac{3000}{1.03^2} + \frac{3000}{1.03^3} - \frac{6000}{1.03^0} = 544.09$$

Discounting

Discounting results in us placing less value on costs and benefits that accrue in the future

A dollar 1 year from now is worth $\beta = \frac{1}{1+r}$ dollars today

The timing of costs and benefits of projects can then sway which project has greater present value

Return to Manne-Richels

We ignored the idea of discounting in our discussion of the Manne-Richels model

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Our new problem with discounting is then:

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Our new problem with discounting is then:

$$\min_{a_1} E[TC] = \underbrace{\frac{1}{2}a_1^2}_{\text{current cost}} + \beta \left[(1-p) \times \underbrace{0}_{\text{good state cost}} + p \times \underbrace{\frac{1}{2}(1-a_1)^2}_{\text{bad state cost}} \right]$$

Return to Manne-Richels

The first-order condition is:

$$\frac{dE[TC]}{da_1} = a_1^* - \beta p(1 - a_1^*) = 0$$

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How does discounting affect our decisionmaking?

Discounting and decisionmaking

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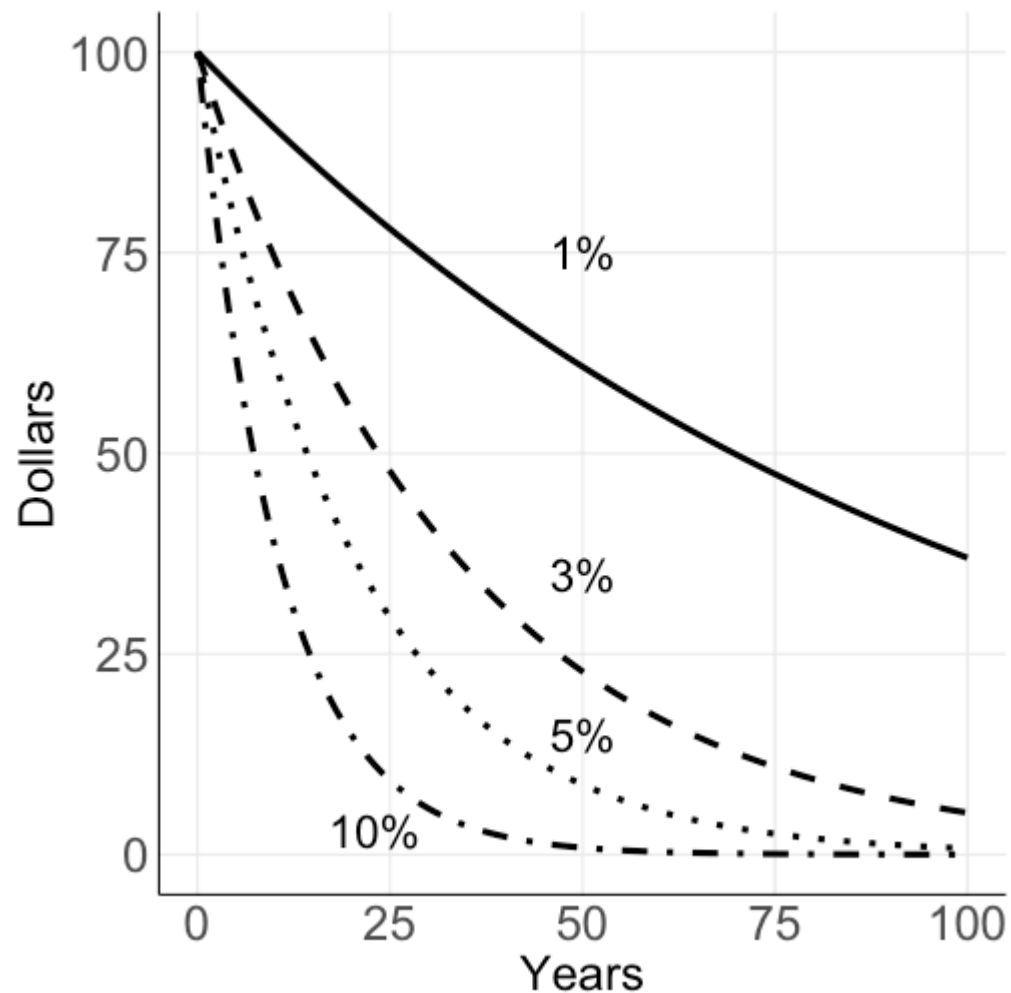
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What is the value of a future payment of \$100?

PV of \$100



Higher discount rates place less value on future benefits

Things > 30 years in the future have basically no value with a 10% discount rate

At a 1% discount rate we value things 100 years in the future at almost half their value today

Discounting

Why does this matter?

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Depending on our choice of discount rate these costs and benefits can be substantial or trivial

Discounting

1 million in damages in 200 years at a discount rate of $r = 2\%$ is worth 19,053 today

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This makes the choice of the discount rate one of the most important (and contentious) things about climate change policy

Discounting: how do we choose?

How do we choose the discount rate?

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Option 1: take the market rate

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Why might this not be the rate we want to choose as a regulator?

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Issues with market rates:

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Market rates don't reflect externalities

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Super-responsibility of government: the government represents future generations as well as current generations (only current ones are represented in the market)

Dual-role of individuals: in political roles, people are more concerned about future generations than in their day-to-day behavior which determines the market rate

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Why should we discount the future?

First, **time**: people are impatient

And **growth/inequality**: all else equal, if someone is richer in 10 years, a dollar is worth more to them today than in 10 years (in utility terms)

Ramsey Discounting

With a decent amount of math we can write the social discount rate r as a function of three terms:

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g is the **growth rate**: how fast does consumption grow over time?

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g : how rich will we / future generations be compared to today?

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What this means is that if we have values for r , η , and g , we can compute the "correct" discount rate from a social perspective

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How do we get values for these terms?

Two common approaches: descriptive and prescriptive

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Most philosophers and economists would probably not prescribe the descriptive approach

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First we decide on the 'correct' level of δ and η

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That gives us r

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The above arguments are ethical arguments, so are typically used by those favoring the prescriptive approach

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η is then often engineered to be between 1 and 4

g is observed and generally between 1 and 3%

Thus the discount rate usually lies between 2 and 7%

Quick example: $\delta = 2\%, \eta = 2, g = 2\% \rightarrow r = 6\%$

What's the discount rate? Prescriptive

The prescriptive approach often results in δ being zero or nearly zero for the ethical reasons described above

What's the discount rate? Prescriptive

Choosing η also conveys ethical choices: how do we weigh the distribution of consumption across generations

Recall: $r = \delta + \eta g$

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- η is large: if there is positive growth, we are **less** likely to invest in the future (future generations will be rich anyway)
- η is large: if there is negative growth, we are **more** likely to invest in the future (future generations will be poorer than today)

Distributive justice

Rawls' theory of justice applied here would set $\delta = 0$ and $\eta = \infty$: fairness for all

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Distributive justice

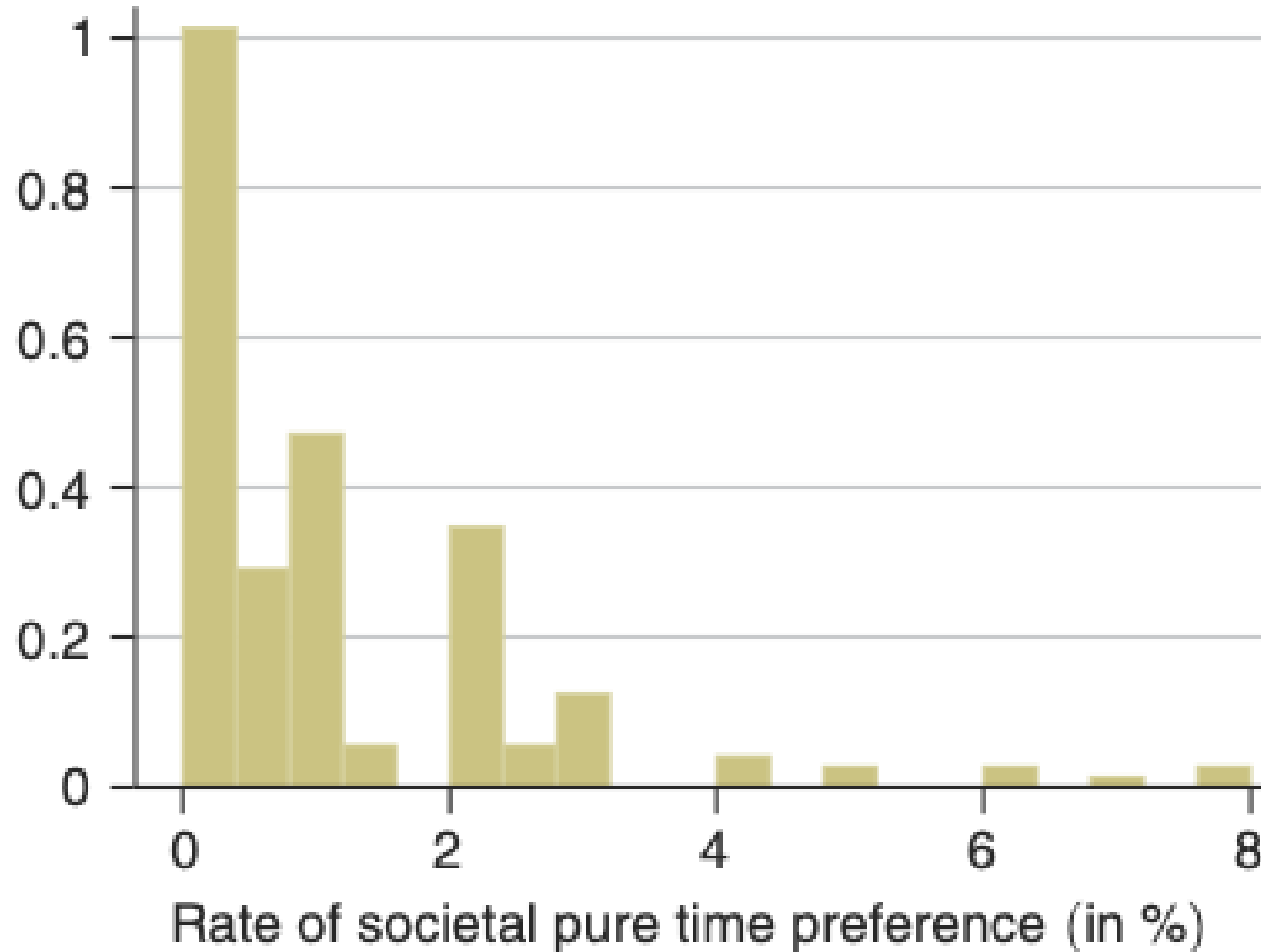
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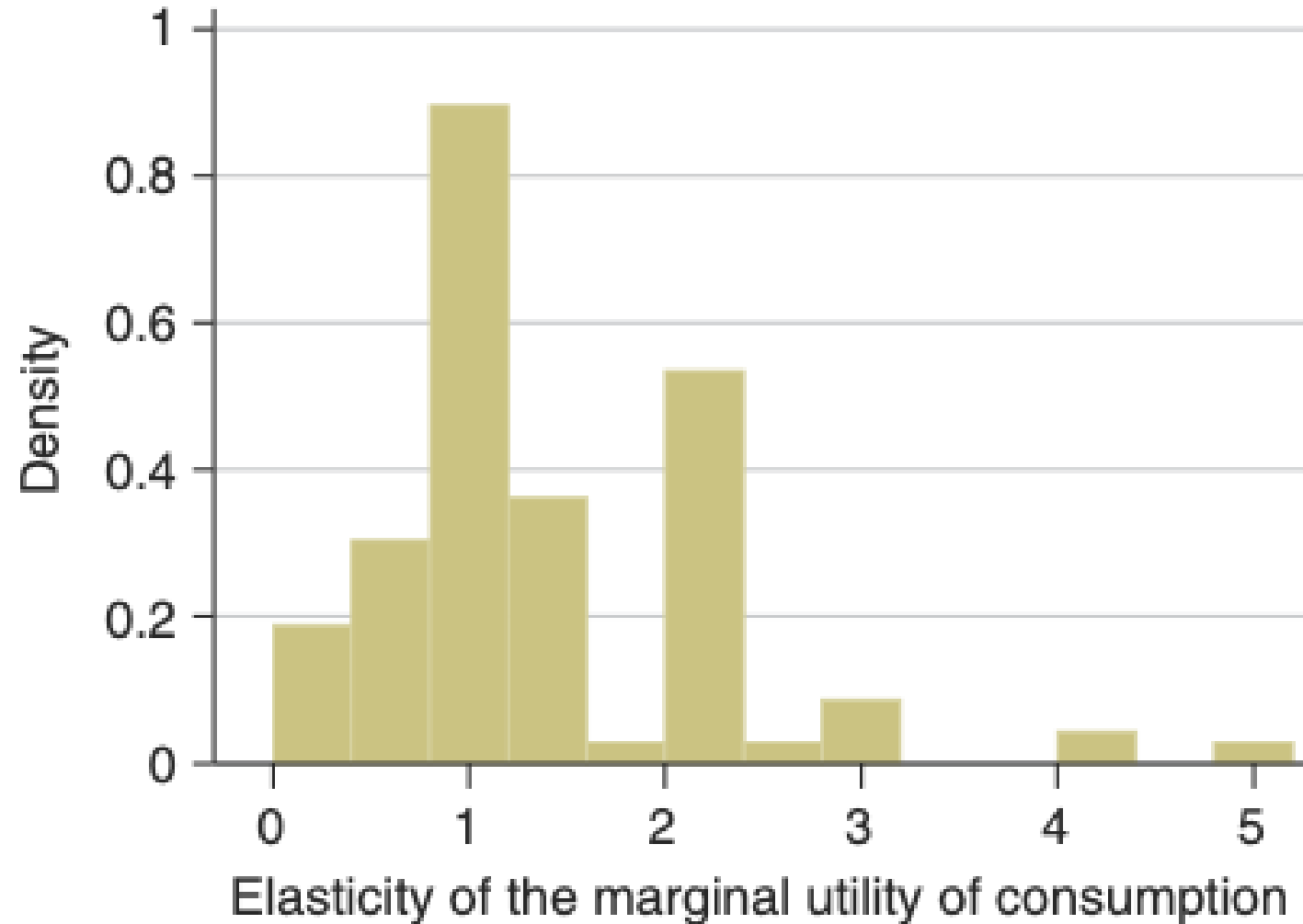
time yields a smaller δ and r

intergenerational inequality yields a larger η and larger r if growth is positive

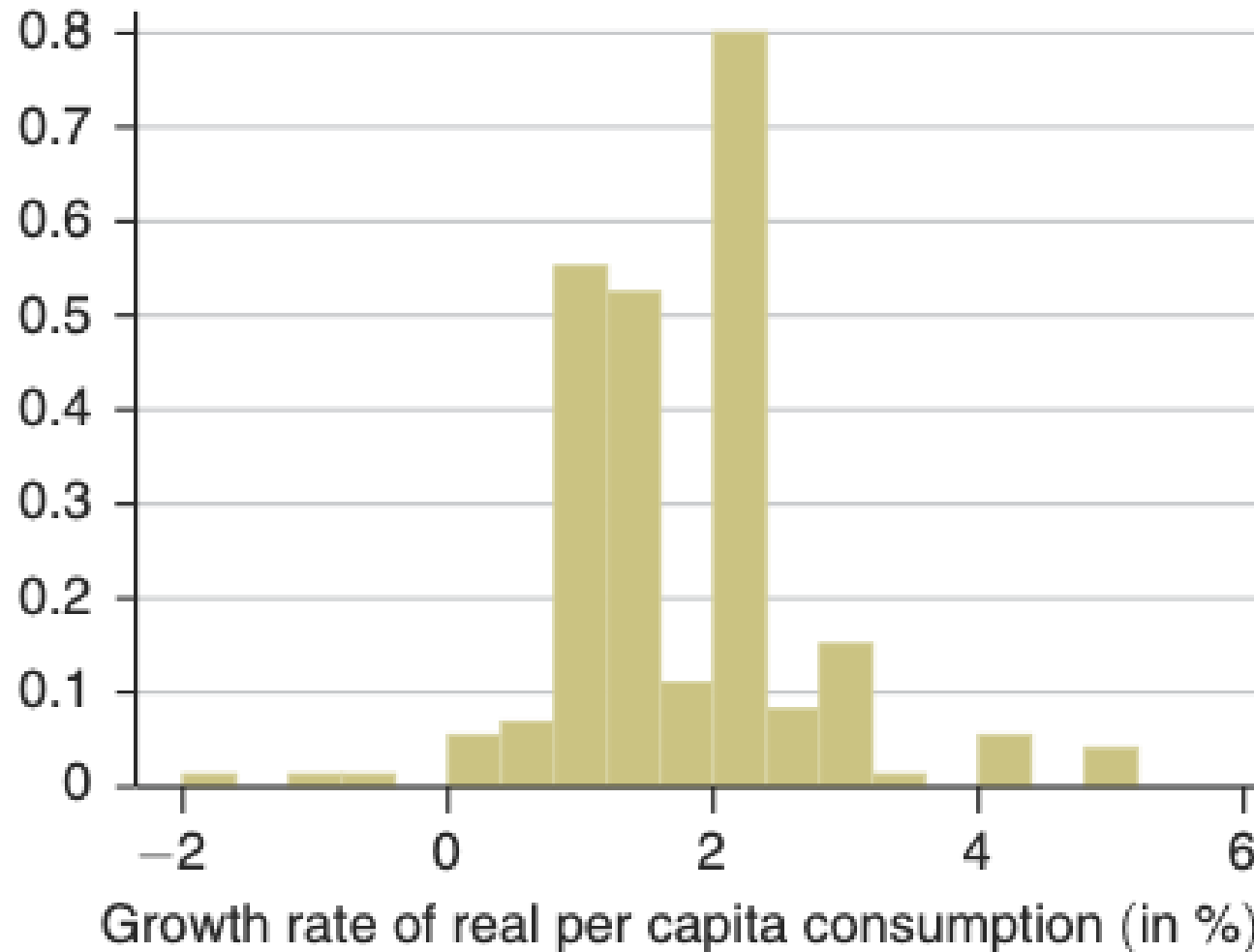
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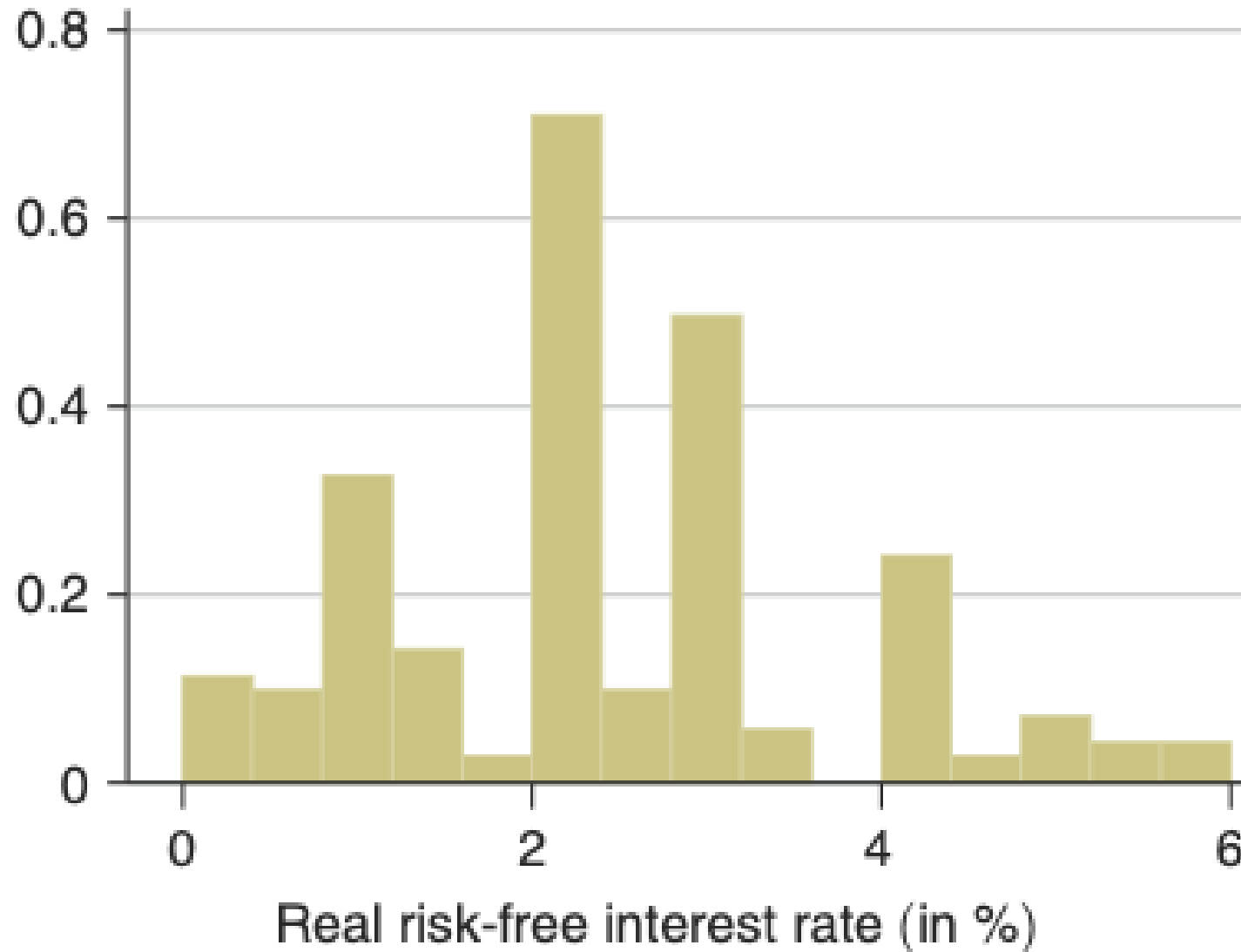
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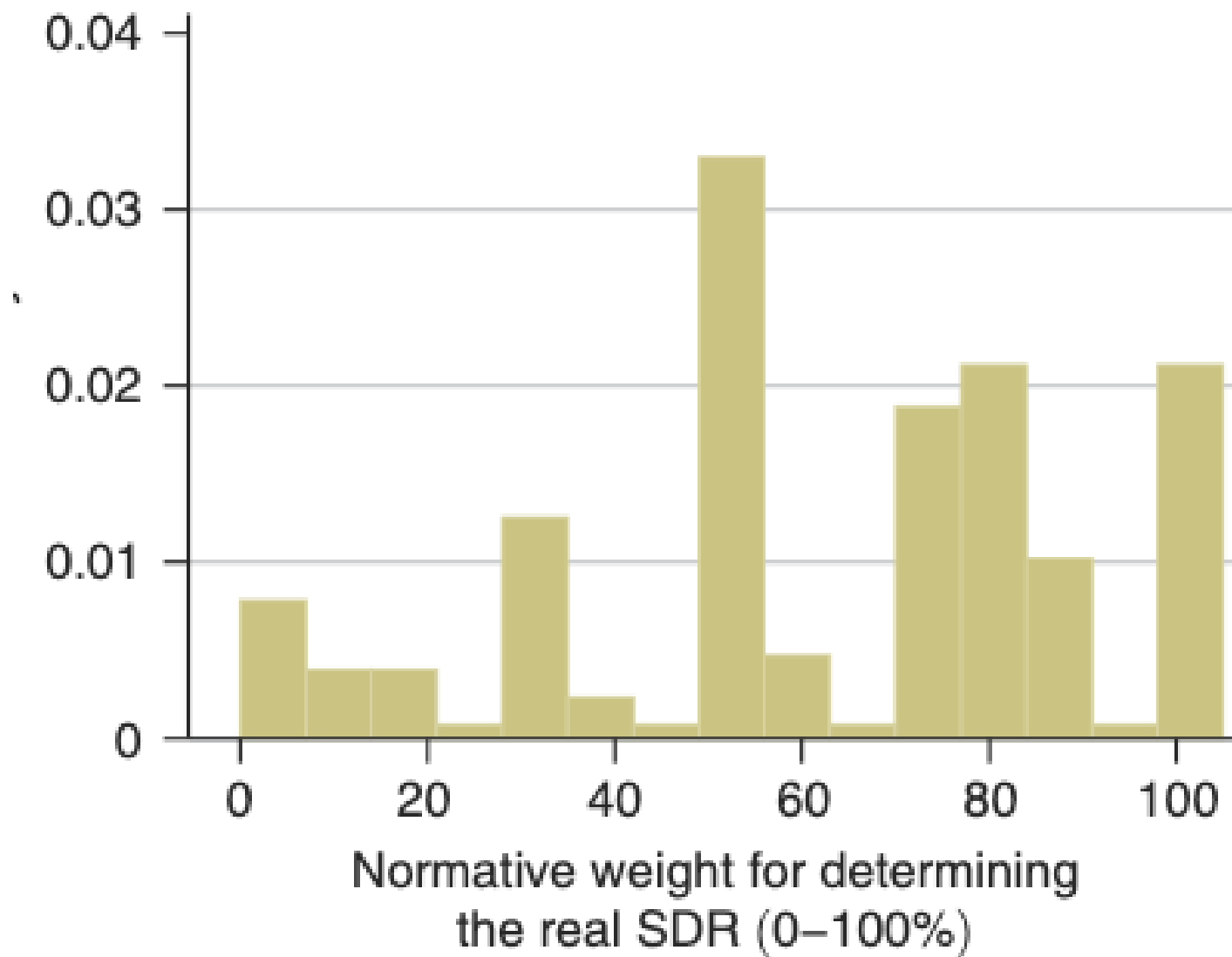
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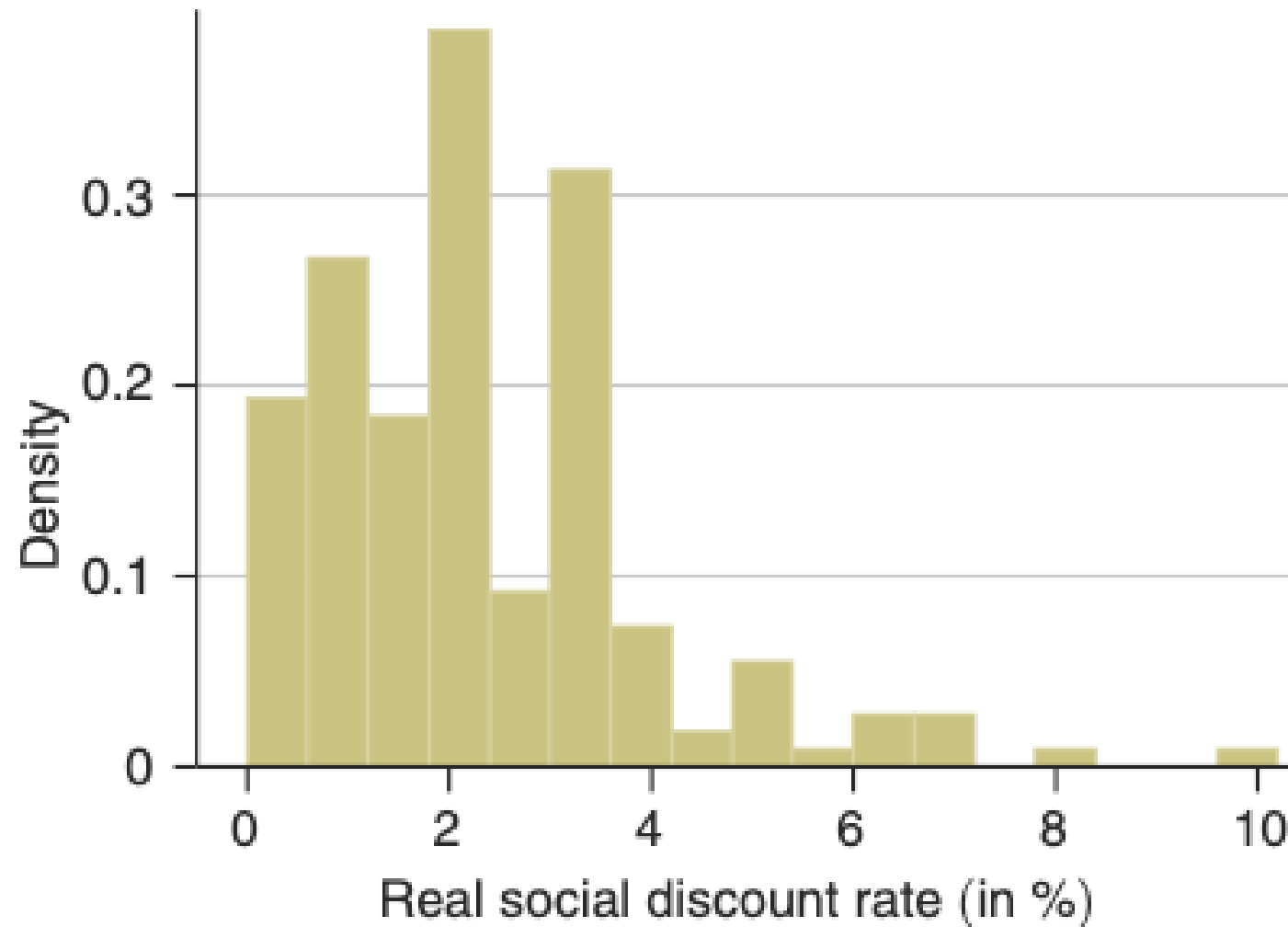
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Discount rates are being significantly revised



Council of Economic Advisers ✓
@WhiteHouseCEA

...

Today, OMB released an important proposed update to Circular A-4, guidance that Federal agencies use to analyze the benefits and costs of proposed Federal regulations. It has not been updated since it was first issued in 2003. 1/



THE WHITE HOUSE
WASHINGTON

Discount rates are being significantly revised



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Council of Economic Advisers  @WhiteHouseCEA · 17h

...

The proposed revision substantially expands guidance on assessing distributional effects. It helps empower agencies to use income-weighted estimates in their analyses by providing them with a weighting methodology if they choose to do so. 6/



2



6



15



2,138



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This option could be especially important in any context where regulations impact disadvantaged communities, which tend to have lower average income & lower property values. Income-weighted analysis can help ensure effects on these communities are not undervalued. 7/



2



2



11



1,246



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...

The proposed revision removes the assumption that individuals affected by regulations are risk neutral. Risk aversion could be consequential for regulations that address areas such as climate change, student loan repayment, health insurance take-up, & pandemic preparedness. 8/



2



2



12



1,238



Discount rates are being significantly revised



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...

Discount rates, which convert future values into present values, are key for analyzing long-term effects. Currently, Circular A-4 recommends two rates for all analyses: 3% and 7%. The proposed revision updates those rates to incorporate new economic data and methods. 9/



2



2



9



1,598



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...

The proposed revision recommends a single primary discount rate & a separate accounting of capital investment effects and risk. Updating the data that produced the original 3% rate produces an updated rate of 1.7%, a critical change for regs with impacts far into the future. 10/



3



16



33



51.9K



Discount rates in the (very) long run

How should we think about discounting in the **very** long run?

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Giglio, Maggiori, and Stroebe (2015) come up with a clever way to think about discount rates in the far future: looking at UK and Singaporean housing markets

Discount rates in the (very) long run

In the UK and Singapore, properties are acquired via **leasehold** or **freehold**

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Discount rates in the (very) long run

In the UK and Singapore, properties are acquired via **leasehold** or **freehold**

- **Leasehold:** temporary, pre-paid, tradable ownership contracts with maturities of 99-999 years, once it expires, you lose the property
- **Freeholds:** same, but **perpetual** ownership, you never lose the property
 - Similar to how things work in the US

Property prices, what do they tell us?

Imagine there are two properties A and B, identical in every way except A is a leasehold with 500 years left until maturity and B is a freehold

Suppose we observe A selling for 900,000 dollars and B selling for 1,000,000 dollars

What do these prices mean? What value do they capture?

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Let's think about a simple example: you are a real estate investor deciding on purchasing a property to add to your rental portfolio in a competitive property market

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\$900,000! investors will compete, bidding higher and higher prices until it reaches the benefits of owning the property (same logic as why prices are the MB of regular goods in competitive markets)

Discount rates in the (very) long run

The price of a house tells us the present value of the future stream of rental payments!

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The price of a house tells us the present value of the future stream of rental payments!

Now let's go back to the original example:

Suppose we observe A selling for 900,000 dollars and B selling for 1,000,000 dollars

What does the price difference between the two properties tell us?

Discount rates in the (very) long run

Both properties are identical until year 500 when **poof**,

Discount rates in the (very) long run

Both properties are identical until year 500 when **poof**, you no longer own property A but you still own property B

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The difference in prices is telling us the present value of property B rental payments starting **500 years from now**

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Both properties are identical until year 500 when **poof**, you no longer own property A but you still own property B

The difference in prices is telling us the present value of property B rental payments starting **500 years from now**

The prices tell us about how the market discounts cash flows very, very far in the future, outside anyone's expected lifespan

Why do discount rates change over time?

Discount rates for cash flows this year versus 500 years in the future may be different for a lot of reasons

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 - The future is getting richer slower, so the future's marginal value of a dollar is higher than if growth did not slow

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- **Changes in growth:** if growth slows down (e.g. from climate change), discount rates fall
 - The future is getting richer slower, so the future's marginal value of a dollar is higher than if growth did not slow
- **Uncertainty:** if we are uncertain about future economic conditions determining the discount rate (e.g. climate change), the discount rate we should use is lower than the average (expected) discount rate

Why do discount rates change over time?

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What are the current expected costs of the project?

Why do discount rates change over time?

The current expected costs are just the costs averaged over either of the potential real discount rates:

$$\frac{1}{2} \frac{\$1 \text{ trillion}}{1.01^{100}} + \frac{1}{2} \frac{\$1 \text{ trillion}}{1.07^{100}} = \$185 \text{ billion}$$

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Now lets compute the value of the damages if we used the expected discount rate, the average of the two: 4%

$$\frac{\$1 \text{ trillion}}{1.04^{100}} = \$20 \text{ billion}$$

Why do discount rates change over time?

The expected discount rate of 4% generated costs that were 10 times smaller than the actual costs!

This means that the expected discount rate is too high compared to the actual discount rate we should be using if we are uncertain about future discount rates

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What discount rate should we use?

$$\frac{\$1 \text{ trillion}}{(1 + r)^{100}} = \$185 \text{ billion} \quad \rightarrow r = .017 = 1.7\%$$

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If we expected the future discount rate to be either 1% or 7%, the proper discount rate to use was actually 1.7%, **not** the average 4%!

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If we expected the future discount rate to be either 1% or 7%, the proper discount rate to use was actually 1.7%, **not** the average 4%!

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1.7% is called the **certainty-equivalent** discount rate: the certain discount rate that delivers the same present value as the possible set of uncertain rates (1% and 7%)

Main takeaway: Uncertainty about the future economic conditions governing the discount rate makes the discount rate we should be using lower than expected

Discount rates in the (very) long run: United Kingdom

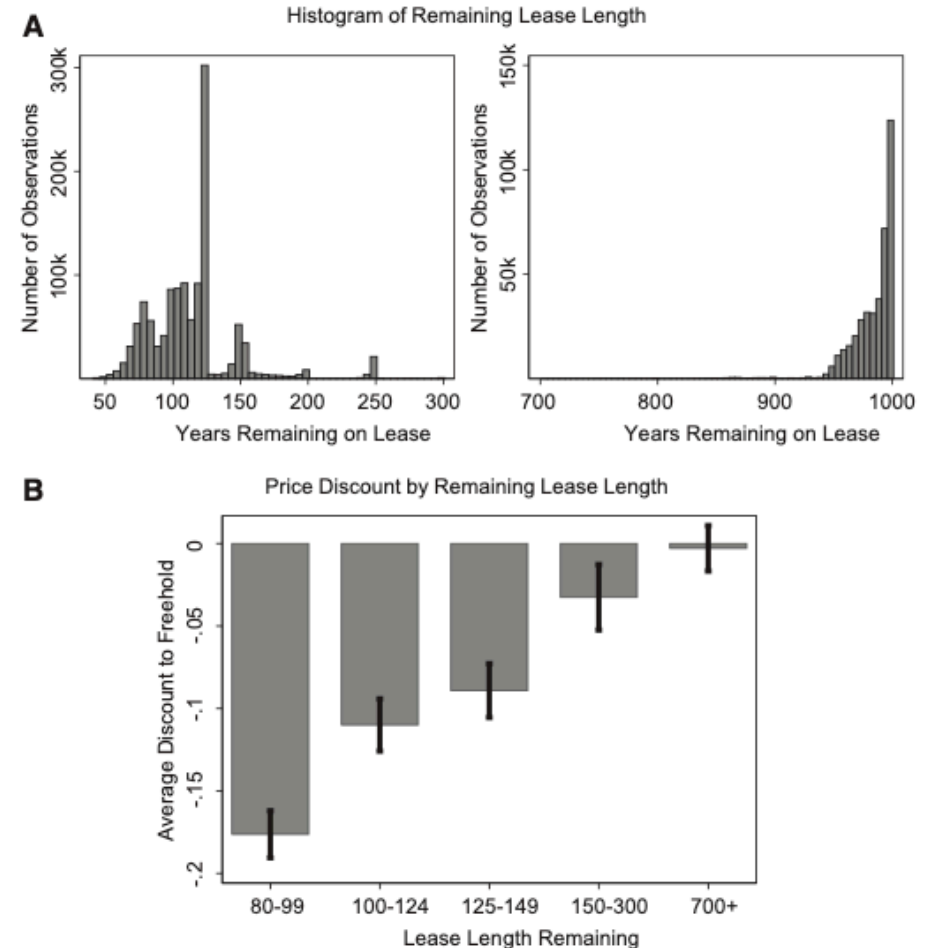
What are these long run discount rate?

In the UK:

- leases expiring within 100 years cost 17% less than a freehold
- leases expiring 150-300 years from now cost 5% less

Implies a discount rate of about

2.6%



Discount rates in the (very) long run: Singapore

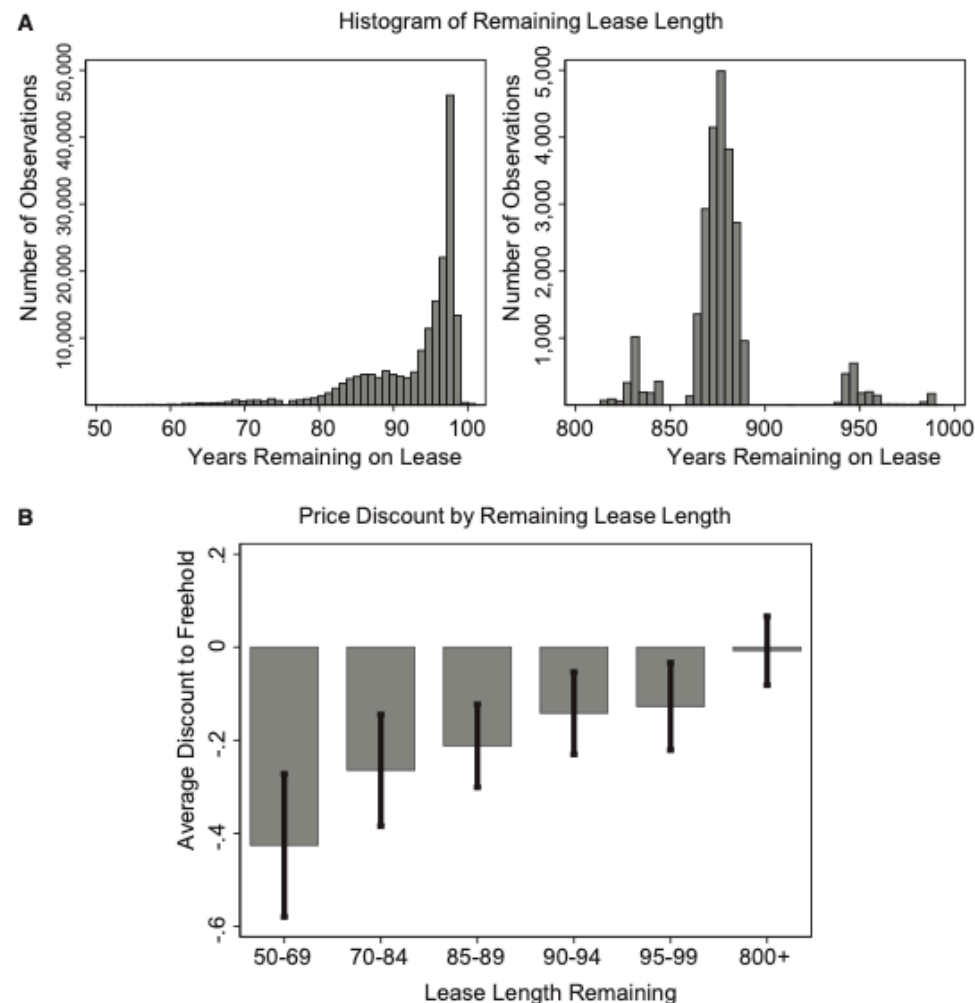
What are these long run discount rate?

In Singapore:

- leases expiring within 70 years cost 40% less than a freehold
- leases expiring 95-99 years from now cost 15% less

Implies a discount rate of about

2.6%



Discount rates on rental payments

We can check the validity of these estimates by seeing whether **rental payments** depend on the length remaining of the contract

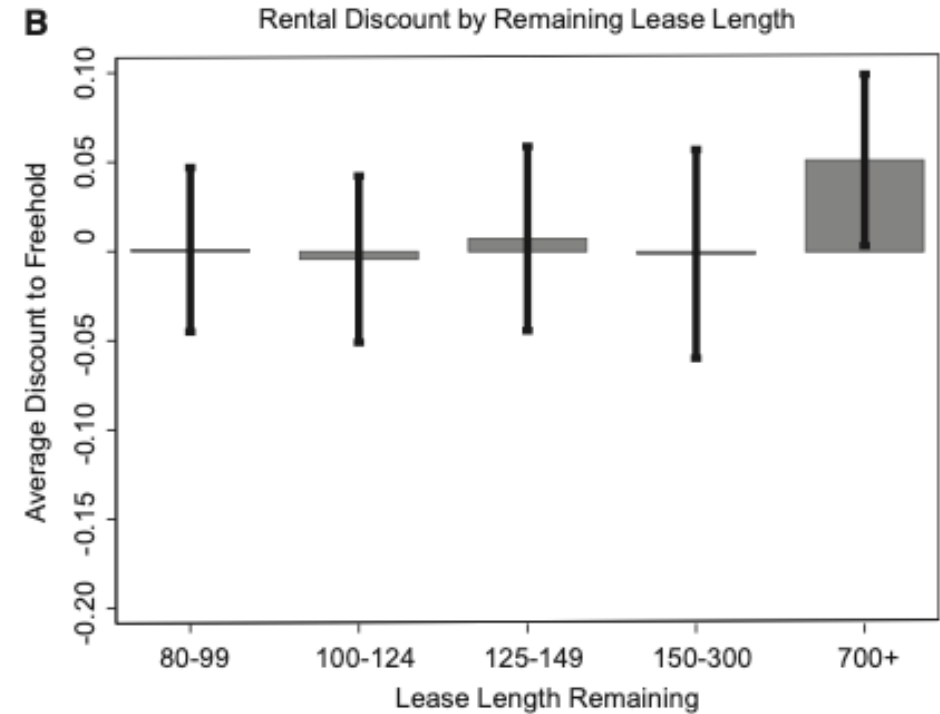
Discount rates on rental payments

We can check the validity of these estimates by seeing whether **rental payments** depend on the length remaining of the contract

There's no reason the rent you pay for your house should depend on how much longer the owner has property rights if your lease is short

Discount rates on rental payments

Rental rates (mostly) do not depend on the remaining lease time!



Municipal bond markets

Bonds

Bonds: what are they?

Bonds

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[Wiki] A bond is a type of security under which the issuer (debtor) owes the holder (creditor) a debt, and is obliged – depending on the terms – to repay the principal (i.e. amount borrowed) of the bond at the maturity date as well as interest (called the coupon) over a specified amount of time.

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Bonds are assets, can be traded on secondary markets

Bonds

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Bonds are a way to raise money:

- At time t , issue bonds with maturity T
- Get money from creditors
- Pay back interest/coupon over time between t and T
- Pay back principal at some future date $t + T$

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We will focus on **municipal bonds (munis)** for pricing climate risk, why?

Municipal bonds

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Suppose Starbucks has coffee packaging plants in Miami

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Miami is expecting disastrous sea level rise, what can Starbucks do to manage it?

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Move its plants somewhere else away from the ocean

Municipal bonds

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Suppose Starbucks has coffee packaging plants in Miami

Miami is expecting disastrous sea level rise, what can Starbucks do to manage it?

Move its plants somewhere else away from the ocean

The city of Miami does not have the same option: it bears the full potential cost of sea level rise

Municipal bonds

How do munis work?

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Local governments issue munis for financing public projects (roads, infrastructure, etc)

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Debt is typically paid back in a pre-specified way

- **General obligation bonds:** paid using tax revenue
- **Revenue bonds:** project-specific revenue (e.g. parking garage revenues)

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Which are less risky for lenders?

Municipal bonds

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They pay out a **coupon** (assume annually) C

They trade on the bond market at some **price** P which will depend on:

- The face value
- The coupon
- When the bond matures
- *Other underlying economic conditions affecting risk of non-payment*

Municipal bonds

The price of a bond should be equal to the expected present value cash flow

If we hold a bond, we discount future payments because:

- Inflation
- Risk of non-payment
- Etc

Municipal bonds

We can define the **yield to maturity** y as:

$$P = \left[\sum_{t=1}^T \frac{C}{(1+y)^t} \right] + \frac{FV}{(1+y)^T}$$

y is the effective interest rate the investor is getting on a price P bond

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Given some price P , a higher yield y means that the flow of coupons C or face value FV must be higher

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Given some coupon C and face value FV , a higher yield y means a lower price P

Municipal bonds

Why would future climate risk be capitalized into munis? Examples for why?

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Why would future climate risk be capitalized into munis? Examples for why?

1. If climate change (e.g. sea level rise) destroys infrastructure, raises municipal costs, raises risk of bankruptcy and non-payment of the bond
2. If climate change induces people to leave, this shrinks the tax base, makes it more difficult for the municipality to pay back the bond, raises risk of non-payment

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Why would future climate risk be capitalized into munis? Examples for why?

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2. If climate change induces people to leave, this shrinks the tax base, makes it more difficult for the municipality to pay back the bond, raises risk of non-payment

Factors like these should be priced into the bond if traders understand climate risk

Municipal bonds

Let's work with a simple one-period zero coupon example: $C = 0$, $T = 1$, $FV = 105$:

$$P = \frac{105}{(1 + y)^1}$$

Suppose there is no climate change and the market yield is 5%, the price of the muni is:

Municipal bonds

Let's work with a simple one-period zero coupon example: $C = 0$, $T = 1$, $FV = 105$:

$$P = \frac{105}{(1 + y)^1}$$

Suppose there is no climate change and the market yield is 5%, the price of the muni is:

$$P = \frac{105}{1 + .05} = 100$$

Municipal bonds

Now suppose we are considering the same muni, but there is a 7% chance that the city will be destroyed by sea level rise before next year

- Additional 7% chance that the bond will not be paid

To bear this additional risk, traders will demand a higher yield (lower price)

Municipal bonds

We can solve for the new price:

$$P = \frac{105}{1 + .05} \times \underbrace{(1 - .07)}_{\approx 1/(1 + .075)} = \frac{105}{1 + .05} \times \frac{1}{1 + .075} = 93$$

and the associated yield:

$$\begin{aligned} \frac{1}{1 + y} &= \frac{1}{1 + .05} \times (1 - .07) = \frac{.93}{1.05} \\ \Rightarrow y &= \frac{1.05}{.93} - 1 = .129 \end{aligned}$$

Municipal bonds

The additional 7% climate risk:

- Decreased the price by 7% from 100 to 93
- Increased the yield by 7.9 percentage points from 5% to 12.9%

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We can measure the financial risks of climate change by looking at how places with different climate risk have munis with different yields

Municipal bonds and sea level rise

Painter (2020) looks at how sea level rise (SLR) risk affects bond yields

- Also looks at other things outside what we're doing in class

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SLR is a slow phenomenon, will matter increasingly over the next century

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How should SLR affect bond yields of different maturities?

SLR is a slow phenomenon, will matter increasingly over the next century

- non-existent in the short run: short-term bonds shouldn't be affected
- only shows up in the long run: long-run bonds should be affected if investors care

Municipal bonds and sea level rise

City	County	Mean Annual Loss (MM\$)	Climate Risk
New Orleans, LA	Orleans	1940	1.479%
Miami, FL	Miami Dade	2964	0.420%
Tampa/St. Petersburg, FL	Hillsborough, Pinellas	948	0.324%
Virginia Beach, VA	Virginia Beach	328	0.173%
Boston, MA	Suffolk	849	0.149%
Baltimore, MD	Baltimore	299	0.104%
LA/Long Beach/Santa Ana, CA	Los Angeles, Orange	217	0.097%
New York, NY/ Newark,NJ	Bronx, Kings, New York, Queens, Richmond, Essex	2159	0.089%
Providence, RI	Providence	135	0.083%
Philadelphia, PA	Philadelphia	309	0.044%
San Francisco/Oakland, CA	San Francisco, Alameda	185	0.042%
Houston, TX	Walker, Montgomery, Liberty, Waller, Austin, Harris, Chambers, Colorado, Wharton, Fort Bend, Galveston, Brazoria, Matagorda	214	0.038%
Seattle, WA	King	90	0.023%
Washington D.C.	Washington	91	0.016%
San Diego, CA	San Diego	14	0.004%
Portland, OR	Multnomah	4	0.002%
San Jose, CA	Santa Clara	2	0.001%

Climate risk: expected percent loss of city GDP

Where is the climate risk?

Municipal bonds and sea level rise

Panel A: Descriptive Statistics by Climate Risk

	<u>Climate Bonds</u>				<u>Non-Climate Bonds</u>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	N	Mean	Median	SD	N	Mean	Median	SD
Total Annualized Cost (%)	40161	3.03	2.93	1.52	210695	2.95	2.85	2.17
Gross Spread (%)	41766	0.54	0.49	0.30	217113	0.60	0.53	0.33
Yield (%)	49309	3.02	3.00	1.42	269820	2.91	2.85	1.37

What does the raw data say about climate/SLR exposed and non-exposed munis?

Municipal bonds and sea level rise

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What does the raw data say about climate/SLR exposed and non-exposed munis?

Climate(-exposed) bonds are 8 basis points more expensive to offer

11 basis points higher yield

Yield

Comparing munis offered in the same state and year, controlling for other factors:

Areas at risk for SLR must offer greater yields by 16pp

<i>Panel B: Yield for Long-Term and Short-Term Bonds</i>				
Dependent Variable:	<u>Long-Term</u>		<u>Short-Term</u>	
	(1)	(2)	(3)	(4)
	Yield	Yield	Yield	Yield
Climate Risk	0.161** (2.219)		0.070 (1.462)	
Ln(Climate Risk)		0.203* (1.816)		0.079 (1.008)
Controls	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes
Observations	27,355	27,355	291,746	291,746
R-squared	0.503	0.503	0.839	0.839

Gross spread

They have higher gross spreads (higher underwriter search costs): 10-15bp higher for long-term bonds

Panel C: Gross Spread for Long-Term and Short-Term Bonds

Dependent Variable:	Long-Term		Short-Term	
	(1)	(2)	(3)	(4)
	Spread	Spread	Spread	Spread
Climate Risk	0.108** (1.972)		-0.004 (-0.072)	
Ln(Climate Risk)		0.152** (2.188)		0.019 (0.222)
Controls	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes
Observations	24,514	24,514	234,321	234,321
R-squared	0.368	0.369	0.326	0.326

Maturities

Effect is larger for longer-maturity bonds: bonds maturing

Panel A: Long-Term Specifications

	(1)	(2)	(3)	(4)	(5)
Issue Maturity:	≥ 20 Years	≥ 30 Years	≥ 2036	≥ 2041	≥ 2046
Ln(Climate Risk)	0.198* (1.876)	0.656** (2.171)	0.205* (1.705)	0.489* (1.714)	1.540*** (2.967)
Controls	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes
Observations	46,191	6,665	25,307	8,495	2,095
R-squared	0.368	0.232	0.339	0.222	0.160

Credit ratings

SLR matters most for
bonds with lower ratings:

Non-high grade munis
costs are 50bp higher with
higher climate risk

Credit Rating:	<u>Long-Term</u>		<u>Short-Term</u>	
	(1)	(2)	(3)	(4)
	< AA-	≥ AA-	< AA-	≥ AA-
Ln(Climate Risk)	0.527** (2.041)	0.141 (0.686)	0.107 (0.878)	0.091 (0.634)
Controls	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes
Observations	5,339	14,095	43,714	187,529
R-squared	0.609	0.238	0.090	0.724

What does this all mean?

We've seen that places more exposed to SLR:

- Must offer higher yields on long-term bonds, with yields increasing in time to maturing
- Incur higher gross spreads
- Must offer even higher yields if they do not have a high credit rating

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Think about the long-run equilibrium of economic activity, where people live, etc

What does this all mean?

We've seen that places more exposed to SLR:

- Must offer higher yields on long-term bonds, with yields increasing in time to maturing
- Incur higher gross spreads
- Must offer even higher yields if they do not have a high credit rating

Think about the long-run equilibrium of economic activity, where people live, etc

What does this suggest will happen?

What does this all mean?

1. Capital is becoming more expensive in SLR-exposed cities

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Does this tell the whole story?

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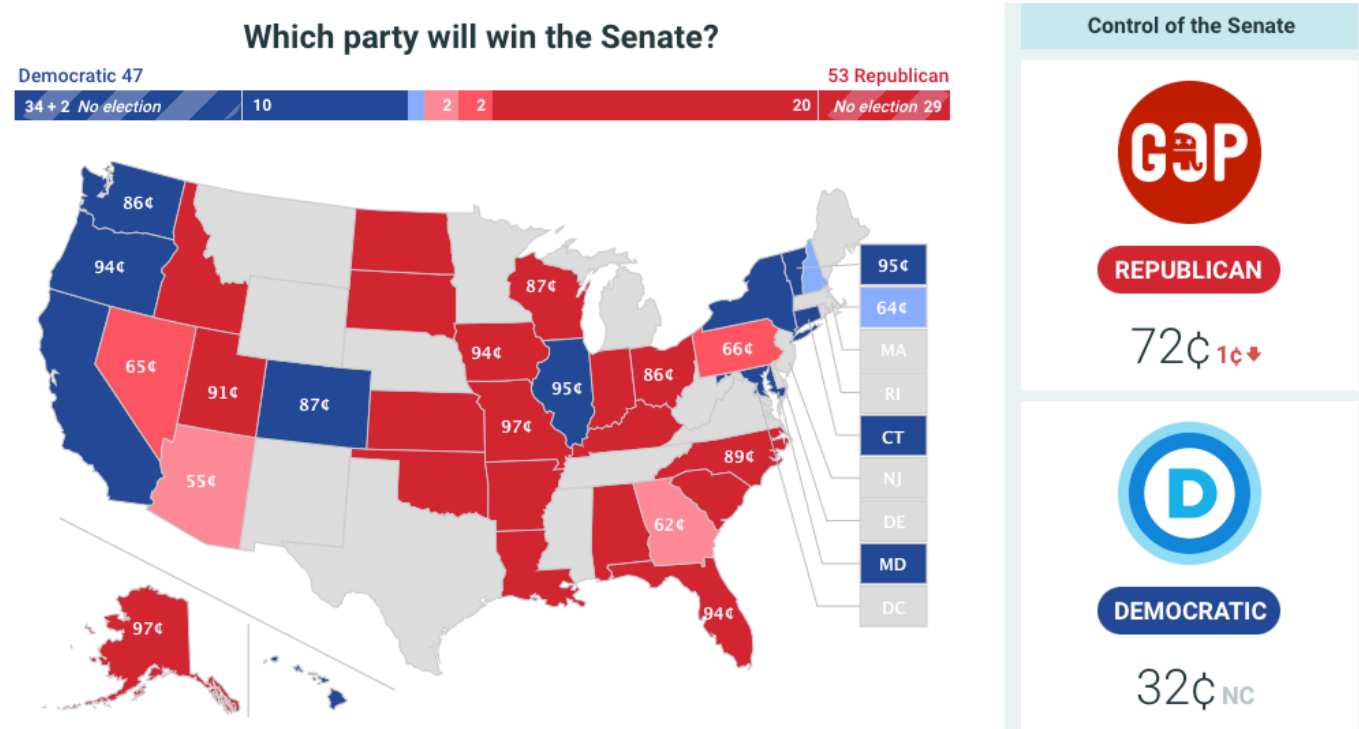
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In both cases, muni markets serve an important function for directing resources and people to the most productive areas

Prediction markets

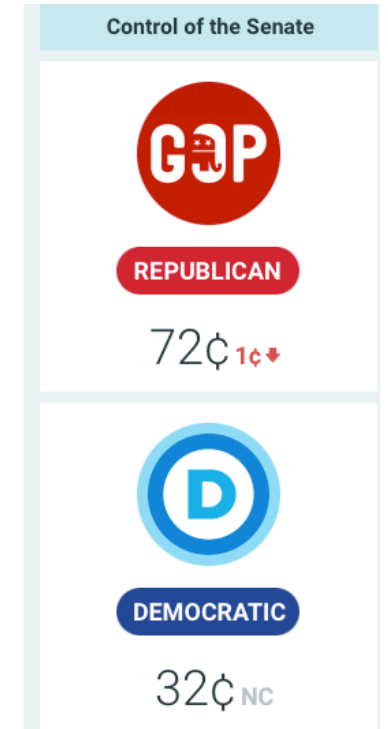
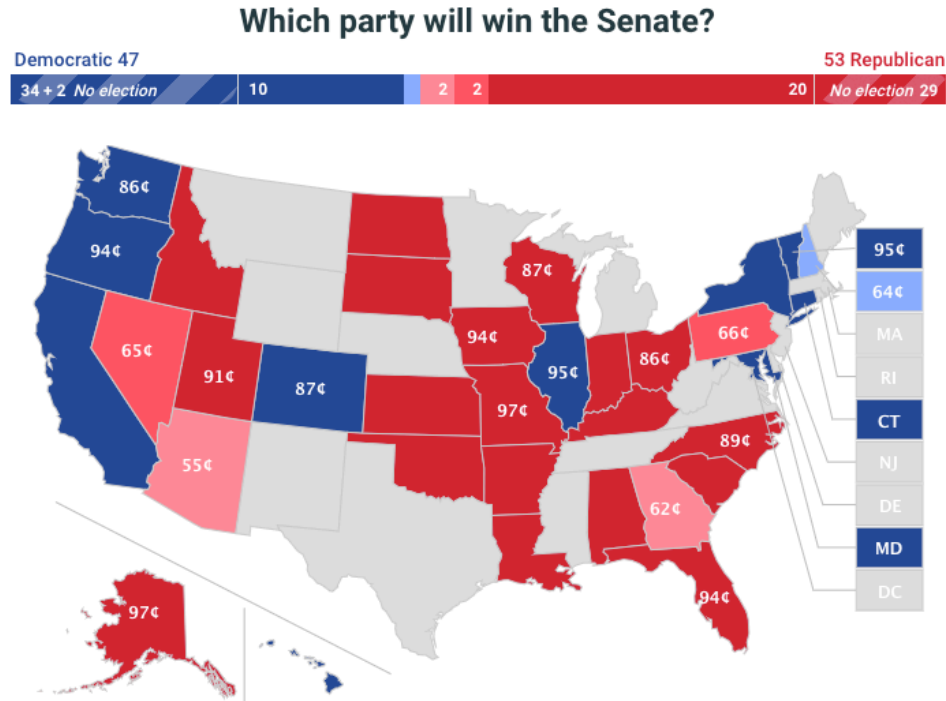
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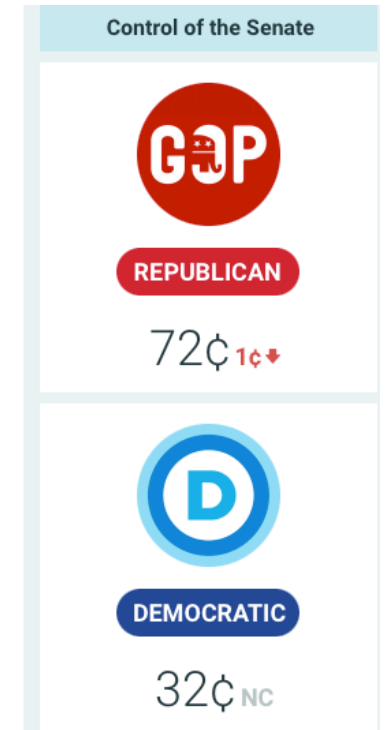
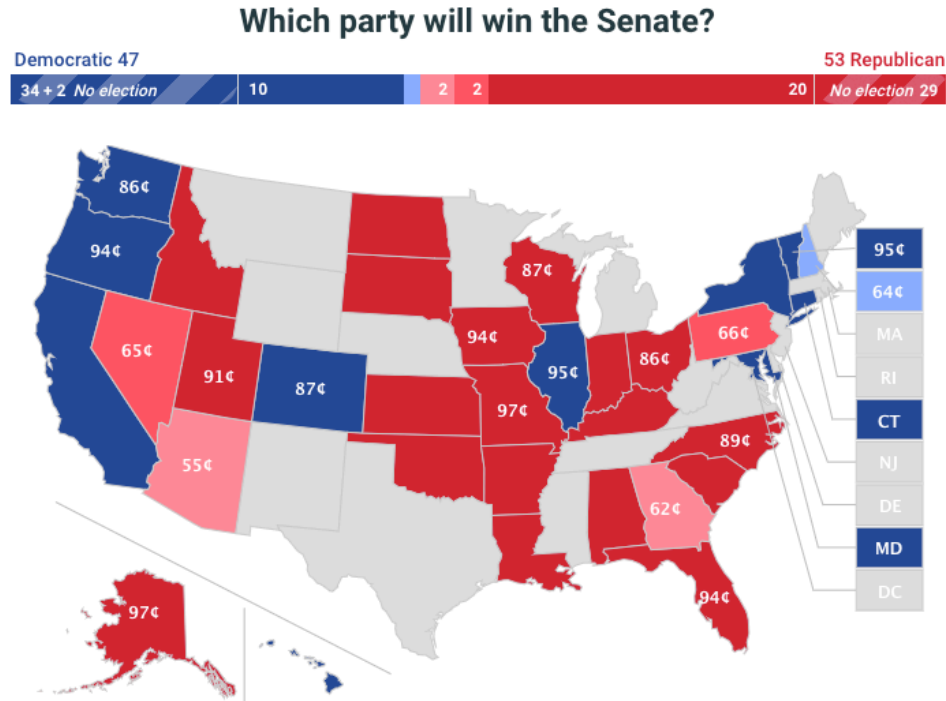


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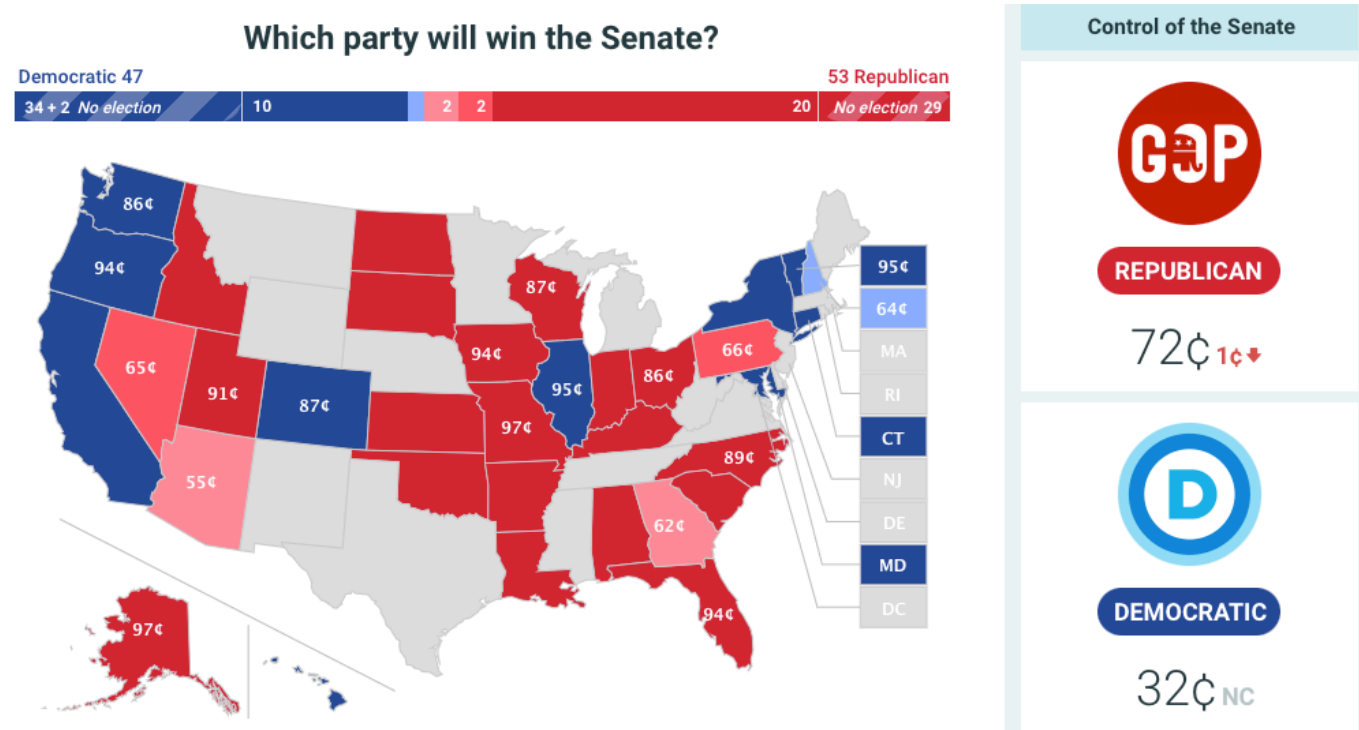
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How does it work?



Prediction markets

You can buy a share (asset) for whether the event will happen or not happen

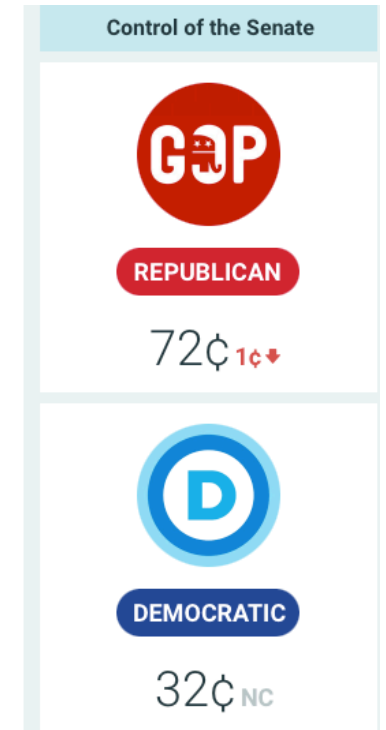
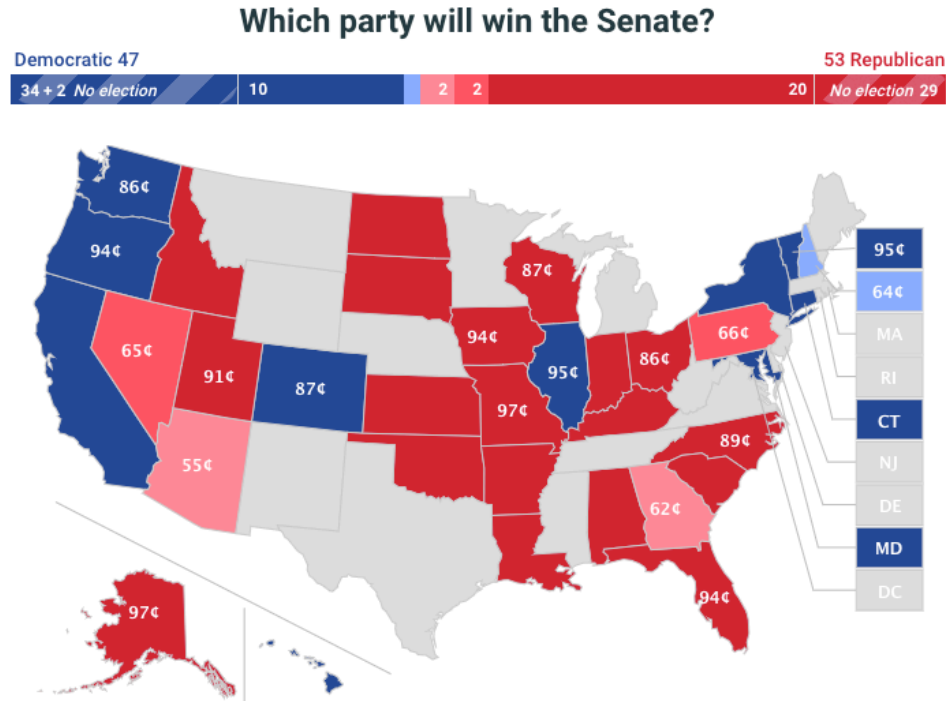


Prediction markets

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(asset) for whether the
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The price of this share is:

- 72c for Republicans winning
- 32c for Democrats winning



Prediction markets

After the election:

- The shares for the winning side pay off \$1 each
- The shares for the losing side are worth \$0

What does the prediction market tell us?

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Let's think about the economics of the market

- The cost of a share be c dollars
- Your belief about the probability of an event happening be p percent

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Based on your beliefs about the event, your expected profit from buying a share is: $[p \times 1 + (1 - p) \times 0] - c$

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Individual profit motives always drive c toward p

The price of the share tells us the market's expectation about the probability of the event!

Waxman-Markey

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What did WM propose to do?

Set an annual cap on CO_2 emissions that starts in 2012 and declines over time to:

- 83% of 2005 levels in 2020
- 58% of 2005 levels in 2030
- 17% of 2005 levels in 2050

Waxman-Markey

WM allowed permits to be **traded**

Waxman-Markey

WM allowed permits to be **traded** and also **banked and borrowed**

- **Banked:** permits not used this year can be saved
- **Borrowed:** can emit more than retired permits today on the promise of retiring the extra necessary permits in the future
- Borrowing had an 8% interest rate

Waxman-Markey

How did WM allocate permits ([info here](#))?

Most were freely allocated:

- **35%** of permits go to electric utilities

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Rest are auctioned or given to different government agencies

Waxman-Markey: the history

June 26, 2009: Waxman-Markey passes the House of Representatives (219-212)

- 211/255 Democrats vote yes, 8/176 Republicans vote yes
- First cap and trade bill to be passed by congress!
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It took until April 2010 to convince one Republican ... who was it?

Waxman-Markey: the history



Waxman-Markey: the history

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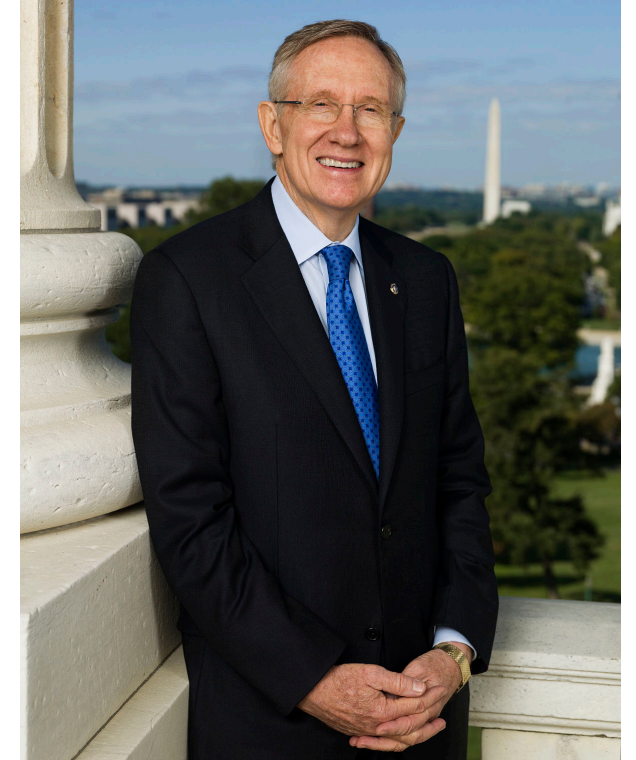
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You won't believe what happens next

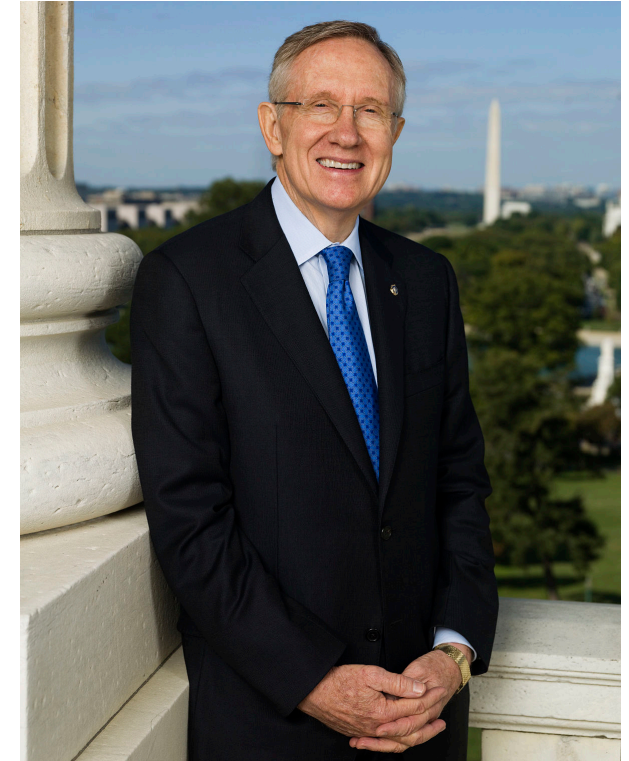


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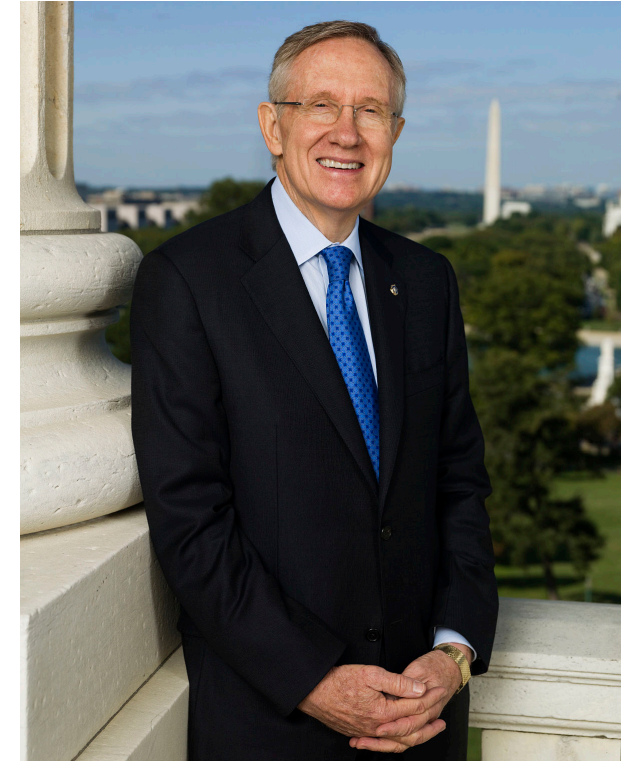
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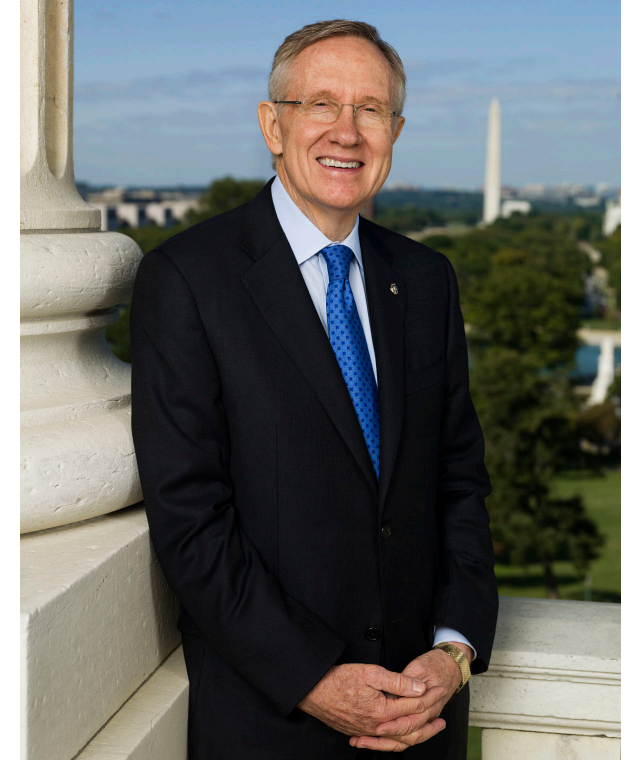


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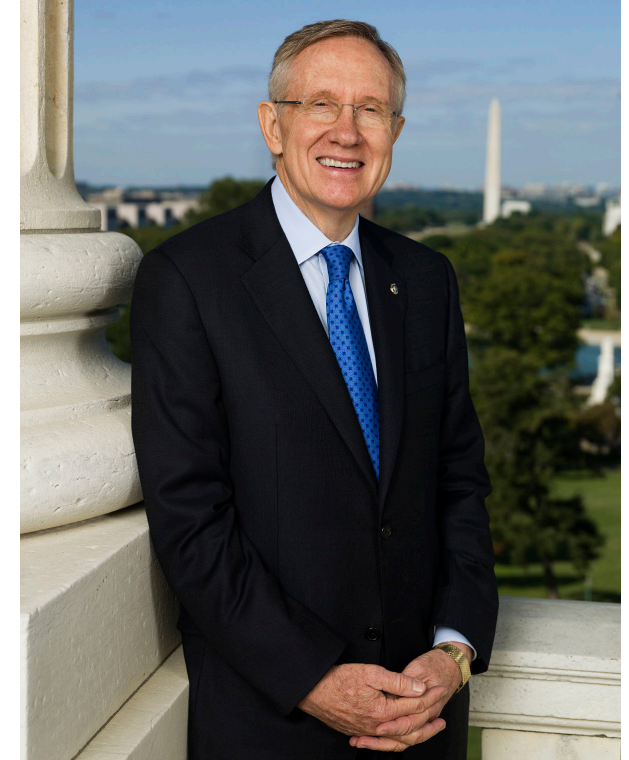
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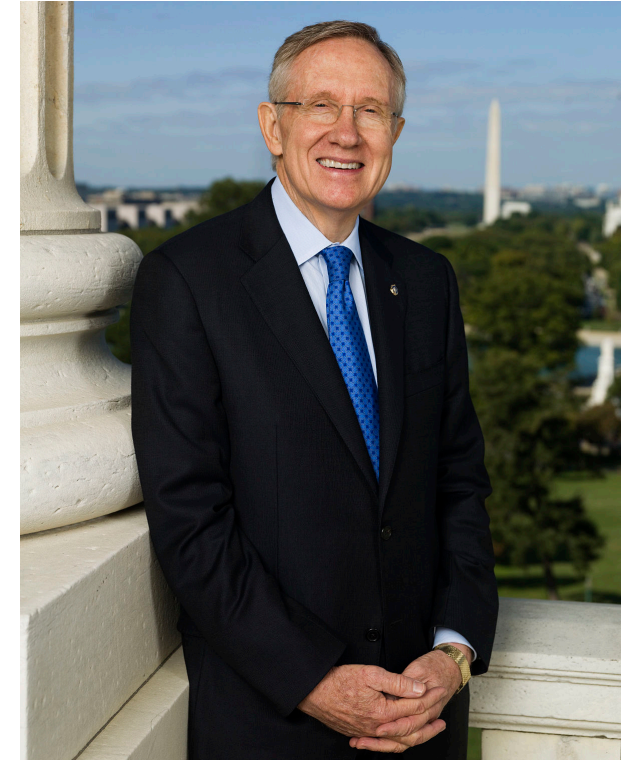
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The Senate calendar couldn't accommodate both climate and immigration legislation: Lindsey Graham thought it was cheap point scoring from Reid



Waxman-Markey: the history



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It was unclear whether governor Jan Brewer would sign it

Waxman-Markey: the history



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Graham formally states he refuses to delay a climate bill for immigration and abandons his legislation



Waxman-Markey: the probabilities

Jun 26: House passes WM

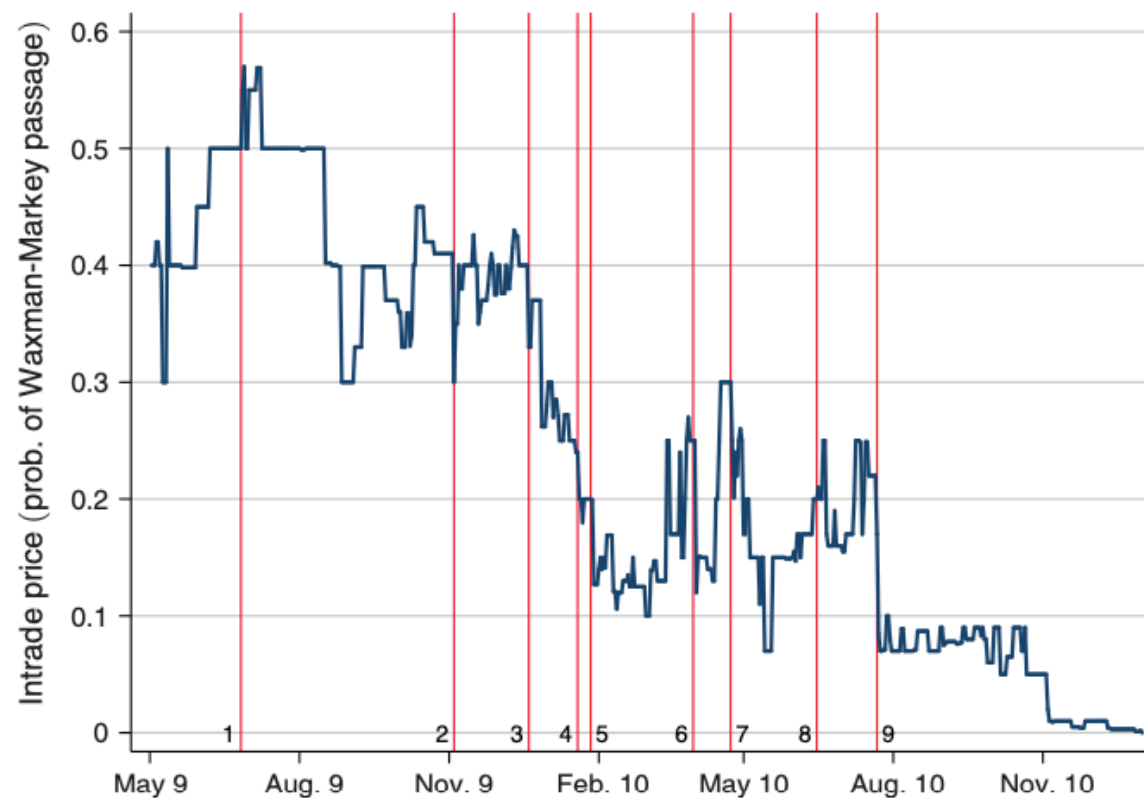
Nov 4: Graham joins senate effort

Dec 20: Copenhagen negotiations

Jan 19: Scott Brown wins Mass. senate seat

Apr 23: Graham drops support

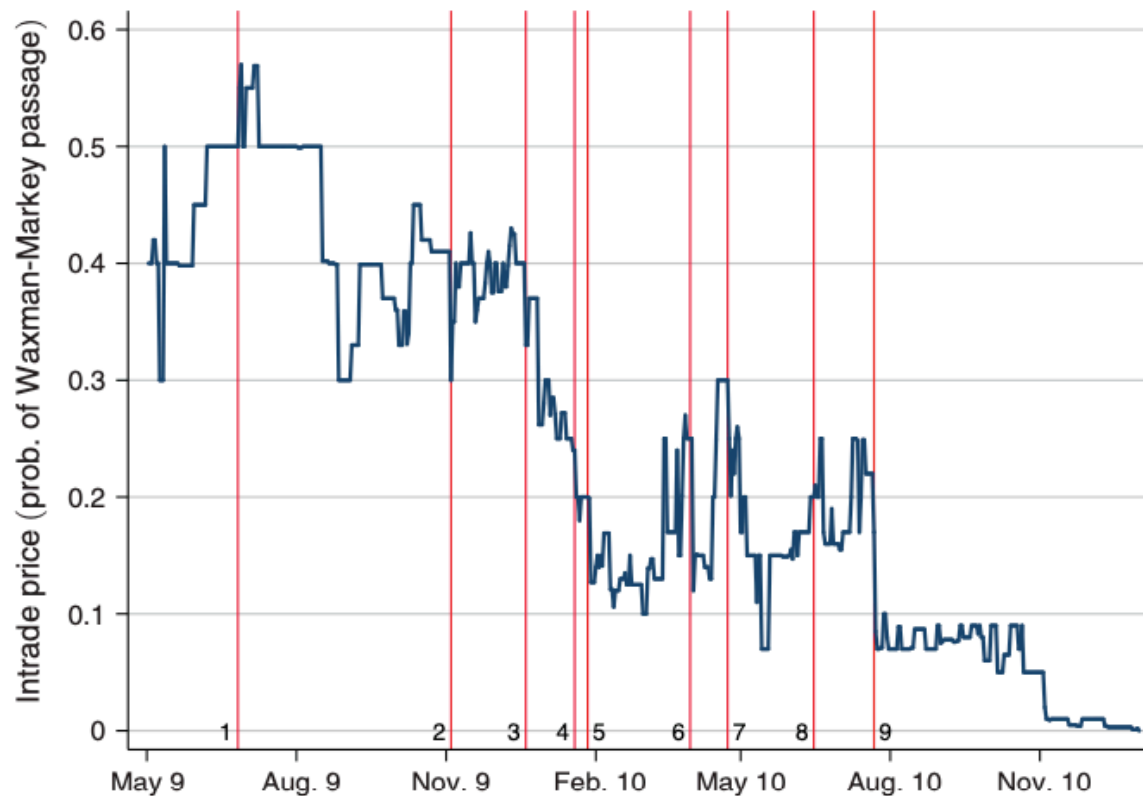
July 22: Senate drops cap and trade



Waxman-Markey: the probabilities

Markets almost never thought WM was a favorite to pass!

Even around Graham-Kerry-Lieberman announcement, probabilities were only 20%

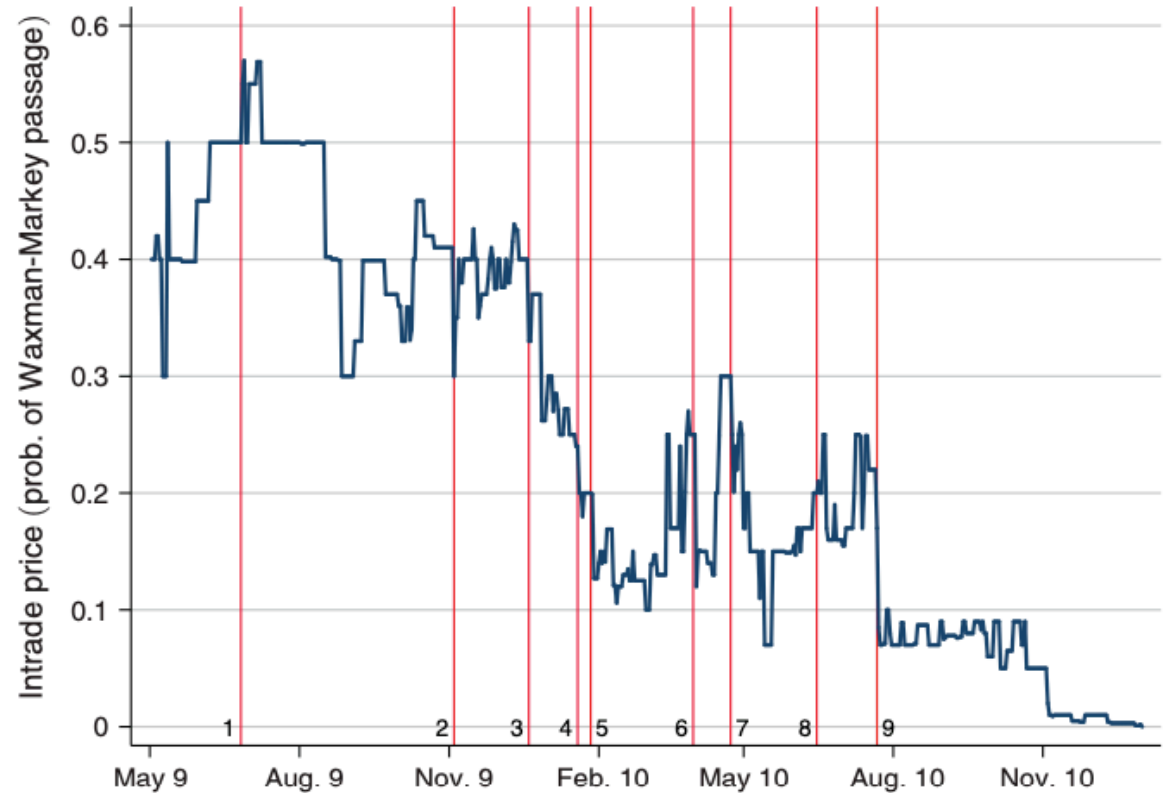


Waxman-Markey: the probabilities

The prediction market prices tell us the market's expectation of climate legislation being implemented

The legislation itself tells us how much emissions must be reduced

With one more piece of information - the **costs** of the legislation - we can back out the CO_2 MAC



Equity markets

We can back out the costs of the legislation from **stock return** data

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Let's write down a simple model of stock returns and firm value to see how

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Let's write down a simple model of stock returns and firm value to see how

- P_t : probability of WM passing
- $V_{i,t}^{WM}$: the value of firm i on day t if WM **passes**
- $V_{i,t}^-$: the value of firm i on day t if WM **does not pass**
- $V_{i,t}$: the expected value of firm i on day t based on the probability of WM passing:
 - $V_{i,t} = P_t \times V_{i,t}^{WM} + (1 - P_t) \times V_{i,t}^-$
 - $V_{i,t}$ is the prediction market weighted average

To keep it simple, assume nothing else is changing besides WM

Equity markets

$$V_{i,t} = P_t \times V_{i,t}^{WM} + (1 - P_t) \times V_{i,t}^-$$

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Define E_i^{WM} as the percentage effect of WM on firm value:

$$E_i^{WM} = \frac{V_{i,t}^{WM} - V_{i,t}^-}{V_{i,t}^-}$$

E_i^{WM} tells us how much firm value changed as a result of WM going into effect

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E_i^{WM} tells us how much firm value changed as a result of WM going into effect

Rewrite V_i in terms of E_i^{WM} :

$$V_{i,t} = V_{i,t}^- \times (1 + P_t E_i^{WM})$$

The firm value on day t is the value of the firm if WM doesn't pass ($V_{i,t}^-$), but

Equity markets

$$V_{i,t} = V_{i,t}^- \times (1 + P_t E_i^{WM})$$

Stock returns $r_{i,t}$ are the change in log firm value:

$$r_{i,t} = \ln V_{i,t} - \ln V_{i,t-1} = \ln \frac{V_{i,t}}{V_{i,t-1}}$$

We can write this as:

$$r_{i,t} = \ln \frac{V_{i,t}^- \times (1 + P_t E_i^{WM})}{V_{i,t-1}^- \times (1 + P_{t-1} E_i^{WM})}$$

Equity markets

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$$r_{i,t} = \ln \frac{V_{i,t}^-}{V_{i,t-1}^-} \ln \frac{(1 + P_t E_i^{WM})}{(1 + P_{t-1} E_i^{WM})}$$

$$r_{i,t} = \ln \frac{V_{i,t}^-}{V_{i,t-1}^-} \times \left[\underbrace{\overbrace{\ln(1 + P_t E_i^{WM})}^{\ln(1+x) \approx x}}_{\approx P_t E_i^{WM}} - \underbrace{\overbrace{\ln(1 + P_{t-1} E_i^{WM})}^{\ln(1+x) \approx x}}_{\approx P_{t-1} E_i^{WM}} \right]$$

$$r_{i,t} = P_t E_i^{WM} - P_{t-1} E_i^{WM} + \underbrace{\ln \frac{V_{i,t}^-}{V_{i,t-1}^-}}_{\substack{\ln 1 = 0 \\ \text{if not changing}}}$$

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$$r_{i,t} = P_t E_i^{WM} - P_{t-1} E_i^{WM} = (P_t - P_{t-1}) \times E_i^{WM}$$

If no other determinants of firm value are changing from day-to-day, the stock return tells us the effect of WM on firm value scaled by the change in the probability that WM happens

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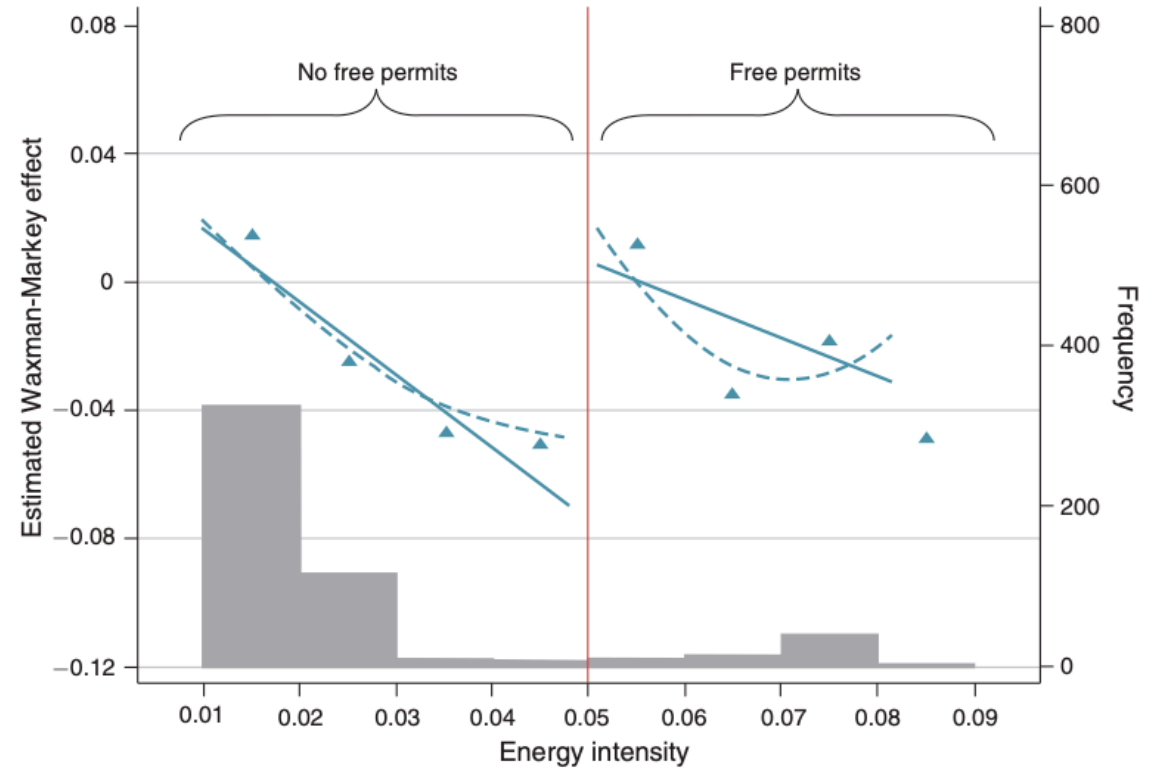
We have data on stock returns $r_{i,t}$, we have data on market expectations P_t, P_{t-1} , we can then get E_i^{WM}

Equity markets: getting MAC

X-axis: energy intensity

Red line: cut off for free permits

Y-axis: change in firm value from WM



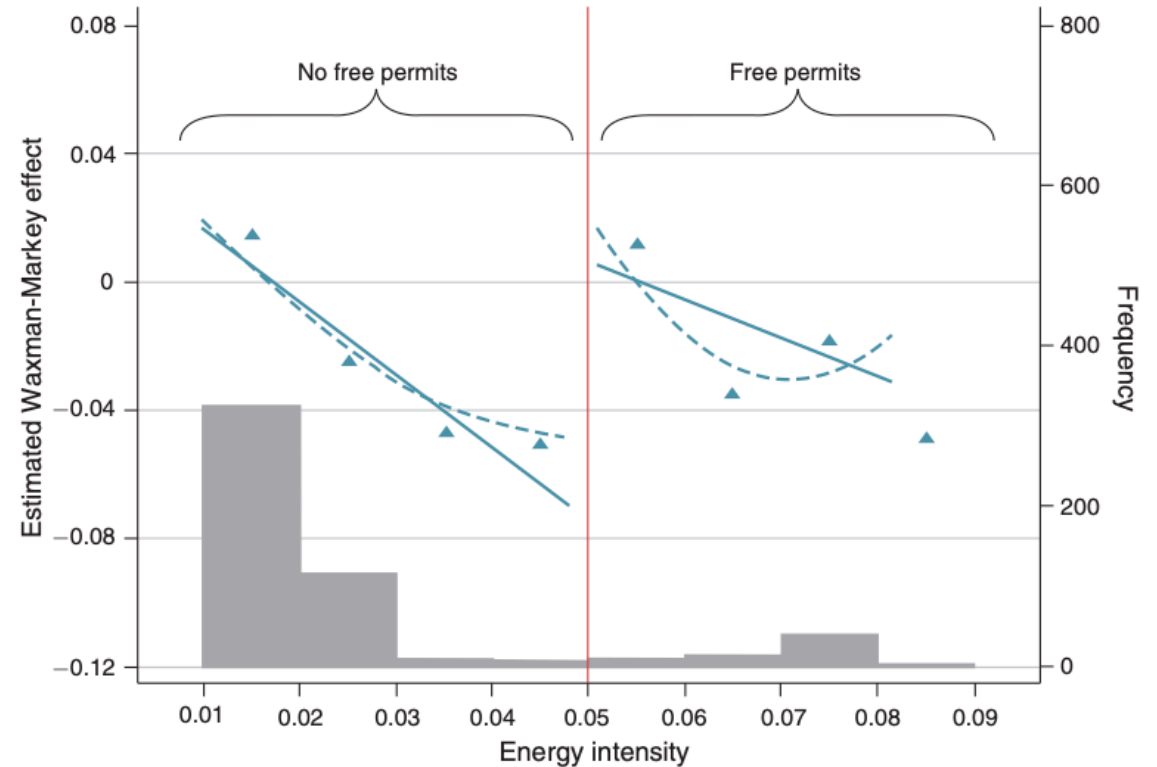
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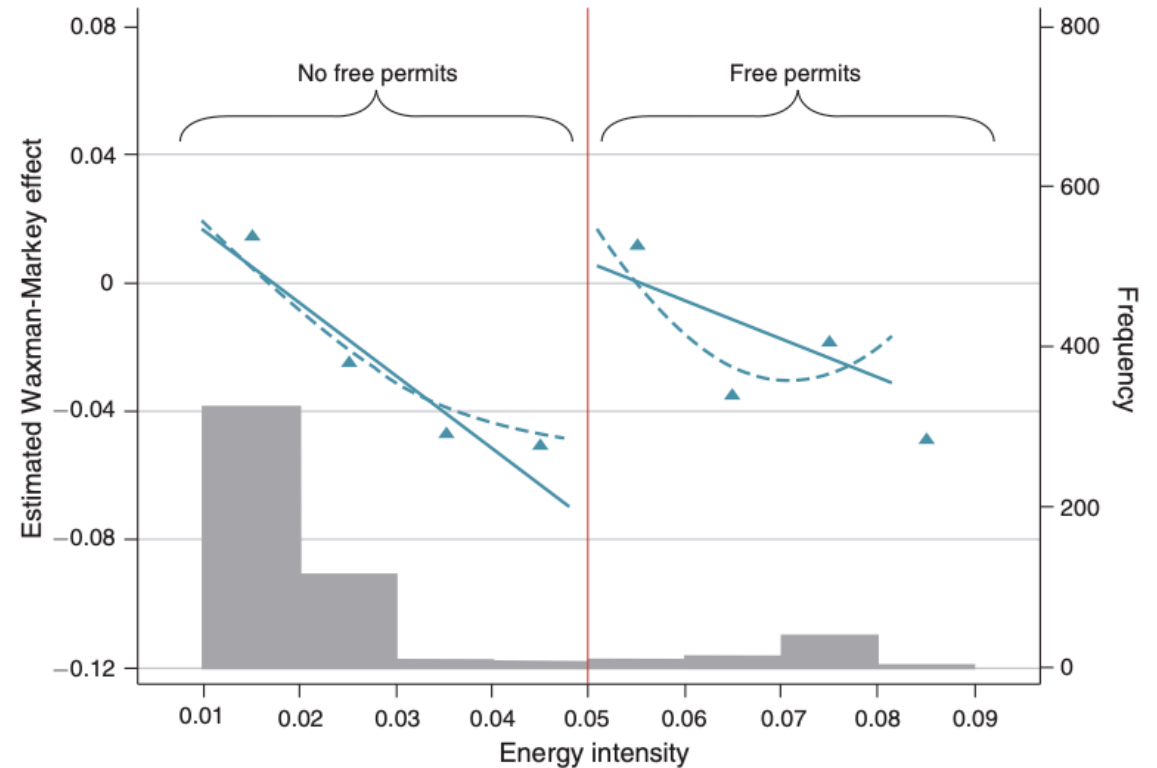
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But: getting free permits mitigates the negative impact!

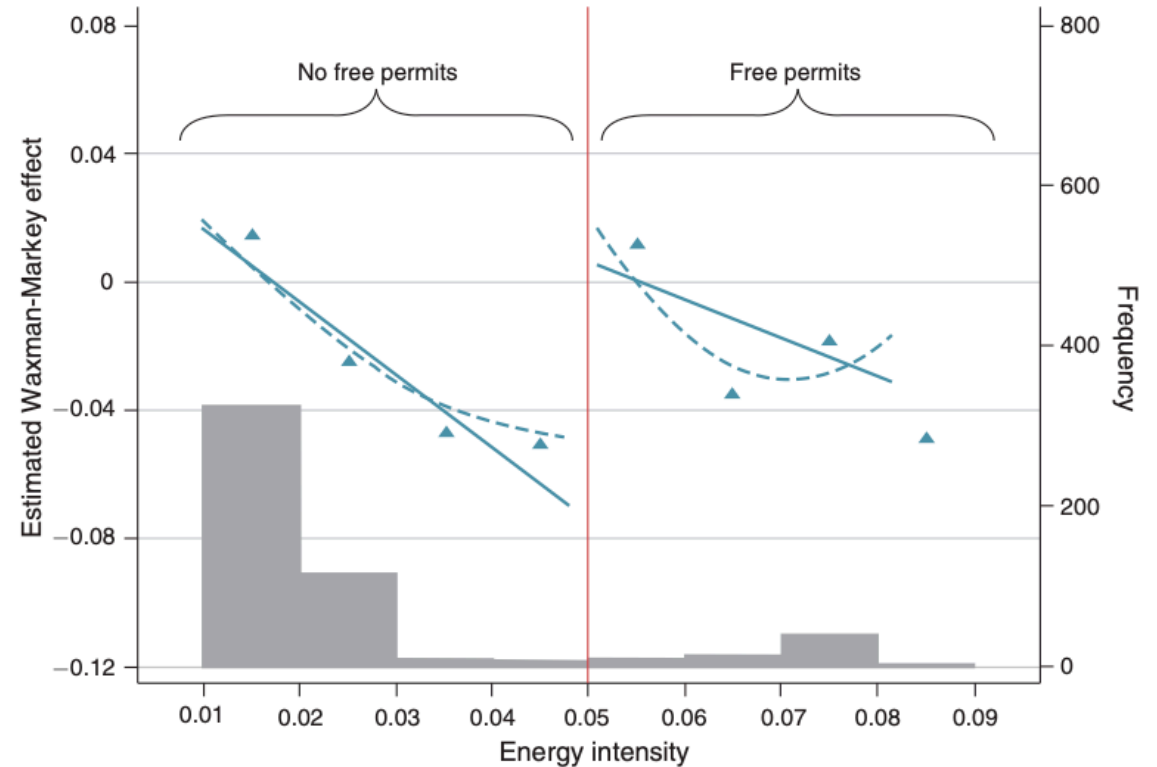


Equity markets: getting MAC

At an energy intensity of 5%, WM reduces firm value by about 6%!

This is the present value of the expected costs of having to buy permits in an auction versus getting them for free

How do we get the MAC (i.e. the implied permit price)?



Equity markets: getting MAC

- τ : permit price / MAC
- E_t : emissions
- r : interest rate

The present value of the expected costs is: $.06 = \sum_{t=0}^{\infty} \frac{\tau \times E_t}{(1+r)^t}$

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If we have forecasts about emissions and the interest rate, we can solve for τ !

Equity markets: getting MAC

Using the listed decline in emissions and a 5% interest rate gives us these MACs

Assumed annual emissions rate (percent)	Corresponding sector (NAICS)	5th percentile	Mean	95th percentile
0	—	1.26	4.59	7.72
−0.70	Petroleum refining (324110)	1.33	4.88	8.19
−1.45	All manufacturing (31–33)	1.42	5.19	8.72
−5.20	Forest products (321, 322)	1.93	7.05	11.84
−7.60	Alumina and Aluminum (3313)	2.32	8.49	14.26
−11.60	Cement (327310)	3.1	11.36	19.08
−12.80	Glass (3272)	3.37	12.34	20.73
−13.90	Transport. equip. (336)	3.63	13.29	22.32
−19.30	Textiles (313–316)	5.09	18.65	31.31
−25	—	6.99	25.57	42.94
−30	—	8.95	32.78	55.04
−35	—	11.25	41.18	69.15

Equity markets: getting MAC

Using the listed decline in emissions and a 5% interest rate gives us these MACs

In 2010 MACs were...quite low!

Assumed annual emissions rate (percent)	Corresponding sector (NAICS)	5th percentile	Mean	95th percentile
0	—	1.26	4.59	7.72
−0.70	Petroleum refining (324110)	1.33	4.88	8.19
−1.45	All manufacturing (31–33)	1.42	5.19	8.72
−5.20	Forest products (321, 322)	1.93	7.05	11.84
−7.60	Alumina and Aluminum (3313)	2.32	8.49	14.26
−11.60	Cement (327310)	3.1	11.36	19.08
−12.80	Glass (3272)	3.37	12.34	20.73
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At most $18/tCO_2$, about 1/10 the current best estimate of MD, 1/3 the best estimate in 2010

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If $MAC < MD$, then the cap is **below** the socially efficient level